



PHASE-III

JEE ADVANCED

GRAND TEST SERIES

GRAND TEST-8



Sri Chaitanya IIT Academy, India

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR-HYD

Sec: Sr.IPLCO_IC_ISB_LIIT

Time:09.00 AM to 12.00 Noon

Date: 01-05-16

Max. Marks:270

Name of the Student: _____

H.T. NO:

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PAPER-I

01-05-16_Sr.IPLCO_IC_ISB_LIIT_GTA-8(P1)_Weekend Syllabus

Mathematics: Total Syllabus

Physics : Total Syllabus

Chemistry : Total Syllabus

PHYSICS

Max Marks : 90

SECTION – A
(TRUE/FALSE ANSWER TYPE)

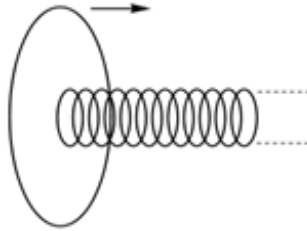
This section contains 5 questions. The answer is a TRUE or FALSE

Marking scheme + 2 for correct answer , 0 if not attempted and -1 in all other cases.

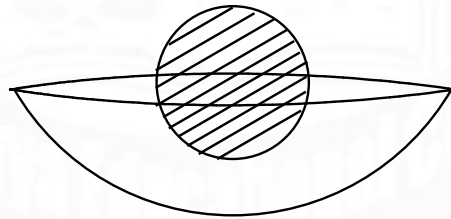
1. A slender uniform rod of mass m and length l is held at rest as an inverted pendulum such that it can freely rotate about a horizontal axis passing through its lower end. It is released with negligible angular velocity. After it has rotated through an angle 45° the torque of force applied by one half of the rod on the other half about axis of rotation is not zero.
2. Two physical pendulums perform small oscillations about the same horizontal axis with angular frequencies ω_1 and ω_2 . Their moments of inertia relative to the given axis are equal to I_1 and I_2 respectively. In a state of stable equilibrium the pendulums were fastened rigidly together. The angular frequency of small oscillations of the compound pendulum is $\frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$

3. A conducting loop having a radius $2a$ is held co-axially near the end of a tightly wound semi-infinite solenoid carrying a current i and having 'n' number of turns per unit length. The radius of the solenoid is a . Now the loop is moved co-axially somewhere towards the middle of the solenoid as shown. If the resistance of the loop is R , the amount of the charge that flows through the loop in the process is

$$\frac{\mu_0 n i \pi a^2}{2R}$$



4. A spherical black body of radius R is arranged concentrically inside a hemispherical shell of a black body of radius $2R$ as shown in the figure. The hemispherical body is maintained at a temperature T_0 . Assuming that the sphere is perfectly conducting and the only mode of heat transfer between the sphere and the rest of the universe is radiation. The temperature of the enclosure around the system is assumed to be zero Kelvin. The steady state temperature of the sphere is $\frac{T_0}{2^{1/4}}$



5. A charged particle of mass m and charge q is projected into a region of uniform field $\vec{E}_0$ with velocity $\vec{v}_0$ perpendicular to $\vec{E}_0$. The particle also experiences a drag force of magnitude qE_0 which is always directed opposite to its instantaneous velocity. If $|\vec{v}_0| = 8m/s$, then its speed when it's velocity vector turns through an angle of 90° is $4m/s$.

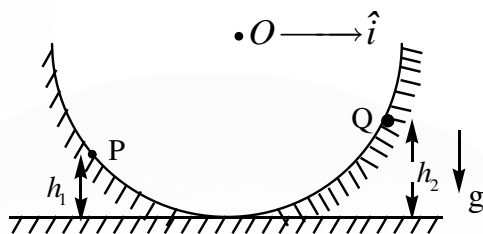
SECTION - B

(COMPREHENSION TYPE)

This section contains **3 or 4 Paragraph of questions**. Each paragraph has 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS correct**. **Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.**

Paragraph for Question Nos. 6 to 8

A point mass moves on a frictionless inner surface of a spherical shell whose inner radius is $R = 1.4m$. Its velocity reaches its minimum and maximum at Q and P at heights $h_2 = 0.3m$ and $h_1 = 0.1m$ respectively. The minimum and maximum velocities achieved are $\bar{v}_1$ and $\bar{v}_2$ respectively. 'O' is centre of curvature of the shell. Velocities at P and Q are $\bar{v}_p$ and $\bar{v}_q$ respectively. Take $g = 10m/s^2$. Take $\hat{i}$ as direction of horizontal in the plane of OPQ as shown in figure.



6. Make the correct statement(s)

a) $\vec{v}_1 = \vec{v}_p$

b) $\overline{OP}$ is perpendicular to $\vec{v}_1$

c) $\vec{v}_2$ is horizontal

d) $\frac{v_2^2 - v_1^2}{2g} = h_2 - h_1$

7. Mark the incorrect statement(s)

a) $v_1 h_1 = v_2 h_2$

b) $v_1 h_1 \sqrt{2R - h_1} = v_2 h_2 \sqrt{2R - h_2}$

c) Plane OPQ must be vertical

d) $v_1 |\widehat{OP} \cdot \hat{i}| = v_2 |\widehat{OQ} \cdot \hat{i}|$

8. Mark the correct statement(s)

a) $v_1 = 4 \text{ m/s}$

b) $v_2 = 3 \text{ m/s}$

c) $v_1 = 2.5 \text{ m/s}$

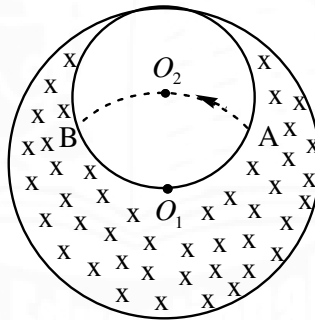
d) $v_2 = 1 \text{ m/s}$

Paragraph for Question Nos. 9 to 10

Inside a long cylinder of radius $R = 10\text{ cm}$, there is a small cylinder of radius $\frac{R}{2}$.

These are arranged in such a way that the two cylinders touch each other along a common generatrix. There is no magnetic field inside the small cylinder, and in the remaining part of the big cylinder there is a uniform magnetic field which is increasing uniformly in time. The rate of change of magnetic field is $\frac{\Delta B}{\Delta t} = 80\text{ V/m}^2$ and the magnetic field is parallel to axis of the cylinder. In the

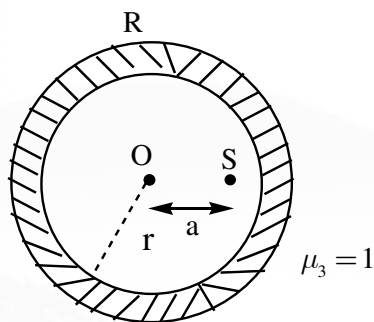
figure shown, a circular cross section is shown. In this plane, O_1 is centre of the bigger cross section, O_2 is centre of the smaller cross section. Arc 'AB' shown is a circular arc of radius $\frac{R}{2}$ with O_1 as centre. A, B lie on smaller cross section.



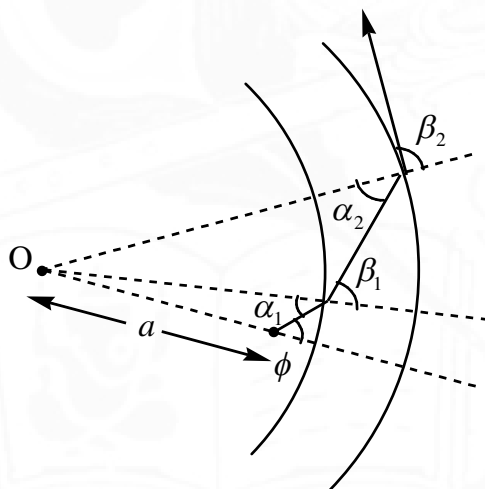
9. Induced electric field at midpoint of O_1O_2 is :
- a) $2V/m$ towards O_2
 - b) $1V/m$ towards right and perpendicular to O_1O_2
 - c) $2V/m$ towards left and perpendicular to O_1O_2
 - d) $1V/m$ towards right and perpendicular to O_1O_2
10. A thin metal wire-1 overlapping arc AB (but not touching the inner cylinder) of resistance $r_1 = \sqrt{3}\Omega$ is placed. Similarly another wire-2 is placed along shortest path connecting AB (but not touching inner cylinder). The resistance of this wire is $r_2 = 2\sqrt{3}\Omega$.
- a) induced emf along wire-1 is $\frac{\pi}{15}$ volt
 - b) induced emf along wire-2 is $\sqrt{3} \times 10^{-1}$ volt
 - c) induced current flowing along wire-1 is zero
 - d) induced current flowing along wire-2 is $0.05A$

Paragraph for Question Nos. 11 to 12

A glass spherical shell of centre O with an outer radius of $R = 7.5cm$ and an inner radius of $r = 6.5cm$ has a refractive index $\mu_2 = 1.5$. The inside of the shell is filled with a material, whose refractive index is $\mu_1 = \frac{25}{16}$. A source of lights S is placed at a distance of $a = 6cm$ from the centre.



11. Consider a light ray starting at S and getting double refracted into surrounding as shown with $\beta_2 = 90^\circ$. Dotted lines represent normals at the points of refraction. Which of the following statement(s) are correct?



- a) $\sin \beta_1 = \frac{1}{\mu_2} \frac{R}{r}$ b) $\sin \alpha_1 = \frac{1}{\mu_1} \frac{R}{r}$ c) $\frac{\sin \phi}{\sin \alpha_1} = \frac{r}{a}$ d) $\sin \phi = \frac{1}{\mu_1} \frac{R}{r}$

12. What percentage of the energy of the light source leaves from right half of the system into medium $\mu_3=1$? (Assuming no absorption of light inside the glass)
- A) 20% b) 40% c) 37.5% d) 75%

Paragraph for Question Nos. 13 to 14

For a sinusoidal transverse mechanical string wave, the path difference between two points in the same phase is 5 m, while the path difference between two points that are out of phase is 1.5 m.

13. Which of the following is/are possible values of wavelength?

- a) $\frac{1}{3}m$ b) $\frac{1}{2}m$ c) 1 m d) 2 m

14. Consider two cases:

Case i: a sinusoidal wave satisfying the condition given in the above passage with maximum possible wavelength. It's wavelength, frequency, Average energy density, energy stored in a distance of 5m are λ_1, f_1, u_1, E_1 respectively.

Case ii: a sinusoidal wave satisfying the condition given in the above passage with second

highest possible wavelength. It's wavelength, frequency, Average energy density, energy stored in a distance of 5m are λ_2, f_2, u_2, E_2 respectively. Assume the string used in both cases is subjected to same tension and is made of same material. Also assume that the amplitudes of waves in both cases is same. Mark the correct option(s):

- a) $\frac{f_1}{f_2} = \frac{1}{2}$ b) $\frac{\lambda_1}{\lambda_2} = 3$ c) $\frac{u_2}{u_1} = 9$ d) $\frac{E_1}{E_2} = 4$

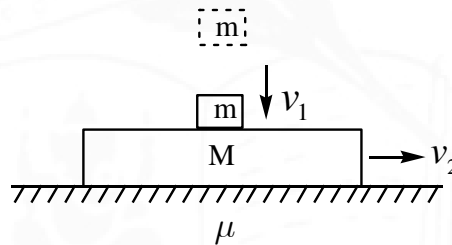
SECTION – C

(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

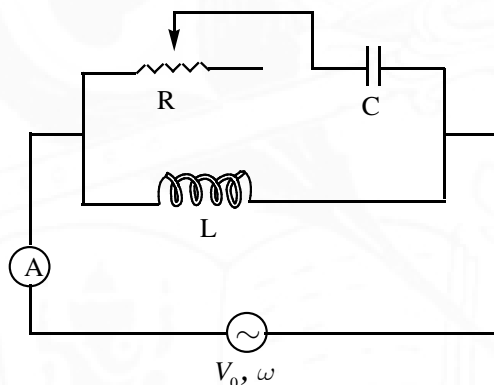
Marking scheme: + 5 for correct answer, 0 if not attempted and --3 in all other cases.

15. A block of mass $M = 5\text{kg}$ is moving on a horizontal plane. An object of mass $m = 1\text{kg}$ is dropped onto the block, hitting it with a vertical velocity of $v_1 = 10\text{ m/s}$. The speed of the block at the same instant is $v_2 = 2\text{ m/s}$ on the floor. The collision is momentary and object sticks to the block upon collision. What will be the speed of the block just after the collision if the coefficient of friction between the block and the horizontal plane is $\mu = 0.4$?



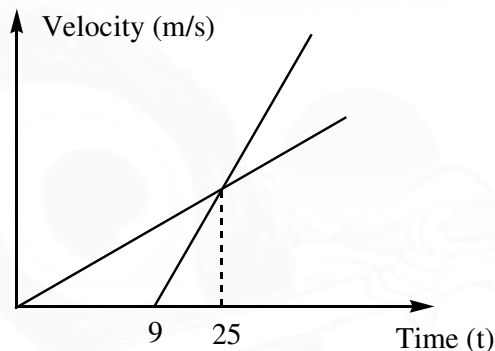
- a) 0.8 m/s b) 1 m/s c) 1.4 m/s d) 1.67 m/s

16. If the cathode of a photocell is illuminated with a light of increasing frequency, the anode current will start at a frequency of $3 \times 10^{14} \text{ Hz}$. A capacitor with capacitance 1 pF is connected between the anode and the cathode of this photocell and the cathode is illuminated with light of frequency $7 \times 10^{14} \text{ Hz}$. Assuming the illumination is long enough, find the approximate number of electrons arriving on the anode? ($h = 6.63 \times 10^{-34} \text{ Js}$, $e = 1.6 \times 10^{-19} \text{ C}$)
- a) 1×10^{10} b) 2×10^{12} c) 3×10^{15} d) 4×10^{17}
17. In the shown 'AC' circuit L , C and V_0 are fixed. What is the value of ω such that reading of hot wire ammeter shown is independent of 'R'?



- a) $\frac{1}{\sqrt{LC}}$ b) $\sqrt{\frac{2}{LC}}$ c) $\sqrt{\frac{3}{LC}}$ d) $\frac{2}{\sqrt{LC}}$

18. A material particle is chasing the other one and both of them are moving on the same straight line. Their motion after they pass a particular point is recorded and the data obtained is shown by velocity-time graphs of the particles. How long after the start will the chase end?

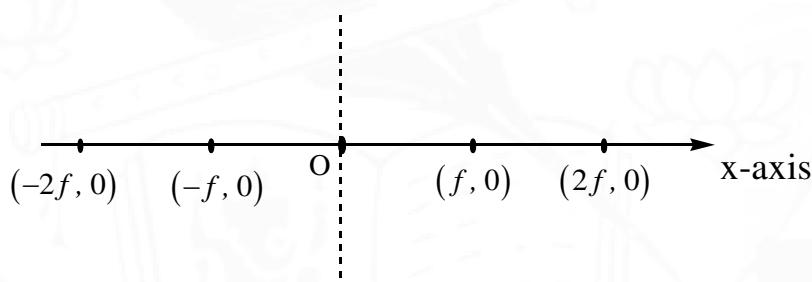


- a) 16 s b) 25 s c) 45 s d) Insufficient information

SECTION - D
(Matching List Type)

Section-D (Matrix-Match Type, Total Marks: 24) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I, Four statements (p, q, r, s) in Column II and Four statements (w, x, y, z) in Column III. Any given statement in Column I can have correct matching with Only ONE statement given in Column II & Column III. For example, if for a given question, statement B matches with the statements given in q and x, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 3 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and -1 marks otherwise. Thus, each question in this section carries a maximum of 12 marks.

19. In the diagram shown the dotted line represents an optical apparatus (kept in vacuum) which is either a convex lens or a concave lens or convex mirror or a concave mirror as mentioned in column I. Focal length of each of these apparatus is of magnitude f . optical centre or pole in each case is $(0, 0)$ and x-axis is their principal axis. Paraxial light rays representing a short linear object (Real or virtual) on the principal axis are incident on these apparatus separately in each case from negative x-direction. The objects in each case are perpendicular to principal axis. Match column-I separately with column-II and column-III



Column-I (Apparatus and object perpendicular)	Column-II (Characteristics of image and relation with object)	Column-III (other characteristics)
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					of image formed)
A)	If the apparatus is convex mirror and the object is virtual and is at $\left(\frac{f}{2}, 0\right)$	P)	Real image is formed	W)	Erect
B)	If the apparatus is concave lens and the object is real and is at $(-3f, 0)$	Q)	Virtual image is formed	X)	Inverted
C)	If the apparatus is convex lens and the object is virtual and is at $(f, 0)$	R)	Magnitude of object distance is greater than that of magnitude of image distance	Y)	Magnified
D)	If the apparatus is concave mirror and the object is real and is at $(-2f, 0)$	S)	Image formed and object are on the same side of apparatus	Z)	Diminished

20. If a physical quantity X has a dimensional formula $X = [M^2 L^{-1} T^5 K^2]$, then the sum of the powers $2-1+5+2=8$ is called the “Richness factor of its dimensions”. Column-I describes a physical quantity you have to match its “richness factor of its dimensions” or order of magnitude or number of significant figures in Column-II (Quantity α) and column-III (Quantity β).

	Column-I (Description of a physical quantity)		Column-II (Quantity α)		Column-III (Quantity β)
A)	The Quantity measured in “Half deflection method of a Galvanometer” was found to be 5.480×10^6 SI units. For this quantity α is “richness factor of its dimensions” and β is the number of significant figures	P)	4	W)	2
B)	A Quantity is measured as product of Reynold’s number for a tube of flow and coefficient of viscosity of the fluid flowing it is equal to 3294.1 SI units. For this Quantity α is “richness factor of its dimensions” and β is the order of magnitude in SI units	Q)	1	X)	3

C)	A Quantity is obtained by dividing surface tension of a liquid with its density and then further dividing it with acceleration due to gravity at that place. It was found to be 2.5×10^{-3} SI units. For this Quantity α is its order of magnitude in C.G.S units and β is the number of significant figures in C.G.S units	R)	-1	Y)	4
D)	A Quantity is obtained by dividing emissive power of a black body of temperature 500K with Stefan's constant in SI units. For this quantity α is "richness factor of its dimensions" and β is the order of magnitude in SI units	S)	-2	Z)	11

CHEMISTRY

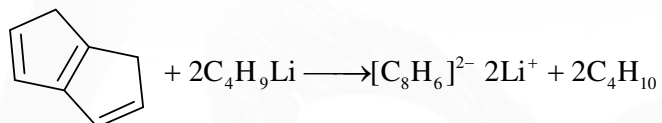
Max Marks : 90

SECTION – A
(TRUE/ FALSE ANSWER TYPE)

This section contains 5 questions. The answer is a TRUE or FALSE

Marking scheme + 2 for correct answer , 0 if not attempted and -1 in all other cases.

21. The reaction shown below



is feasible

22. Decrease in freezing point (ΔT_f) of ethanoic acid is more in benzene than in water.
23. Due to Frenkel defect, there is no effect on density of the crystalline solid .
24. Acid catalyzed hydration of allene ($H_2C = C = CH_2$) yields acetone.
25. Nucleophiles always attack carbonyl groups at carbon and attack not from the direction perpendicular to the plane of carbonyl group but at about 107° (obtuse angle).

SECTION - B

(COMPREHENSION TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS correct**. Marking scheme: **+4 for correct answer, 0 if not attempted and -2 in all other cases**.

Paragraph for Question Nos. 26 to 28

A metallic chloride (A) gives a white ppt with limited amount of another metal chloride (B) and grey ppt with excess amount of (B). (A) when treated with KI gives a scarlet ppt, which dissolve in excess of KI forming an important reagent (C).

26. What is the hybridization of central atom of the reagent?

- A) sp^3 B) dsp^2 C) sp^3d D) dsp^3

27. Reagent (C) is used for detection of

- A) Mg^{2+} B) NH_4^+ C) Na^+ D) Al^{3+}

28. $A + Na_2S \longrightarrow D + NaCl$

Which is true about D?

- A) yellow coloured ppt, in soluble in ammonium poly sulphide and soluble in HNO_3 .
B) Black coloured ppt, soluble in ammonium poly sulphide and in HNO_3
C) Yellow ppt, soluble in ammonium poly sulphide and insoluble in HNO_3
D) Black ppt, insoluble in ammonium poly sulphide and insoluble in HNO_3

Paragraph for Questions Nos. 29 to 31

Consider a system and its surroundings at the normal transition temperature, T_{trs} is the temperature at which two phases are in equilibrium at 1 atm. At the transition temperature, any transfer of heat between the system and its surroundings is reversible because the two phases in the system are in equilibrium. Because at constant pressure $q = \Delta H_{\text{trs}}$, the change in molar entropy of the system is

$$\Delta_{\text{trs}} S = \frac{\Delta_{\text{trs}} H}{T_{\text{trs}}}$$

Therefore,

$$\Delta S_{\text{vap}} = \frac{\Delta H_{\text{vap}}}{T_b}$$

Although heats of vapourisation of liquids vary over a considerable range, an interesting generalisation was pointed out by Trouton in 1884 (and is known as Trouton's rule) is that a wide range of liquids have the same standard entropy of vapourisation (about $88 \text{ K}^{-1} \text{ mol}^{-1}$). Liquids that show significant deviations from Trouton's rule do so on account of the molecules in the liquid being arranged in a partially ordered manner or the liquids having very low boiling

temperature and the liquids whose molecules are partially associated in the vapour phase (e.g. CH_3COOH).

29. Which of the following liquid is expected to follow Trouton's rule

- A) H_2O B) PhCOOH C) H_2S D) CH_3COOH

30. Match the following for the isomeric liquids X, Y and Z are same

Boiling point	K_b
X. 100	i) 0.68
Y 27	ii) 0.53
Z 253	iii) 0.48

Codes

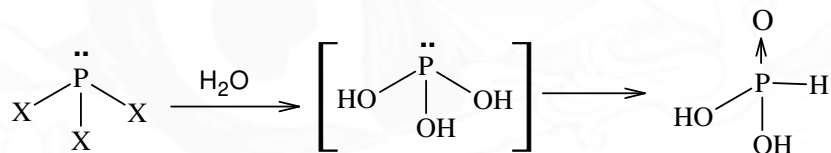
- A) (X – i), (Y – ii), (Z – iii) B) (X – iii), (Y – ii), (Z – i)
C) (X – ii), (Y – i), (Z – iii) D) (X – ii), (Y – iii), (Z – i)

31. If Mol. wt. and ΔH_v of a solvent is 50gm and 32R/gm (R = gas constant) respectively then the entropy of vaporization of the liquid in cal K^{-1} will be approximately (solvent boils at 47°C)

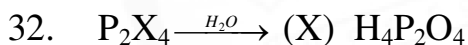
- A) 20 B) 44.4 C) 5.3 D) 10

Paragraph for Question Nos. 32 to 34

It is fact that second-row elements, Li to F, show a maximum covalency of four, corresponding to maximum hybridization of sp^3 . In contrast, third row and heavier elements show 5, 6 or 7 coordination consistent with use of d-orbitals. The chemistry of sulphur and phosphorous are dominated by their special ability to form $p_\pi - d_\pi$ bonding. Many physical and chemical properties are affected by this kind of bonding. For example, trimethylamine is pyramidal, while trisilylamine is planar. The lower phosphorous acid exists in 4-coordinate structures even though they are prepared by the hydrolysis of 3-coordinate halides.



After reading above paragraph, now answer the following questions.

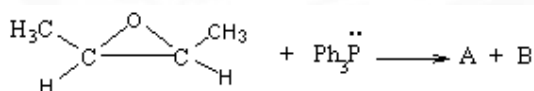


Which of the following statement are INCORRECT for the oxyacid (X) of phosphorous.

- A) (X) is a tetrabasic acid
- B) (X) is a dibasic acid having all the P – O bond lengths equal.
- C) (X) is a dibasic acid and also reducing in nature having all P-O bond lengths equal.
- D) (X) is a dibasic acid, also reducing in nature and having two types of P-O bond lengths.

33. Which of the following statement are INCORRECT for methyl isothiocyanate and silyl isothiocyanate?

- A) Both are linear, but silyl isothiocyanate is more basic than methyl isothiocyanate.
- B) Both molecules are bent, but silyl isothiocyanate is more basic than methyl isothiocyanate.
- C) Methyl isothiocyanate is bent, while silyl isothiocyanate is linear (around Nitrogen atom) and latter is less basic than former.
- D) Methyl isothiocyanate is bent while silyl isothiocyanate is linear and latter is more basic than former.



34.

A and B may be are

- A) Cis-2-butene + triphenylphosphine oxide
- B) trans-2-butene + triphenylphosphine oxide
- C) 1-butene + triphenylphosphine oxide
- D) Isobutylene + triphenylphosphine oxide

SECTION – C**(SINGLE CORRECT ANSWER TYPE)**

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: + 5 for correct answer, 0 if not attempted and --3 in all other cases.

35. A radioactive material decays by simultaneous emission of two particles with half lives 1620 and 810 years. The time after which one fourth of the material remains, is
- A) 1080 years B) 2430 years C) 3240 years D) 4840 years
36. Which of the following statements are true
- i) Sucrose does not reduce Fehling solution although its both the constituent monosaccharide glucose and fructose reduces
 - ii) One molecule of glucose can react with more than one mols of phenylhydrazine
 - iii) glucose and fructose are empimers
 - iv) fructose shows mutarotation
- A) (i), (ii), (iii) B) (ii), (iii), (iv) C) (i), (ii), (iv) D) (i), (ii), (iii), (iv)

37. Kaolin $[\text{Al}_2(\text{OH})_4\text{Si}_2\text{O}_5]$ is an example of
A) ortho silicate B) pyro silicate C) sheet silicate D) cyclic silicate
38. Which of the following state is/are false?
i) Except glycine all α -amino acids are optically active.
ii) All α -amino acids have primary structure
iii) At isoelectric point α -amino acid mostly exist in zwitterionic form
iv) Reactions of amino acids are similar to primary amines and carboxylic acids
A) (i), (iii), (iv) B) (ii), (iv) C) (ii) D) (i), (ii) and (iv)

SECTION - D
(Matching List Type)

Section-D (Matrix-Match Type, Total Marks: 24) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I, Four statements (p, q, r, s) in Column II and Four statements (w, x, y, z) in Column III. Any given statement in Column I can have correct matching with Only ONE statement given in Column II & Column III. For example, if for a given question, statement B matches with the statements given in q and x, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 3 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and -1 marks otherwise. Thus, each question in this section carries a maximum of 12 marks.

39.

	Column – I		Column – II		Column – III
A)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{C}-\text{OH} \\ \\ \text{CH}_3 \end{array}$	P)	on treatment with NaOH/I ₂ produces yellow ppt	W)	On heating with Cu – 300°C produces propanal
B)	CH ₃ -CH ₂ -CH ₂ -OH	Q)	cannot decolourise acidified K ₂ Cr ₂ O ₇	X)	On heating with Cu-300°C produces propanone
C)	$\begin{array}{c} \text{OH} \\ \\ \text{H}_3\text{C}-\text{CH} \\ \\ \text{CH}_3 \end{array}$	R)	can be oxidized by MnO ₂	Y)	On heating with Cu – 300°C produces 2-methyl propene
D)	CH ₃ -CH=CH-CH ₂ -OH	S)	cannot give Lucas test at room temperature	Z)	can decolourise bromine water

40.

	Column – I		Column – II		Column – III
A)	Bakelite	P)	polytetrafluoro ethylene	W)	used in electrical fittings
B)	Teflon	Q)	semi-synthetic polymer	X)	used in tyre manufacture
C)	Nylon-6	R)	thermo-setting polymer	Y)	used as explosive
D)	Gun cotton	S)	polyamide	Z)	used in nonstick utensils

MATHEMATICS

Max Marks : 90

SECTION – A
(TRUE/ FALSE ANSWER TYPE)

This section contains 5 questions. The answer is a TRUE or FALSE

Marking scheme + 2 for correct answer , 0 if not attempted and -1 in all other cases.

41. The number of Independent arbitrary constants involved, in the equation of a straight line in three dimensional space, which is not perpendicular to any of the coordinate axes, can exceed four (T/F)
42. $A_1, A_2, A_3, \dots, A_{18}$ are the vertices of a 18 sided regular polygon "B" is an external point such that $A_1 A_2 B$ is an equilateral and if $A_{18} A_1$ and $A_1 B$ are adjacent sides of another regular polygon of n sides, then $n=9$ (T/F)
43. Two parabolas with same axis, focus of each being exterior to the other and the latusrectum being $4a$ and $4b$ are given. The locus of the midpoints of the intercepts between the parabolas made on the lines parallel to the common axis is Parabola if a is not equal to b (T/ F)
44. Consider $f(x) = x^2 - |x^2 - 1| + 2||x| - 1| + 2|x| - 7$ and the function $f(x)$ is differentiable at every point on $\mathbb{R}$ (T/ F)

45. A man firing a distant target has 20% chance of hitting the target in one shoot. If p be the probability of hitting the target in 'n' attempts, where $20p^2 - 13p + 2 \leq 0$ then the possible value of n is 2 (T/ F)

SECTION - B
(COMPREHENSION TYPE)

This section contains **3 Paragraph of questions**. Each paragraph has 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE IS correct**. **Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.**

Paragraph for Question Nos. 46 to 48

Let $f : R \rightarrow R$ be a function as $f(x) = (x-1)(x+2)(x-3)(x-6) - 100$. If $g(x)$ is a polynomial of degree ≤ 3 such that $\int \frac{g(x)}{f(x)} dx$ does not contain any logarithmic function and $g(-2) = -10$ then

46. Which of the following is/are correct
- A) $f(x) = 0$ has all four distinct real roots
 - B) $f(x) = 0$ has two real and two imaginary roots
 - C) $f(x)$ is many one and in to function
 - D) real roots of the equation $f(x) = 0$, have opposite sign

47. Which of the following is/are correct

- A) Degree of $g(x)$ is 2
- B) Minimum value of $f(x)$ is -84
- C) Degree of $g(x)$ is 3
- D) Minimum value of $f(x)$, occurs when $x = 2$

48. Which of the following is/are correct

- A) Minimum value of $g(x)$ is -42 When $x = 2$
- B) Minimum value of $g(x)$ is -21 when $x = 2$
- C) $\int_0^4 \frac{g(x)}{f(x)} dx = \frac{\pi}{2}$
- D) $\int_0^4 \frac{g(x)}{f(x)} dx = \pi$

Paragraph for Question Nos. 49 to 51

Let OABC be a tetrahedron with edges length

$BC = \sqrt{2}$, $CA = \sqrt{3}$, $AB = \sqrt{4}$, $OA = \sqrt{5}$, $OB = \sqrt{6}$, $OC = \sqrt{7}$ and G be centroid of the triangle ABC. MN are the midpoints of OB & CA

49. Which of the following is/are correct

A) $MN = \frac{3}{2}$ B) $(OG)^2 = 5$ C) $(OG)^2 + 2(MN) = 8$ D) $MN = \frac{3}{4}$

50. Which of the following is/are correct

A) volume of the tetrahedron OABC is $\frac{\sqrt{11}}{4}$

B) volume of the tetrahedron OABC is $\sqrt{\frac{11}{4}}$

C) perpendicular distance between the point O (origins) and the plane ABC is $\frac{3\sqrt{11}}{\sqrt{23}}$

D) perpendicular distance between the point O (origins) and the plane ABC is $\sqrt{\frac{11}{92}}$

51. Which of the following is/are correct

A) angle between the face OAB and edge BC is $\sin^{-1}\left(\frac{3\sqrt{11}}{\sqrt{142}}\right)$

B) angle between the edges OA and BC is 90°

C) angle between the face OAB and the edge BC is $\cos^{-1}\left(\frac{3\sqrt{11}}{\sqrt{142}}\right)$

D) angle between the edges OA and BC is $\cos^{-1}\left(\frac{1}{\sqrt{10}}\right)$

Paragraph for Question Nos. 52 to 54

Consider $f_1(x) = \sum_{r=1}^n \sec\left(x + \frac{r\pi}{6}\right) \sec\left(x + (r-1)\frac{\pi}{6}\right)$ & $f_1(x) - 2f_2(x) = 2\tan\left(x + \frac{n\pi}{6}\right)$ &

$$f_2(x) + f_3(x) = 0 \text{ \& } f_4(x) = \begin{cases} \exp.\left(\frac{e^x + f_2(x) + \tan x - 1}{2(e^x - 1)}\right), & x < 0 \\ k_1, & x = 0 \\ (1 + |f_2(x)|)^{\frac{k_2}{f_3(x)}}, & x > 0 \end{cases}, \text{ then}$$

52. Which of the following is/are correct

A) $f_4(x)$ is continuous at $x=0$, **then** $k_1 = \sqrt{e}$, $k_2 = \frac{1}{2}$

B) $f_4(x)$ is continuous at $x=0$, **then** $k_1 = \frac{\sqrt{e}}{2}$, $k_2 = 1$

C) $f_4(0^-) = \sqrt{e}$

D) $f_4(0^+) = e^{k_2}$

53. Which of the following is/are correct

A) $f_3(x) = \tan x$

B) $f_3(x) = -\tan x$

C) $f_3(x)$ is continuous and differentiable in $\left(0, \frac{\pi}{2}\right)$

D) $f_3(x)$ is continuous but not differentiable at $x = \frac{2\pi}{5}$

54. Which of the following is/are true?

A) $f_1(x) = 2\tan\left(x + \frac{n\pi}{6}\right) - 2\tan x$

B) $f_1(x) = 2\tan\left(x + \frac{n\pi}{6}\right) - \tan x$

C) for $n=3$, the number of solutions of the equation $f_1(x)+4=0$, in $\left(0, \frac{\pi}{2}\right)$ is 1

D) for $n=3$, the number of solution of the equation $f_1(x)+4=0$, in $\left(0, \frac{\pi}{2}\right)$ is 3

SECTION – C

(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +5 for correct answer, 0 if not attempted and --3 in all other cases.

55. Consider the circle $|Z-5i|=3$ and two points Z_1 and Z_2 on it such that

$|Z_1| < |Z_2|$ and $\arg(Z_1) = \arg(Z_2) = \frac{\pi}{3}$. A tangent is drawn at Z_2 to the circle, which cuts the real axis at Z_3 , then $|Z_3| = \text{-----}$

A) 3

B) $\sqrt{13}$

C) $\sqrt{11}$

D) 4

56. Consider a circle C_1 with equation $(x-1)^2 + y^2 = 1$ and a shrinking circle C_2 with radius r and centre at the origin. P is the point $(0, r)$, “Q” is the upper point of intersection (above x-axis) of the two circles and “R” is the point of intersection of the line PQ with x-axis. then x-coordinate of the point R, as C_2 shrinks, (i.e. $r \rightarrow 0$)?

A) 4

B) 2

C) 0

D) 1

57. $\frac{1}{\sin 2^\circ} + \frac{1}{\sin 4^\circ} + \frac{1}{\sin 8^\circ} + \frac{1}{\sin 16^\circ} + \frac{1}{\sin 32^\circ} + \frac{1}{\sin 64^\circ} + \frac{1}{\sin 128^\circ} = \text{-----}$

A) $\sin 127^\circ \sin 1^\circ \sin 128^\circ$

B) $\cos ec 127^\circ \cos ec 1^\circ \cos ec 128^\circ$

C) $\sin 127^\circ \sin 1^\circ \cos ec 128^\circ$

D) $\sin 127^\circ \cos ec 1^\circ \cos ec 128^\circ$

58. Let $a = 3^{\frac{1}{223}} + 1, \forall n \geq 3$

& $f(n) = {}^nC_0 a^{n-1} - {}^nC_1 a^{n-2} + {}^nC_2 a^{n-3} + \dots + (-1)^{n-1} {}^nC_{n-1} a^0$

If the value of $f(2007) + f(2008) = k \cdot 3^4, k \in N$ then $K = \text{-----}$

A) 243

B) 81

C) 729

D) 0

SECTION - D (Matching List Type)

Section-D (Matrix-Match Type, Total Marks: 24) contains 2 questions. Each question has four statements (A, B, C and D) given in Column I, Four statements (p, q, r, s) in Column II and Four statements (w, x, y, z) in Column III. Any given statement in Column I can have correct matching with Only ONE statement given in Column II & Column III. For example, if for a given question, statement B matches with the statements given in q and x, then for the particular question, against statement B, darken the bubbles corresponding to q and r in the ORS. For each question you will be awarded 3 marks for each row in which you have darkened ALL the bubble(s) corresponding to the correct answer(s) ONLY and -1 marks otherwise. Thus, each question in this section carries a maximum of 12 marks.

59.

COLUMN-I	COLUMN-II	COLUMN-III
a) If $\alpha, \beta, \gamma, \delta$ are real numbers such that $\alpha^2 + \beta^2 = 9, \gamma^2 + \delta^2 = 4$ & $\alpha\delta - \beta\gamma = 6$	p) $m = 2$	x) $M = 5$

then maximum value of $\alpha\gamma$ is M minimum value of $\alpha\gamma$ is m		
b) If range of the function $f(x) = \sin^{-1}x + 2\tan^{-1}x + x^5 - 5x^4 + 5x^3 + 1$ is $[a, b]$ & $[a] = m, [b] = M$	q) $m = 6$	y) $M = 3$
c) If for each r, $p(r)$ be the number of points $(x, y), (x, y \in I)$ which lie inside or on the curve $ xy + r^2 = r x + r y , r \in N$ and $\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \sqrt{p(r)}}{an^3 + bn^2 + c} = \frac{1}{2}$ then the value of $ a - b = m, a + b = M$	r) $m = -3$	z) $M = 2$
d) Let a continuous function $f(x)$ satisfies the equation $\int_0^2 f(x)(6x - f(x))dx = 24$ then $m = \int_0^2 f(x)dx$ & $M = f(2)$	s) $m = -14$	w) $M = 6$

60. Matching

COLUMN-I	COLUMN-II	COLUMN-III
a) Consider two points A and B which are 4cm apart. The vertex "C" of a $\Delta^{le} ABC$ moves such that $\sin(A+B) = 2 \sin A \sin B$, then m = minimum distance between the locus of "C" & the point A M = number of point(s) of intersection of a circle whose radius is 4, centre at A with locus of C.	p) $m = -1$	x) $M = 5$
b) If $\vec{v}_1 = i - 2j + 3k$, $\vec{v}_2 = ai + bj + ck$, $\vec{v}_2$ is non-zero vectors & $a, b, c \in \{-1, 0, 1, 2, 3\}$ a,b,c are chosen such that $\vec{v}_1 \cdot \vec{v}_2 = 0$ then	q) $m = 5$	y) $M = 2$

$m = \min \{a+b+c\}, M = \max \{a+b+c\}$		
c) If the distance between one extremity major axis and one extremity of minor axis of an ellipse whose director circle, discribed on a normal chord of $y^2 = 4x$ as diameter and passes through vertex of the parabola is $\frac{a\sqrt{b}}{c} (a, b, c, \in N)$ then minimum value of $a+b+c$ is m & radius of diameter circle r & $[r] = M$. ([.] denotes G.I.F)	r) $m = 2$	Z) $M = 6$
d) Consider $f(x) = \sin^{-1}\left(\frac{x+3}{2x+5}\right)$ $g(x) = \sin^{-1}\left(\frac{ax^2+b}{x^2+5}\right)$ & $\lim_{x \rightarrow \infty} (f(x) - g(x)) = 0,$ $\lim_{x \rightarrow 0} (f(x) + g(x)) = \frac{\pi}{4}$ then $10a = m, 10b^2 = M$	s) $m = 7$	w) $M = 7$



PHASE-III

JEE ADVANCED

GRAND TEST SERIES

GRAND TEST-8



Sri Chaitanya IIT Academy, India

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR-HYD

Sec: Sr.IPLCO_IC_ISB_LIIT
Time:02.00 PM to 05.00 PM

Date: 01-05-16
Max. Marks:246

Name of the Student: _____

H.T. NO:

PAPER-II

01-05-16_Sr.IPLCO_IC_ISB_LIIT_GTA-8(P2)_Weekend Syllabus

Mathematics: Total Syllabus

Physics : Total Syllabus

Chemistry : Total Syllabus

PHYSICS

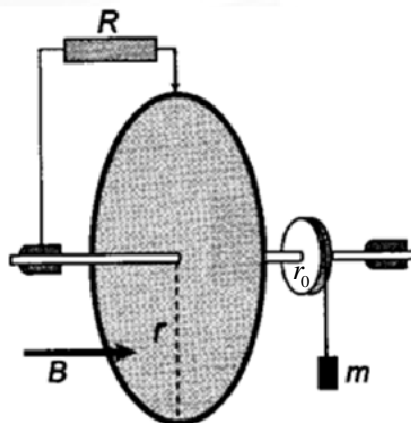
Max Marks : 82

SECTION – I
(INTEGER ANSWER TYPE)

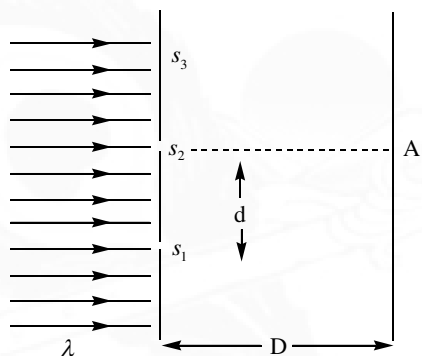
This section contains 4 questions. The answer is a Three digit integer ranging from 000 to 999 (both inclusive-Ex: If answer is 1 you should mark as 001).

Marking scheme + 4 for correct answer , 0 if not attempted and 0 in all other cases.

1. A conducting disc of radius r can rotate about a horizontal conducting shaft. A smaller non conducting disc of radius r_0 is fixed onto the same shaft and has an ideal mass less cord wrapped around it, which is attached to a small object of mass m hanging vertically as shown. Two ends of the resistor 'R' are connected to the perimeter of the disc and to the shaft by wiping contacts. The system is then placed into a uniform horizontal magnetic field B and mass m is released. If $r = 10\text{ cm}$, $r_0 = 2\text{ cm}$, $R = 0.01\Omega$, $B = 0.2\text{ T}$, $m = 50\text{ gm}$ then find the constant angular velocity (in rad/s) with which the disc will rotate after a certain time?



2. In standard YDSE ($D \gg d \gg \lambda$) with identical slits S_1 and S_2 light reaching a point A on the screen opposite to slit S_2 has an intensity I. It was also found that when only one of the two slits S_1 or S_2 was illuminated by same light beam, the intensity at A is still I. Now when a third slit S_3 of four times the slit width is made as shown ($S_1 S_2 = S_2 S_3 = d$) and all the three slits are illuminated then intensity of light reaching A is nI where $n =$ _____



3. Moseley's law for K_α photon is given by $\sqrt{\gamma} = a(Z - b)$ where a is a universal constant and b is a screening constant. Moseley's logic helps us assume $b = 1$ for K_α photon. If there is a percentage error $= 10^{-2}$ in the measurement of ' b ' due to actual electronic orbital configuration of an atom, then the relative error in the measurement of γ if $Z = 51$ is $a \times 10^{-b}$ in scientific notation where a is a rational number and b is a positive integer. The value of $a + b$ is _____

4. If three sources of frequency 400Hz, 402Hz, 405 Hz of similar loudness are sounded together, the number of beats produced per second is _____

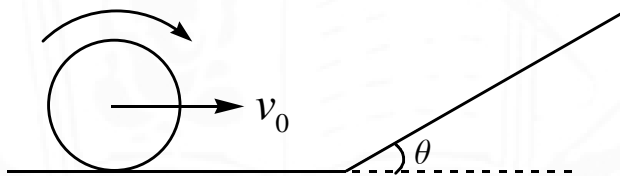
(Assume the minimum intensity detectable is nearly zero)

SECTION – II
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

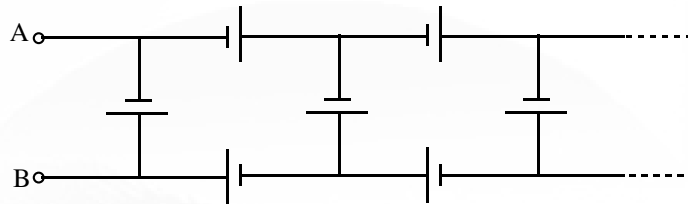
Marking scheme: + 4 for correct answer, 0 if not attempted and -2 in all other cases.

5. In a model experiment, a rigid surface consists of a rough horizontal plane and an inclined plane connecting to it. A thin hoop of radius r is rolling towards the slope without slipping at a velocity v_0 , perpendicular to the base of the slope. The slope connects to the horizontal plane with a smooth curve of radius $R > r$, which is considered part of the slope, such that the hoop will move up the slope without collision. In three different models of the experiment, everything is kept same except the friction coefficient of slope which is $\mu_1 = 0$, $\mu_2 = \tan \theta$, $\mu_3 = 2 \tan \theta$ respectively. In each case the height to which hoop rises on the incline before stopping momentarily is Δh_1 , Δh_2 , Δh_3 respectively. Then



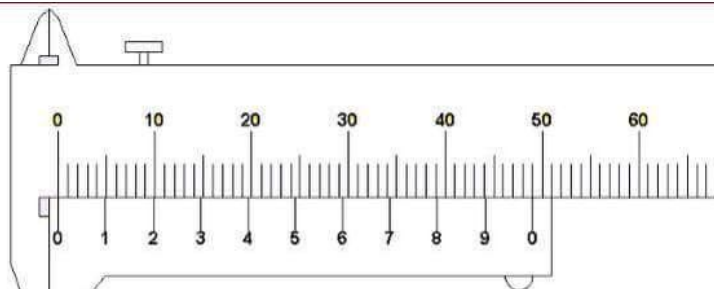
- a) $\Delta h_1 < \Delta h_2 < \Delta h_3$ b) $\Delta h_1 > \Delta h_2 > \Delta h_3$
c) $2\Delta h_1 = \Delta h_2$ d) $\Delta h_2 = \Delta h_3$

6. All the batteries in the infinite chain shown in the figure have emf ε and internal resistance r .



- effective internal resistance of battery system across AB is $\sqrt{3}r$
 - effective emf of battery system across AB is $\sqrt{3}\varepsilon$
 - If a resistance of value r is connected across AB the terminal voltage across the battery system is ε
 - Short circuit current across AB for the above system is $\frac{\varepsilon}{r}$
7. A charged particle (charge q , mass m) is placed at the centre of a circular region containing a magnetic field. The field is given as $\vec{B} = \vec{B}(r)$ ie; it is a function of radial distance and it is perpendicular to the plane of the circle. The total magnetic flux through the circle is zero. The particle is given a radial impulse. At a later instant the radial and tangential components of velocity of the particle are v_r and v_θ respectively.
- The tangential force on the particle at an instant is $qv_\theta B$
 - The torque about centre on the particle is $qBrv_r$
 - If the particle leaves the circular region then at the instant it leaves, its velocity is only in radial direction
 - If the particle leaves the circular region then at the instant it leaves, its velocity is only in tangential direction

8. In the diagram shown take $1\text{MSD} = 1\text{mm}$ and there is no zero error. The left most zero and the right most zero in the vernier scale are the only divisions coinciding with a MSD. Which of the following statement(s) regarding the shown vernier calipers is (are) true?



- a) Its least count is 0.1 mm
- b) Its least count is 4.9 mm
- c) Its vernier constant ($|MSD - VSD|$) is 2.9 mm
- d) Its vernier constant ($|MSD - VSD|$) is 3.9 mm

SECTION - III

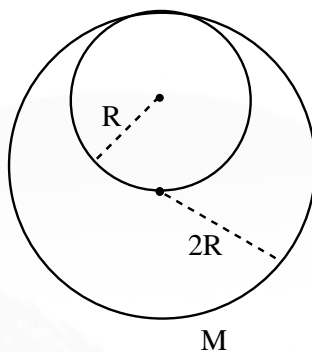
(COMPREHENSION TYPE)

This section contains **2 or 3 Comprehensions**. Each comprehension has 2 or 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +5 for correct answer , 0 if not attempted and -2 in all other cases.

Paragraph for Question Nos. 9 to 10

On a fixed cylindrical rod of radius R hangs a thin uniform bangle of radius $2R$ and mass m . The bangle can swing freely in a vertical plane parallel to the cross section of the cylinder. Now the bangle is given a slight disturbance in its plane.



9. What will be time period of small oscillations of the bangle if the rod is frictionless?

- a) $2\pi\sqrt{\frac{R}{g}}$ b) $2\pi\sqrt{\frac{2R}{g}}$ c) $2\pi\sqrt{\frac{4R}{g}}$ d) $2\pi\sqrt{\frac{8R}{g}}$

10. What will be time period of small oscillations of the bangle if the rod has sufficient friction to prevent slipping?

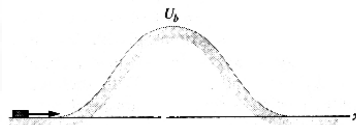
- a) $2\pi\sqrt{\frac{R}{g}}$ b) $2\pi\sqrt{\frac{4R}{g}}$ c) $2\pi\sqrt{\frac{8R}{g}}$ d) $2\pi\sqrt{\frac{2R}{g}}$

Paragraph for Question Nos. 11 to 12

Suppose you slide a puck over frictionless ice toward an ice-covered hill as shown in figure 1 below. As the puck climbs the hill, kinetic energy K is transformed into

gravitational potential energy U . If the puck reaches the top, its potential energy is U_b .

Figure 1: A puck slides over frictionless ice toward a hill. The puck's gravitational potential at the top of the hill will be U_b



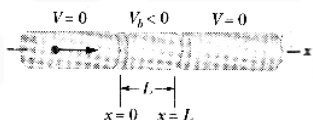
Thus, the puck can pass over the top only if its initial mechanical energy $E > U_b$.

Otherwise, the puck eventually stops its climb up the left side of the hill and slides back to the left. For instance, If $U_b = 20\text{ J}$ and $E = 10\text{ J}$, you cannot expect the puck to pass over the hill. We say that the hill acts as a potential energy barrier (or, for short, a potential barrier) and that, in this case, the barrier has a height of $U_b = 20\text{ J}$.

Figure 2 shows a potential barrier for a nonrelativistic electron traveling along an idealized wire of negligible thickness. The electron, with mechanical energy E , approaches a region (the barrier) in which the electric potential V_b is negative. Because it is negatively charged, the electron will have a positive potential energy $U_b (= qV_b)$ in that region. If $E > U_b$, we expect the electron to pass through the

barrier region and come out to the right of $x=L$ in figure 2. Nothing surprising there. If $E < U_b$, we expect the electron to be unable to pass through the barrier region. Instead, it should end up traveling leftward, much as the puck would slide back down the hill in figure 1 if the puck has $E < U_b$.

Figure 2 : The elements of an idealized thin wire in which an electron (the dot) approaches a negative electric potential V_b in the region $x=0$ to $x=L$



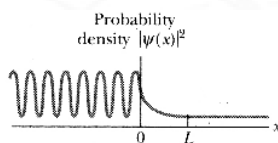
However, something astounding can happen to the electron when $E < U_b$. Because it is a matter wave, the electron has a finite probability of leaking (or, better, tunneling) through the barrier and materializing on the other side, moving rightward with energy E as though nothing (strange or otherwise) had happened in the region of $0 \leq x \leq L$.

The wave function $\psi(x)$ describing the electron can be found by solving Schrodinger's equation. Separately for the three regions in figure 2 : (1) to the left of the barrier, (2) within the barrier, and (3) to the right of the barrier. The arbitrary constants that appear in the solutions can then be chosen so that the

values of $\psi(x)$ and its derivative with respect to x join smoothly (no jumps, no kinks) at $x=0$ and at $x=L$. Squaring the absolute value of $\psi(x)$ then yields the probability density.

Figure 3 shows a plot of the result. The oscillating curve to the left of the barrier (for $x<0$) is a combination of the incident matter wave and the reflected matter wave (which has a smaller amplitude than the incident wave). The oscillations occur because these two waves, travelling in opposite directions, interfere with each other, setting up a standing wave pattern.

Figure 3 : A plot of the probability density $|\psi|^2$ of the electron matter wave for the situation of Fig. The value of $|\psi|^2$ is nonzero to the right of the potential barrier.



Within the barrier (for $0<x<L$), the probability density plot describes a transmitted (through the barrier) wave with low but constant amplitude. Thus, the electron can be detected in this region but with a relatively small probability.

We can assign a transmission coefficient T to the incident matter wave and the barrier. This coefficient gives the probability with which an approaching electron will be transmitted through the barrier- that is, that tunneling will occur. As an example, if $T=0.020$, then of every 1000 electrons fired at the barrier, 20 (on average) will tunnel through it and 980 will be reflected. The transmission coefficient T is approximately

$$T \approx e^{-2bL},$$

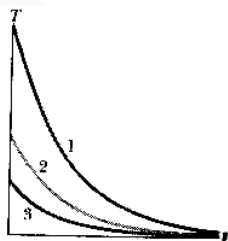
In which $b = \sqrt{\frac{8\pi^2m(U_b - E)}{h^2}}$ and e is the exponential function. Because of the exponential form of Eq. the value of T is very sensitive to the three variables on which it depends: Particle mass m , barrier thickness L , and energy difference $U_b - E$. (Because we do not include relativistic effects here, E does not include mass energy.)

11. A plot of the probability density $|\psi|^2$ of the electron matter wave for the situation in above passage is as shown in figure 3. Consider the following statements:
- i) The minima in the values of $|\psi|^2$ at the left of $x=0$ are greater than zero.
 - ii) The value of $|\psi|^2$ is non-zero to the right of the potential barrier.
 - iii) The amplitude of the reflected wave is not equal to the incident wave
 - iv) The uncertainty in the measurement of the deBroglie wavelength of electron to the left of $x=0$ is zero

The number of above statements which is/are true

- a) 1 b) 2 c) 3 d) 4

12. For three experiments, Fig gives the transmission co-efficient T for electron tunneling through a potential barrier, plotted versus barrier thickness L . The de Broglie wavelengths of the electrons are identical in the three experiments. The only difference in the physical setups is the barrier heights U_b . Rank the three experiments according to U_b greatest first.



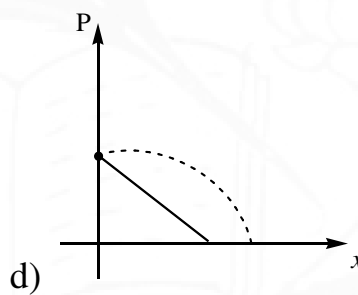
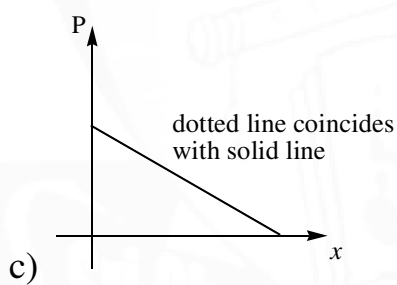
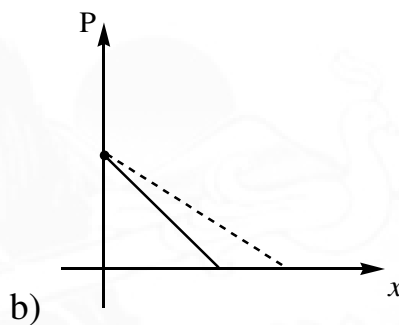
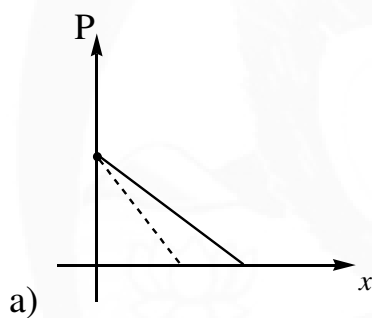
- a) 1, 2, 3 b) 3, 2, 1 c) $1 = 2 = 3$ d) Cannot be determined

Paragraph for Question Nos. 13 to 14

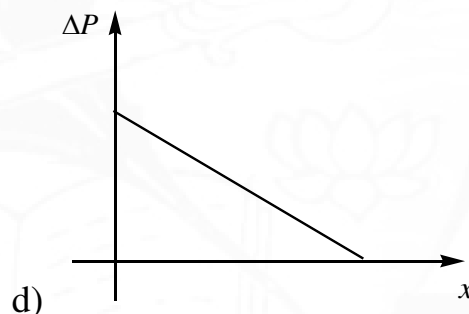
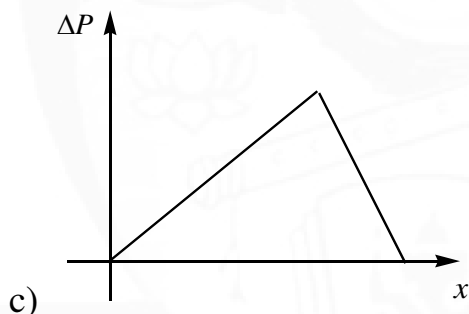
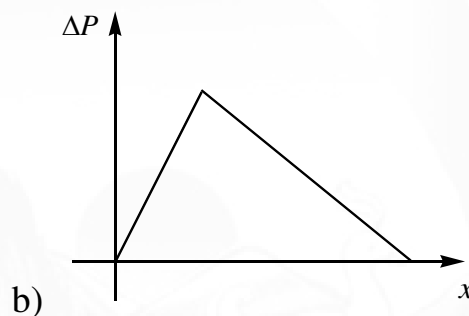
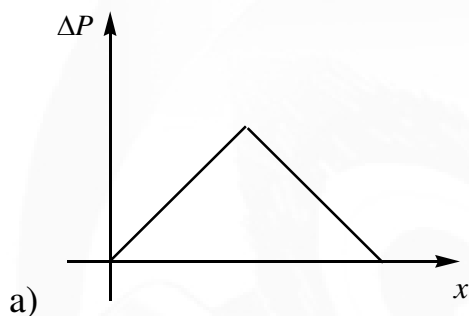
A long cylindrical container whose base area $A = 10 \text{ cm}^2$ contains $h = 60 \text{ cm}$ high water column. Taking base of the vessel as origin and upward vertical as direction of positive x-axis, gauge pressure P at any point x inside the vessel is graphed first using a solid line. The entire water column is now heated uniformly to a temperature rise $\Delta T = 80^\circ\text{C}$. Assume coefficient of volume expansion of water in the temperature range $= 0.00013 / \text{K}$, density of water at initial temperature

$= 10^3 \text{ kg / m}^3$, the expansion of the container is negligible. Again the process of drawing P vs x graph as mentioned above is repeated on same graph paper with a dotted line.

13. Which of the following best represents the graphs mentioned above?
(Not drawn to scale)



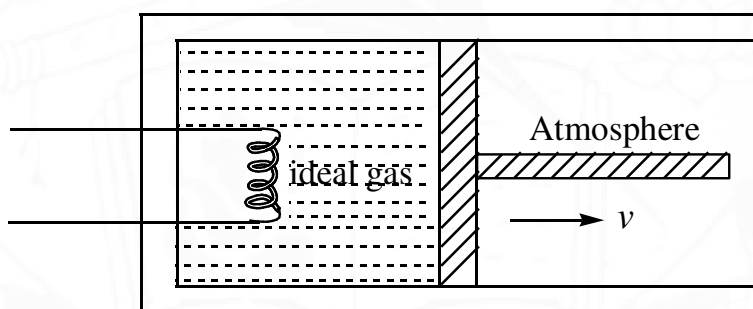
14. Which of the following best represents the graph of ΔP (increase in gauge pressure at a point in the liquid due to ΔT rise) versus x graph? (Not drawn to scale)

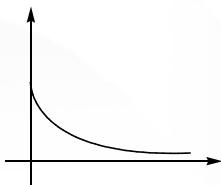
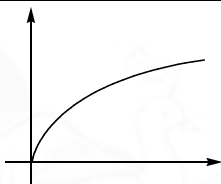
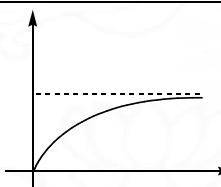
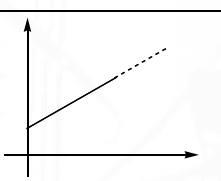


SECTION - IV
(MATRIX MATCHING ANSWER TYPE)

This section contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and four statements (P, Q, R and S) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question darken the bubbles corresponding to Q and R in the OMR sheet. For each correct matching **will be awarded + 2 marks ONLY** and 0 if not attempted and -1 in all other cases. If all are correct a bonus of 2 marks will be awarded

15. An ideal gas at initial pressure P_0 (= atmospheric pressure) and volume V_0 is enclosed by a light piston in a cylinder as shown. Initially the piston is in static equilibrium and the gas is in thermodynamic equilibrium. Now we start to move the piston outwards at a constant slow velocity v . The cross sectional area of piston is A . While the piston is moving we can deliver heat to the gas through a heating filament and pull the piston in such a way that temperature of gas remains constant. Apart from the heat transfer between the gas and the heating filament all other heat exchange can be neglected and the atmospheric pressure is constant. Mark the quantities in column-I with their, appropriate variation in column-II (time t on x-axis)



	COLUMN-I		COLUMN-II
a)	Pressure P of gas on y-axis	A)	
b)	Volume V of gas on y-axis	B)	
c)	Force applied by pulling agent (apart from atmosphere) on the piston on y-axis	C)	
d)	Amount of cumulative work done by gas on piston from start till time 't' on y-axis	D)	

16. A non-conducting sphere of radius R is given a volume charge density $\rho = kr$ where r is the distance of a point in the spherical volume from its centre and k is a positive constant. Match the physical quantities pertaining to the sphere in column-I with the values given in column-II

	Column-I		Column-II
a)	If electrostatic pressure at distance r from the centre is given by $P(r) = a(b - r^n)$ where a , b and n are constants then $n =$	A)	7
b)	Electric field at a distance r from the centre is $\frac{kr^2}{m\epsilon_0}$ where m is a positive constant. The value of m is	B)	4
c)	Electro static force of interaction between the left and right hemispheres is $\frac{k^2\pi R^6}{4l\epsilon_0}$ where l is a positive constant. The value of l is	C)	5
d)	Self energy (Electrostatic) of the sphere is given by dR^e where d and e are constants then value of e is	D)	6

CHEMISTRY**Max Marks : 82****SECTION – I
(INTEGER ANSWER TYPE)**

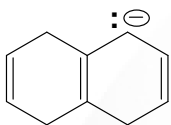
This section contains 4 questions. The answer is a Three digit integer ranging from 000 to 999 (both inclusive-Ex: If answer is 1 you should mark as 001).

Marking scheme + 4 for correct answer , 0 if not attempted and 0 in all other cases.

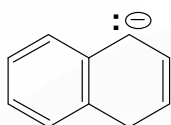
17. How many of the following statements are correct

- (A) Nascent hydrogen is a powerful reducing agent than molecular hydrogen.
- (B) White phosphorous is more reactive than red phosphorous.
- (C) The reactivity of K_2O towards water is more than that of Na_2O towards water
- (D) Nitrate salt of silver is soluble in ammonia.
- (E) Liquid nitrogen does not stick to the poles of magnet whereas liquid oxygen does
- (F) NO_2 is brown in colour at room temperature but colourless below $0^\circ C$.
- (G) Bond order of $\bar{C}N$ and N_2 are equal but $\bar{C}N$ is more reactive
- (H) NO^+ is having lower bond strength than NO

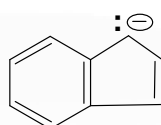
18. How many are the correct stability order of the following carbanions



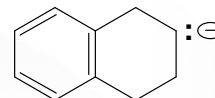
I



II



III



IV

- A) III > I B) III > II C) II > I D) I > IV

19. Van't Hoft factor (i) for three different aqueous solution of 0.1 molal $K_4[Fe(CN)_6]$, 0.01 molal K_2SO_4 and 1 molal NaCl are 4.5, 2.7 and 1.8 respectively. Then the number of INCORRECT statement are

- A) Boiling point of $K_4Fe(CN)_6$ solution is maximum
 B) Freezing point of NaCl solution is minimum
 C) Boiling point of K_2SO_4 solution is maximum
 D) All the three solution have same freezing point

20. How many of the following statements are correct

STATEMENT – 1: CH_3NO_2 forms $CHCl_3$, when it is reacting with $Cl_2 + KOH$

STATEMENT – 2: $Cl_2 + KOH$ forms hypochlorite ion

STATEMENT – 3: p-chloro phenol is less acidic than p-fluorophenol

STATEMENT – 4: Fluorine is more electronegative atom than chlorine.

STATEMENT – 5: Bond length C-O bond decreases when CO forms the complex with metal

STATEMENT – 6: The non bonded pair of electrons of metal involves in back bonding with vacant antibonding molecular orbital of CO.

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: + 4 for correct answer, 0 if not attempted and -2 in all other cases.

21. $\text{Ca}_2\text{B}_6\text{O}_{11} + \text{Na}_2\text{CO}_3 \xrightarrow{\Delta} [\text{X}] + \text{CaCO}_3 + \text{NaBO}_2$ (Unbalanced equation)

Correct choice for [X] is /are

- A) Structure of anion of crystalline [X] has one boron atom sp^3 hybridised and other three boron atoms sp^2 hybridised.
- B) [X] with NaOH (aq.) gives a compound which on reaction with hydrogen peroxide in alkaline medium yields a compound used as **brightness** in soaps.
- C) Hydrolysis of [X] with HCl or H_2SO_4 yields a compound which on reaction with HF gives fluoroboric acid.
- D) [X] on heating with chromium salt in oxidizing flame gives green coloured bead

22. Which of the following is true for 3-methyl-2-butanone?

- A) It may be prepared by CrO_3 oxidation of 2-methyl-2-butanol.
- B) Reaction with NaBH_4 gives a secondary alcohol.
- C) It may be prepared by acidic Hg^{2+} catalyzed hydration of 3-methyl-1-butyne.
- D) It forms a silver mirror on treatment with $\text{Ag}(\text{NH}_3)_2^+$.

23. The four elements A, B, C and D form diatomic molecules and also form singly charged negative ions. The following observations are made in the series of experiments

- (i) $2\text{B}^- + \text{C}_2 \longrightarrow 2\text{C}^- + \text{B}_2$
- (ii) $2\text{A}^- + \text{C}_2 \longrightarrow \text{No reaction}$
- (iii) $2\text{D}^- + \text{B}_2 \longrightarrow 2\text{B}^- + \text{D}_2$

Which of the following statements are correct ?

- A) $E_{\text{A}_2/\text{A}^-}^0 > E_{\text{C}_2/\text{C}^-}^0 > E_{\text{B}_2/\text{B}^-}^0 > E_{\text{D}_2/\text{D}^-}^0$
- B) $E_{\text{D}_2/\text{D}^-}^0 > E_{\text{C}_2/\text{C}^-}^0 > E_{\text{B}_2/\text{B}^-}^0 > E_{\text{A}_2/\text{A}^-}^0$
- C) A_2 can oxidize D^- into D_2 , C^- into C_2 and B^- into B_2 .
- D) A_2 is the strongest oxidizing agent.

- 24 . You are about to undertake kinetic comparisons of rates and intimate mechanism of fluoride ion exchanges of SF_6 and free F^- versus SeF_6 and free F^- , in a suitable inert solvent. Which of the following statements are likely to be correct?
- A) SF_6 is more likely to react by dissociative mechanism.
 - B) SeF_6 is more likely to react by associative mechanism
 - C) Both react by dissociative mechanism
 - D) SeF_6 is more likely to react rapidly

SECTION - III**(COMPREHENSION TYPE)**

This section contains **2 or 3 Comprehensions**. Each comprehension has 2 or 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +5 for correct answer , 0 if not attempted and -2 in all other cases.

Paragraph for Question Nos. 25 to 26

A sodium salt [X] on treatment with dil HCl gives a colourless gas [Y], which gave white precipitate when passed through lime water. When salt [X] was heated with sulphur in boiling water gave [Z], which dissolves AgBr, when [Z] is added in excess. When salt [Z] solution was treated with AgNO_3 solution it gave a white precipitate of [Q], which slowly gives a black precipitate [R]

25. Which of the following is correct for the salt [Z] solution

- A) It on treatment with dil HCl gives an acid which is unstable, decomposes in water.
- B) It can be used for iodine estimation
- C) The salt [Z] can act as a reducing agent.
- D) All of them

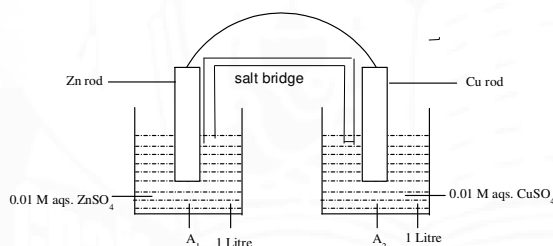
26. $2\text{Na}_2\text{S}_3 + 3\text{O}_2 \xrightarrow{\text{Heat in air}} (\dots\dots\dots) + \text{S}$

in the given reactions the products are

- A) salt [X] only
- B) gas [Y] only
- C) salt [Z] only
- D) salt [Z] and gas [Y]

Paragraph for Questions Nos. 27 to 28

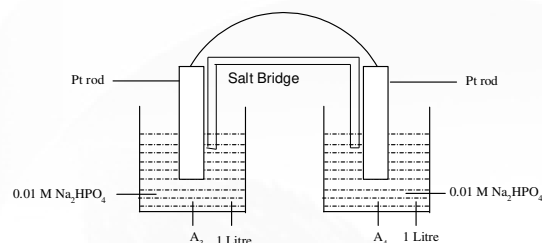
These are two presentation of reversible cell.



Cell - I

$$E_{\text{Zn}^{++}/\text{Zn}}^{\circ} = -0.76\text{V}$$

$$E_{\text{Cu}^{++}/\text{Cu}}^{\circ} = +0.34\text{V}$$



Cell – II

Given H_3PO_4

$$K_{a1} = 10^{-5}$$

$$K_{a2} = 10^{-8}$$

$$K_{a3} = 10^{-12}$$

Passage of 9.65 amp. of current for 20 second in both cell, charging reaction take place in cell – I and electrolysis of water take place during electrolysis O_2 liberated at chamber A_4 and H_2 liberated at chamber A_3 . (Consider no change in volume after electrolysis).

27. Calculate the cell potential after passage of current.

$$\text{A) } E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log_{10} \frac{0.01}{0.011}$$

$$\text{B) } E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log_{10} \frac{0.009}{0.011}$$

$$\text{C) } E_{\text{cell}} = E_{\text{cell}}^{\circ}$$

$$\text{D) } E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{2} \log_{10} \frac{0.08}{0.012}$$

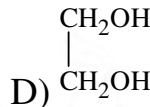
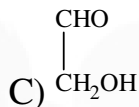
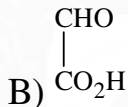
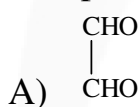
28. Difference in pH value of chamber A_3 & A_4 after electrolysis

- A) 4 B) $4 - 4 \log_{10} 2$ C) $4 + \log_{10} 2$ D) $4 + 4 \log_{10} 2$

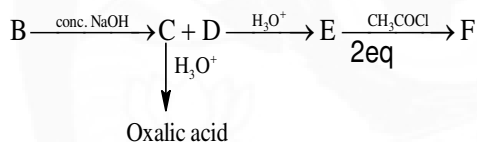
Paragraph for Questions Nos. 29 to 30

A aldehyde A ($C_{11}H_{18}O$) which does not undergo self aldol condensation, gives benzaldehyde and two moles of B on ozonolysis. Compound B which can reduce ammoniacal silver nitrate and itself oxidizes to oxalate salt.

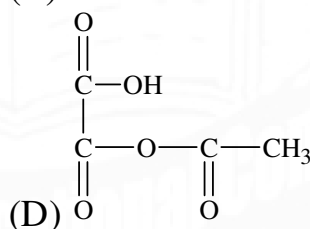
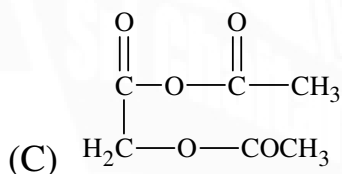
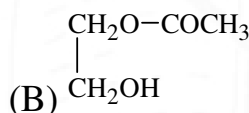
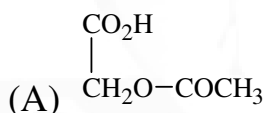
29. Compound B would be



30.



F is



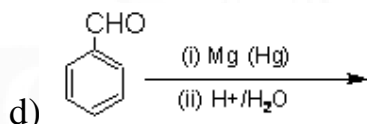
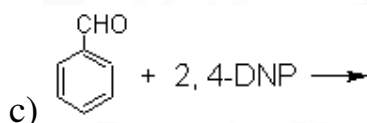
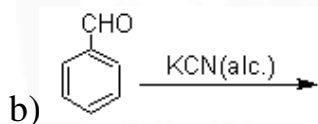
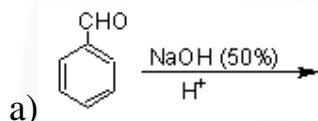
SECTION - IV

(MATRIX MATCHING ANSWER TYPE)

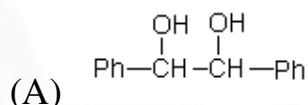
This section contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and four statements (P, Q, R and S) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R then for the particular question darken the bubbles corresponding to Q and R in the OMR sheet. For each correct matching **will be awarded + 2 marks ONLY and 0 if not attempted and -1 in all other cases**. If all are correct a bonus of 2 marks will be awarded

31. Match Column - I and Column - II

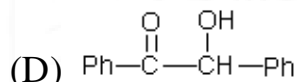
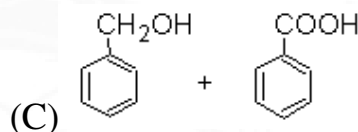
Column - I



Column - II



(B) yellow or orange-red ppt



32. Match the column I and column II

	<i>Column – I</i>		<i>Column – II</i>
a)	Precipitate with KCN, which is soluble in excess of reagent	A)	Fe^{2+}
b)	Precipitate with NaOH and NH_4OH , which is insoluble in excess of both the reagents separately	B)	Hg^{2+}
c)	Coloured ppt. with KI which is soluble in excess of reagent	C)	Pb^{2+}
d)	Black precipitate with H_2S , which is soluble in hot and dilute HNO_3	D)	Ag^+

MATHS**Max Marks : 82****SECTION – I
(INTEGER ANSWER TYPE)**

This section contains 4 questions. The answer is a Three digit integer ranging from 000 to 999 (both inclusive-Ex: If answer is 1 you should mark as 001).

Marking scheme + 4 for correct answer , 0 if not attempted and 0 in all other cases.

33. Let $I_n = \int_0^1 \frac{x^n}{x^{2012}-1} dx$, $J_n = \int_0^1 \frac{x^n}{x^{2013}+1} dx$, $\forall n > 2012$, $n \in N$ & the matrix $A = (a_{ij})_{3 \times 3}$ where

$$a_{ij} = \begin{cases} I_{2012+i} - I_i, & i = j \\ 0, & i \neq j \end{cases} \text{ \& the matrix } B = (b_{ij})_{3 \times 3} \text{ where } b_{ij} = \begin{cases} J_{2016+j} + J_{j+3}, & i = j \\ 0, & i \neq j \end{cases}$$

Then the value of trace $(A^{-1}) + \det(B^{-1}) = \text{-----}$

34. 'A box contains 10 balls numbered through 1 to 10. Five balls are drawn successively, one by one at random without replacement. Let A be the event that exactly two odd numbered balls are drawn and that they occur on odd numbered draws from the box. Then $P(A) = \frac{m}{n}$ (m, n are coprime) where

m+n=_____

35. Area enclosed by the curves given by $|x-60|+|y|=\left|\frac{x}{4}\right|$

36. Let $A(\omega^2)$ $B(2i\omega)$ $C(-4)$ be 3 points lying on the argand plane. Now a point P is taken on the circumcircle of the triangle ABC such that $|PA|.|BC|=|PC|.|AB|$ where P, A, B, C are in order). If Z is the complex number associated with the midpoint of PB, then the value of $|Z|^2$ is.....(ω is non real cube root of 1)

SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: + 4 for correct answer, 0 if not attempted and -2 in all other cases.

37. In the non-decreasing sequence of odd integers

$(a_1, a_2, a_3, \dots) = (1, 3, 3, 3, 5, 5, 5, 5, \dots)$ each positive odd integer k, appears k times. It is a fact that there are integers b, c and d such that for all positive integer n,

$$a_n = b \left[\sqrt{n+c} \right] + d \quad ([.] \text{ G.I.F}) \text{ then}$$

A) Possible value of $b+c+d$ is 2 B) possible value of $b+c+d$ is 1

C) Possible value of $\frac{b-2d}{8}$ is 0 D) possible value of $\frac{c+d}{2b}$ is 0

38. Which of the following is/are true

A) If $f: R \rightarrow R$ satisfying $f(x - f(y)) = f(f(y)) + x f(y) + f(x) - 1 \quad \forall x, y \in R$, then

$$f(x) \text{ can be } 1 - \frac{x^2}{2}$$

B) If $f(x) = \frac{ax+b}{cx-a}$, $x \neq \frac{a}{c}$, $a, b, c \in R - \{0\}$ then $f(f(x)) = x$

C) Let f be a real valued integrable function satisfying

$$f(x) + f(x+4) = f(x+2) + f(x+6) \text{ and } g(x) = \int_x^{x+8} f(t) dt, \forall x \in R \text{ then } g(x) \text{ can be a}$$

constant function

D) domain of the function $\sqrt{1-f(x)}$ is $[0, \infty)$., where $(f(x))^3 = 1 - x^3 - 3xf(x)$, $x \in R$ and $f(-1) \neq -1$

39. Given: $f(x) - \mu \int_0^{\pi/4} \sin^2 x \cos^{-5} x \cos t f(t) dt = \sin^2 x \cos^{-5} x$, ($\mu = \text{constant}$) then

A) $f(x) = \frac{3}{3-\mu} \sin^2 x \cos^{-5} x$

B) $f(x) = \frac{3}{\mu} \sin x \cos x$

C) $f(x)$ is periodic with period π D) $f(x)$ is periodic with period 2π

40. Suppose $f(z)$ is possibly complex valued function of a complex number z , which satisfies a functional equation $af(z) + bf(\omega^2 z) = g(z)$, $\forall z \in C$ where a and b are some fixed complex numbers and $g(z)$ is some function of z , " ω " is a cube root of 1 ($\neq 1$), then $f(z)$ can be determined uniquely if
- A) $a + b \neq 0$ B) $a^2 + b^2 \neq 0$ C) $a^3 + b^3 \neq 0$ D) $a^3 + b^3 = 0$

SECTION - III**(COMPREHENSION TYPE)**

This section contains **2 or 3 Comprehensions**. Each comprehension has 2 or 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct. **Marking scheme +5 for correct answer, 0 if not attempted and -2 in all other cases.**

Paragraph for Question Nos. 41 to 42

A point P is moving in XY -plane such that if it lies inside the triangles formed by the lines $|x| = 3$ & $y = \pm x$ then its distance from the origin is always r , otherwise it always lies on the curve $y^2 = 4|x|$

41. Range of r , such that there exist exactly eight points on the locus of P , from which if we draw two tangents on the parabola part, then they are at right angle is.....
- A) $r > 1$ B) $1 < r < \sqrt{2}$ C) $1 < r \leq \sqrt{2}$ D) $r \geq 1$

42. When $r=1$ equation of the largest circle which touches the locus of P, exactly at three distinct points, can be

A) $(x-5)^2 + y^2 = 16$

B) $(x+5)^2 + y^2 = 20$

C) $x^2 + (y-5)^2 = 16$

D) $\left(x - \frac{1}{2}\right)^2 + y^2 = \frac{1}{4}$

Paragraph for Question Nos. 43 to 44

Two vectors $\vec{a}$ and $\vec{b}$ are having unit modulus and angle between them is θ . Now

$\phi(\theta) = \int_{-|\vec{a} \cdot \vec{b}|}^{|\vec{a} \times \vec{b}|^2} f^2(x) dx$ and $f(x)$ satisfies $f(x) + f(y) = \frac{x+y}{xy}$ for all $x, y \in \mathbb{R} - \{0\}$ &

$h(\theta) = -\phi(\theta) + |\vec{a} \times \vec{b}_1|^2 |\vec{a} \cdot \vec{b}_1|^2$ where $\vec{b}_1 = 2\vec{b}$ & $f^2(x) = (f(x))^2$

43. Which of the following is correct.

A) Fundamental period of $\phi(\theta)$ is π

B) Angle between $\vec{a}$ and $\vec{b}$ is $\theta = \frac{\pi}{3}$ for which $h(\theta)$ is minimum

C) Volume of the parallelepiped formed by $\vec{a}, \vec{b}$ and $\vec{a} \times \vec{b}$ (where angle between $\vec{a}$ and $\vec{b}$ is taken as θ) for which $h(\theta)$ is minimum is $\frac{1}{2}$

D) Volume of the parallelepiped formed by $\vec{a}, \vec{b}$ and $\vec{a} \times \vec{b}$ (where angle between $\vec{a}$ and $\vec{b}$ is taken as θ) for which $h(\theta)$ is minimum is $\frac{1}{4}$

44. If $\vec{a}$ and $\vec{b}$ are non-collinear vectors, then number of solution of the equation

$$4\phi(\theta) + \sqrt{3}\phi^1(\theta) = 0 \text{ in } [0, 2\pi] \text{ are } \dots\dots\dots$$

A) 2

B) 4

C) 6

D) 8

Paragraph for Question Nos. 45 to 46

Let $f(x) = e^{(p+1)x} - e^x$ for some real $p > 0$

45. Let $g(t) = \int_t^{t+1} f(x)e^{t-x} dx$, the value of $t = t_p$ for which $g(t)$ is minimum is

A) $-\ln\left(\frac{e^p - 1}{p}\right)$

B) $\frac{-1}{p} \ln\left(\frac{l^p - 1}{p}\right)$

C) $\frac{-1}{p} \ln\left(\frac{(p+1)(e^p - 1)}{p}\right)$

D) $-\ln((1+p)(e^p - 1))$

46. Use the fact that $1 + \frac{p}{2} \leq \frac{e^p - 1}{p} \leq 1 + \frac{p}{2} + p^2$ ($0 < p \leq 1$). The value of $\lim_{p \rightarrow 0^+} (s_p - t_p) = \underline{\hspace{2cm}}$

where $x = s_p$, for which $f(x)$ is minimum

A) 0

B) $\frac{1}{2}$

C) 1

D) does not exist

SECTION - IV

(MATRIX MATCHING ANSWER TYPE)

This section contains 2 questions. Each question has four statements (A, B, C and D) given in Column I and four statements (P, Q, R and S) in Column II. Any given statement in Column I can have correct matching with ONE or MORE statement(s) given in Column II. For example, if for a given question, statement B matches with the statements given in Q and R, then for the particular question darken the bubbles corresponding to Q and R in the OMR sheet. For each correct matching **will be awarded + 2 marks ONLY and 0 if not attempted and -1 in all other cases**. If all are correct a bonus of 2 marks will be awarded

47. For $x > 0$, let $A = \begin{pmatrix} x + \frac{1}{x} & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & 16 \end{pmatrix}$, $B = \begin{pmatrix} \frac{5x}{x^2+1} & 0 & 0 \\ 0 & \frac{3}{x} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix}$

& define, three other matrices $X = (AB)^{-1} + (AB)^{-2} + (AB)^{-3} + \dots + (AB)^{-n}$. $Y = \lim_{n \rightarrow \infty} X$,

$Z = Y^{-1} - 2I$ ($I =$ Identity matrix of order 3)

Then

COLUMN-I	COLUMN-II
a) minimum value of $[tr(AY)] = \dots$ where $([.]$ G.I.F)	A) 24
b) $\det(Y^{-1}) = \dots$	B) 12
c) If $tr(Z + Z^2 + Z^3 + \dots + Z^{10}) = 2^a + b$, ($a, b \in N$) Then the value of $a + b = \dots$	C) 6
d) The value of $ adj(\sqrt{5}Y^{-1}) = K$, then number of positive divisors of k , which are odd.	D) 19

48.

Consider the curve $\sin x + \sin y = 1$, lying in the first quadrant then

COLUMN-I	COLUMN-II
a) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{d^2 y}{dx^2} = \dots\dots$	A) 0
b) $\lim_{x \rightarrow 0^+} x^\alpha \frac{d^2 y}{dx^2} = \dots\dots$ where $\alpha = \frac{3}{2}$	B) 1
c) $\lim_{x \rightarrow 0^+} x^\alpha \frac{d^2 y}{dx^2} = \dots\dots$ where $\alpha = 2$	C) $\frac{1}{\sqrt{2}}$
d) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{dy}{dx} = \dots\dots$	D) $\frac{1}{2\sqrt{2}}$



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Sec: Sr.N-Super Chaina
Time: 3hr

Date: 14-06-20
Max.Marks:186

Name of the Student: _____

H.T. NO:

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14-06-20_SR.N-SUPER CHAINA _Jee-Adv_GTA-18(P-1)_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEAMTICS : TOTAL SYLLABUS

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

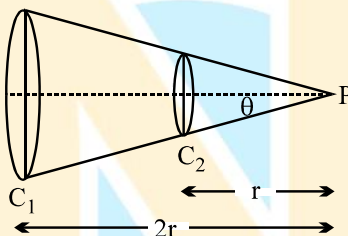
This section contains **4 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

1. Two long wires have uniform charge density λ per unit length each. The wires are non-coplanar and mutually perpendicular. Shortest distance between them is d . The interaction force between them is:

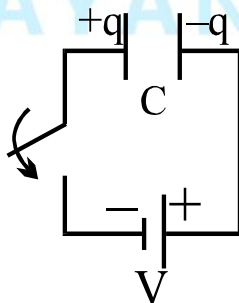
A) $\frac{\lambda^2}{2\epsilon_0}$ B) $\frac{\lambda^2}{2\pi\epsilon_0}$ C) $\frac{2\lambda^2}{\epsilon_0}$ D) $\frac{\lambda^2}{4\epsilon_0}$

2. The coils C_1 and C_2 have same number of turns and carry equal currents in the same sense. They subtend the same angle θ to P. If the magnetic field produced by C_1 at P is B then that produced by both C_1 and C_2 will be



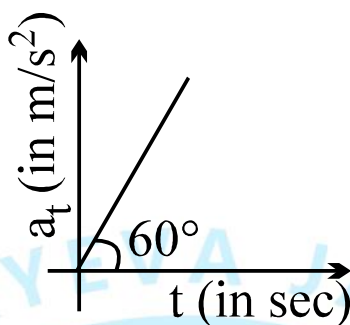
A) $3B$ B) $2B$ C) $B/2$ D) $\frac{B}{3\sqrt{2}}$

3. A capacitor having capacitance 'C' is charged by a battery having e.m.f. $2V$ (Battery is disconnected after charging). Now this charged capacitor is connected to another battery having e.m.f. V . The positive plate of capacitor is connected with $-ve$ terminal of battery & vice-versa. The amount of heat produced after connecting it with battery.



A) $4.5 CV^2$ B) $1.5 CV^2$ C) $4 CV^2$ D) $2.5 CV^2$

4. Tangential acceleration of a particle starting from rest and moving in a circle of radius 1 metre varies with time as in graph. Time after which total acceleration of particle makes an angle of 30° with radial acceleration is :



- A) 4 sec B) $4/3$ sec C) $2^{2/3}$ sec D) $\sqrt{2}$ sec

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

5. A thin non-conducting ring of mass m , radius a carrying a charge q can rotate freely about its own axis which is vertical. At the initial moment the ring was at rest and no magnetic field was present. At instant $t = 0$, a uniform magnetic field is switched on which is vertically downwards and increases with time according to the law

$B = B_0 t$. Neglecting magnetism induced due to rotational motion of the ring, which of the following statements is/are correct:

- A) angular acceleration of the ring is $\frac{qB_0}{2m}$
- B) angular acceleration of the ring is $\frac{qB_0}{m}$
- C) power developed by the forces acting on the ring as a function of time is $\frac{B_0^2 q^2 a^2 t}{4m}$
- D) power developed by the forces acting on the ring as a function of time is $\frac{B_0^2 q^2 a^2 t}{2m}$

6. Nuclei of radioactive element A are being produced at a constant rate α . The element has a decay constant λ . At time $t = 0$, there are N_0 nuclei of the element. Which of the following statements are correct:

A) The number of nuclei of A at time t is $\frac{1}{\lambda} \{ \alpha - (\alpha - \lambda N_0) e^{-\lambda t} \}$

B) The number of nuclei of A at time t is $\frac{1}{\lambda} \{ (\alpha - \lambda N_0) e^{-\lambda t} \}$

C) If $\alpha = 2N_0\lambda$, the number of nuclei of A after one half life of A is $\frac{3}{2}N_0$

D) The number of nuclei of A after a very long time is $3N_0$

7. The x-coordinate of a particle moving on x-axis is given by

$x = 3\sin 100t + 8\cos^2 50t$, where x is in cm and t is time in seconds. Which of the following is/are correct about this motion.

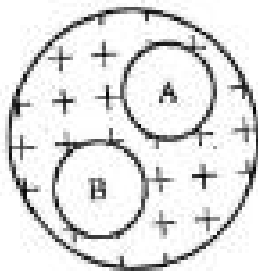
A) the motion of the particle is not S.H.M.

B) the amplitude of the S.H.M. of the particle is 5 cm

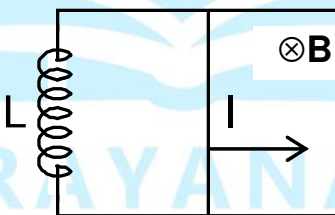
C) the amplitude of the resultant S.H. M. is $\sqrt{73}$ cm

D) the maximum displacement of the particle from the origin is 9 cm

8. A solid non conducting sphere have two spherical cavities A and B of radius ' r_A ' and ' r_B ' respectively, as shown. Radius of sphere is ' R '. Select the correct statement(s):



- A) Electric field in cavity A is uniform
 B) Electric field at the centre of cavity A depends on radius of cavity A
 C) Electric field at the centre of cavity A depends on radius of cavity B
 D) Electric field at the centre of cavity A will depend on distance between centers of cavity A and B
9. Two parallel resistanceless rails are connected by an inductor of inductance L at one end as shown in figure. A magnetic field B exists in the space which is inward perpendicular to the plane of the rails. Now a conductor of length l , mass m and negligible resistance is placed transverse on the rail and given an impulse J towards the rightward direction. Then select the correct option(s) :



- A) Displacement of the conductor when velocity of the conductor is half of the initial

velocity is $\sqrt{\frac{3J^2L}{4B^2l^2m}}$

- B) Displacement of the conductor when velocity of the conductor is half of the initial

velocity is $\sqrt{\frac{3J^2L}{B^2l^2m}}$

C) Current flowing through the inductor at the instant when velocity of the conductor is

half of the initial velocity is $\sqrt{\frac{3J^2}{4Lm}}$

D) Current flowing through the inductor at the instant when velocity of the conductor is

half of the initial velocity is $\sqrt{\frac{3J^2}{Lm}}$

10. A blast breaks a body initially at rest of mass 0.5 kg into three pieces, two smaller pieces of equal mass and the third double the mass of either of small piece. After the blast the two smaller masses move at right angles to one another with equal speed. Find the statements that is/are true for this case assuming that the energy of blast is totally transferred to masses.

A) All the three pieces share the energy of blast equally

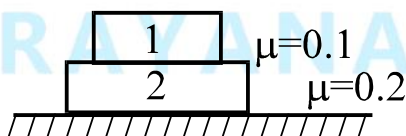
B) The speed of bigger mass is $\sqrt{2}$ times the speed of either of the smaller mass

C) The direction of motion of bigger mass makes an angle of 135° with the direction of smaller pieces

D) The bigger piece carries double the energy of either piece

11. For the two blocks initially at rest under constant ext. forces of mass 1 kg and 2kg shown, $a_1 = 5 \text{ m/s}^2 \rightarrow$, $a_2 = 2 \text{ m/s}^2 \leftarrow$. Which of the following is correct?

(Take $g = 10 \text{ m/s}^2$)



A) Force of friction on 1 due to 2 is 1 N to right.

B) Force of friction on 2 due to 1 is 1 N to right.

C) Force of friction on 2 due to ground is 4 N to right.

D) Force of friction on 2 due to ground is 6 N to right

12. $Y(x,t) = 0.05 / [(4x + 2t)^2 + 5]$ represents a moving wave pulse, where x and y are in meters and t is in seconds. Then which statement(s) are correct:

- A) pulse is moving in $-x$ direction
- B) wave speed is 0.5 m/s
- C) maximum particle displacement is 1 cm
- D) it is a symmetric pulse

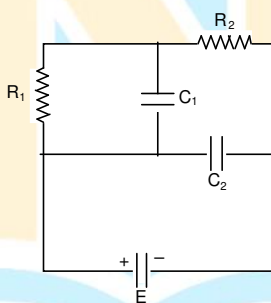
SECTION – III
(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place).
Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

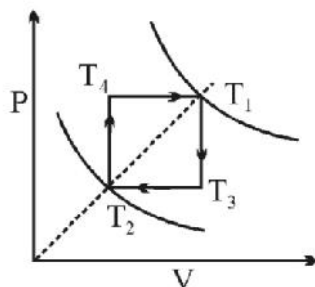
Marking scheme : +3 for correct answer, 0 in all other cases.

13. In the circuit shown in figure, $R_1 = 1\Omega$, $R_2 = 2\Omega$, $C_1 = 1\mu F$, $C_2 = 2\mu F$ and $E = 6V$.

The sum of magnitude of charges on the two capacitors in steady state in μC is

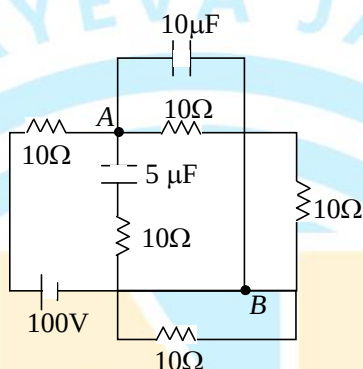


14. Between two isotherms we have a cycle as shown. The work done by the gas during the cycle is $n J$. Find the value of n . (Take $T_1 = 127^\circ C$, $T_2 = 16^\circ C$, $n = 1$ mole)

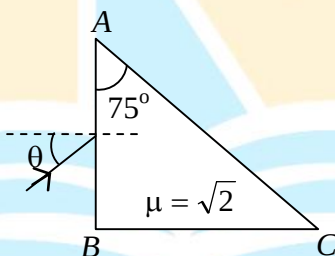


15. A container filled with air under pressure P_0 contains a soap bubble of radius R . The air pressure has been reduced to half isothermally and the new radius of the bubble becomes $\frac{5R}{4}$. If the surface tension of the soap water solution is S , P_0 is found to be $\frac{nS}{R}$ SI unit. Find n .

16. In the circuit shown, find the potential difference between points A and B in V.



17. In the figure, if ray does not emerge from side AC then maximum value of θ is $n\pi$ radian. Then n is



18. If stress in a stretched wire of a material (whose Young's modulus is Y) is λY , in order that the speed of longitudinal waves is equal to 10 times the speed of transverse waves then find value of λ ?

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

19. Find the equilibrium constant for the given reaction, $In^{+2} + Cu^{+2} \rightarrow In^{+3} + Cu^{+}$ at 298 K

$$[E^0 Cu^{+2} / Cu^{+} = 0.15V, E^0 In^{+3} / In^{+} = -0.42V, E^0 In^{+2} / In^{+} = -0.40V]$$

- A) 10^{10} B) 10^8 C) 10^9 D) 10^6

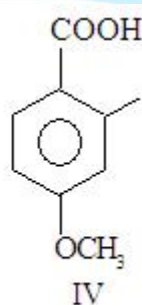
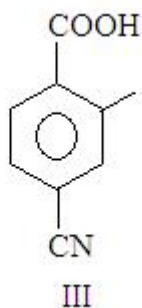
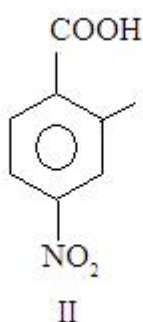
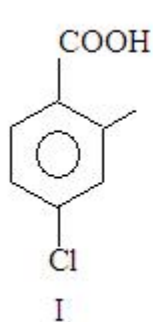
20. Fe^{+3} can be distinguished by using a reagent (A) which develops a red colour with Fe^{+3} due to the formation of a compound (B). Fe^{+2} in pure state does not develop any colour with the reagent. The reagent (A) and the product (B) are respectively

- A) $K_4[Fe(CN)_6]$ and $Fe_4[Fe(CN)_6]_3$ B) $K_3[Fe(CN)_6]$ and $K_2Fe[Fe(CN)_6]$
C) NH_4CNS and $[Fe(SCN)]^{+2}$ D) Na_2HPO_4 and $FePO_4$

21. Willemite, cryolite, Pyrolusite and Argentite respectively are

- A) Zn_2SiO_4 , Na_3AlF_6 , Ag_2S , MnO_2 B) Zn_2SiO_4 , Na_3AlF_6 , MnO_2 , Ag_2S
C) MnO_2 , Ag_2S , Zn_2SiO_4 , Na_3AlF_6 D) MnO_2 , Ag_2S , Na_3AlF_6 , Zn_2SiO_4

22. The correct acidic order of the following carboxylic acids is



- A) II > IV > III > I B) III > II > IV > I
C) III > I > II > IV D) II > III > I > IV

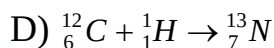
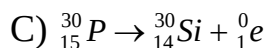
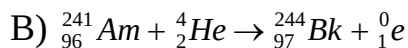
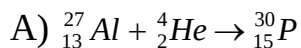
SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

23. Nuclear reactions accompanied with emission of neutron(s) are



24. Identify the correct statements

A) 1 mole of MnO_4^- ions can oxidized 5 moles of Fe^{+2} ions in acidic medium

B) 1 mole of $\text{Cr}_2\text{O}_7^{2-}$ ions can oxidized 6 moles of Fe^{+2} ions in acidic medium

C) 1 mole of Cu_2S can be oxidized by 2 moles of MnO_4^- ions in acidic medium

D) 1 mole of Cu_2S can be oxidized by 1.66 moles of $\text{Cr}_2\text{O}_7^{2-}$ ions in acidic medium

25. Which of the following are correct

A) Aqueous solution of CoCl_2 is pink while $[\text{CoCl}_4]^{-2}$ is deep blue

B) When CaCl_2 is added to a aqueous CoCl_2 pink colour changes to blue

C) When HgCl_2 is added to $[\text{CoCl}_4]^{-2}$ blue colour changes to pink

D) Both pink and blue solution of aqueous CoCl_2 and $[\text{CoCl}_4]^{-2}$ contain same magnetic moment value

26. NH_3 can be obtained by

A) Urea with NaOH

B) Heating of NH_4Cl (or) $(\text{NH}_4)_2\text{CO}_3$

C) Heating of NH_4NO_3 with NaOH

D) Reaction of AlN or Mg_3N_2 with H_2O

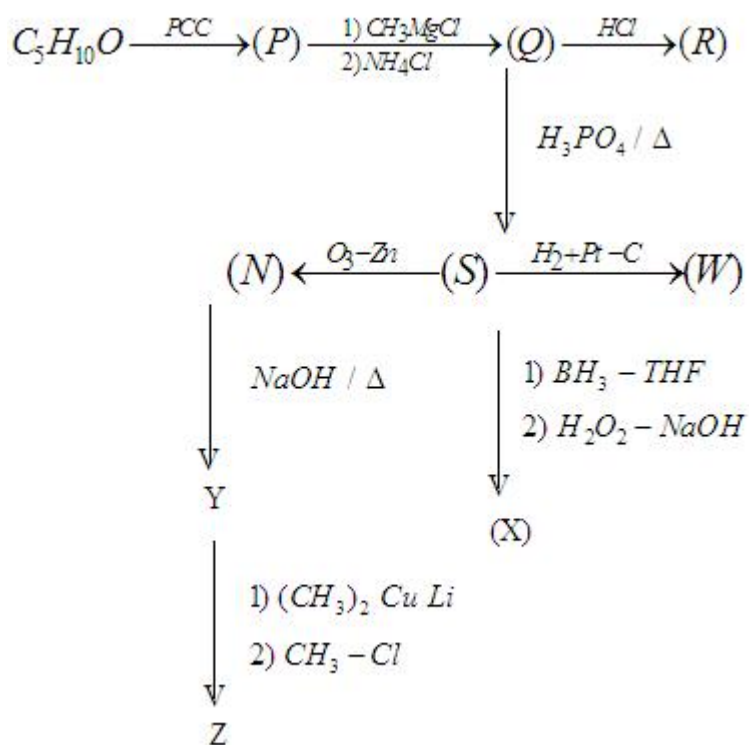
27. Which of the following molecules LUMO is ungerade

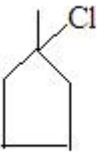
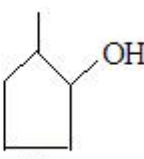
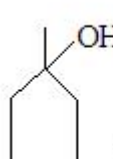
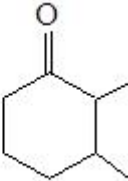
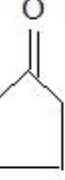
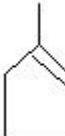
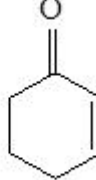
- A) N_2^{-2} B) O_2^{-2} C) Be_2 D) Li_2^{-2}

28. Which of the following are non reducing sugars

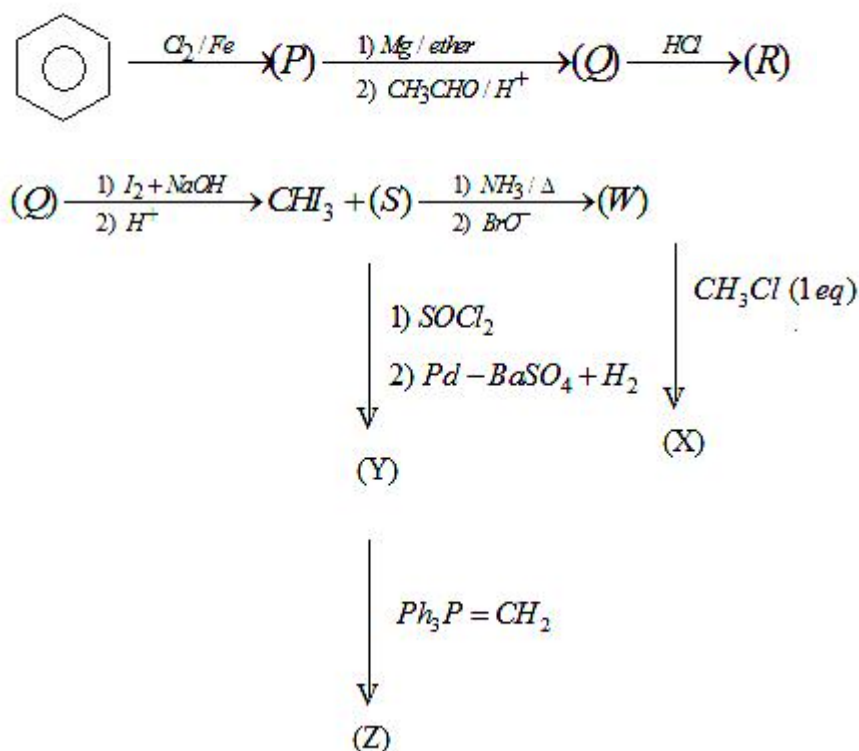
- A) Ethyl- β -D-glycopyranoside B) Ethyl- α -D-fructofuranoside
C) Ethyl- β -D-manoside D) Ethyl- α -D-galactoside

29. Identify the correct option(s) for the following set of reaction.



A)	 	B)	 
C)	 	D)	 

30. Which of the following are correct for the given reaction.



- A) 'P' and 'R' can be distinguished by $AgNO_3$
- B) 'Q' and 'S' can be distinguished by $NaHCO_3$
- C) 'W' and 'X' can be distinguished by $CHCl_3 + KOH$
- D) 'Y' and 'Z' can be distinguished by Tollen's reagent

SECTION – III (Numerical Value Answer Type)

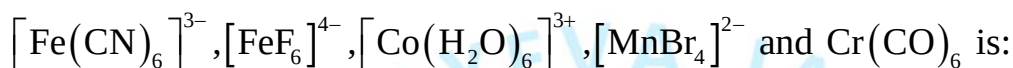
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Marking scheme : +3 for correct answer, 0 in all other cases.

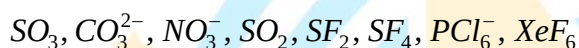
31. Two litre of N_2 at $0^\circ C$ and 5 atm pressure are expanded isothermally against a constant external pressure of 1 atm until the pressure of gas reaches 1 atm. Assuming gas to be ideal find out work of expansion in joule. (Given 1Ltr.atm=101.325 J)

32. Two gm of benzoic acid is dissolved in 25gm of C_6H_6 and its depression in freezing point is equal to 1.62 K. Molal depression constant of C_6H_6 is $4.9K \text{ mole}^{-1} \text{ kg}$. Find the percentage association of acid if it forms double molecule in solution.

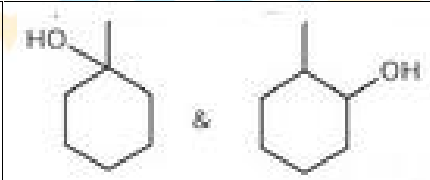
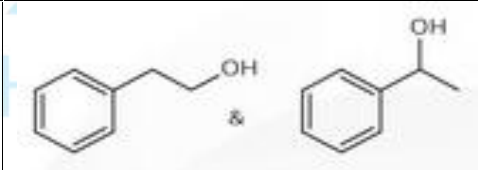
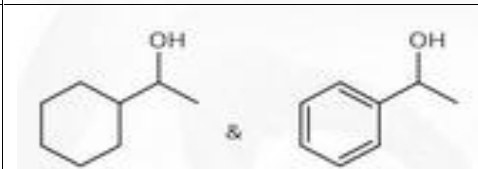
33. The sum of unpaired electrons in the complexes

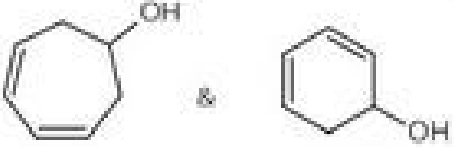


34. In how many of the following species the structure is distorted due to the presence of non bonding electron pairs ?

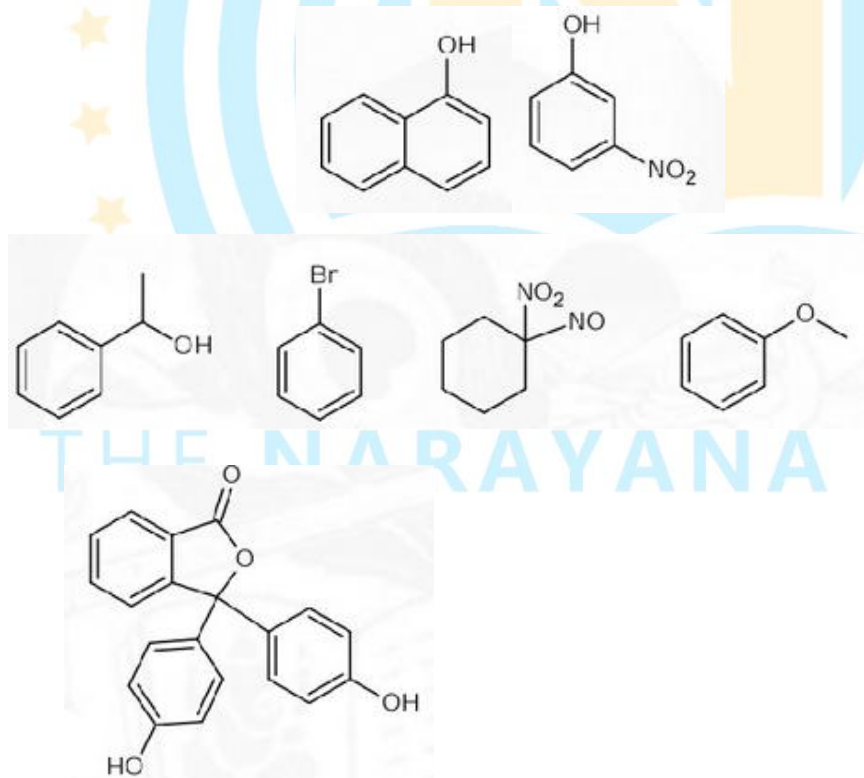


35. In how many of the following cases, the second molecule undergoes dehydration faster than the first molecule when heated with concentrated sulphuric acid to form a double bond ?

a)	
b)	
c)	

d)	
e)	
f)	
g)	

36. How many of the given compounds are either dissolve or give colour (or change colour) on addition of aqueous solution of NaOH.



MATHEMATICS

Max.Marks:62

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains **4 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

37. PQ is a chord of parabola $x^2 = 4y$ which subtends right angle at vertex. Then locus of centroid of triangle PSQ, where S is the focus of given parabola, is

- A) $x^2 = 4(y+3)$ B) $x^2 = \frac{4}{3}(y-3)$ C) $x^2 = \frac{-4}{3}(y+3)$ D) $x^2 = \frac{4}{3}(y+3)$

38. An electric component manufactured by a company owned by Mr. Kumar Satyam is tested for its defectiveness by a sophisticated device. Let 'A' denote the event "the device is defective" and 'B' the event "the testing device reveals the component to be defective". Suppose $P(A) = \alpha$ and $P(B/A) = P(\bar{B}/\bar{A}) = 1 - \alpha$. Where $0 < \alpha < 1$. If it is given that the testing device reveals it to be defective, then the probability that the component is not defective is

- A) $\frac{1}{4}$ B) $\frac{3}{4}$ C) 0.7 D) 0.5

39. If $p + q + r = 0 = a + b + c$, then the value of the determinant $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix}$ is

- A) 0 B) $pq + qb + rc$ C) 1 D) none of these

40. A plane passing through (1, 1, 1) cuts +ve direction of co-ordinate axes at A, B & C, then the range of volume of tetrahedron OABC (V) is

- A) $[27, \infty)$ B) $[9, \infty)$ C) $[4.5, \infty)$ D) $[18, \infty)$

SECTION – II

(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

41. Let $S = \{x_i : i \in \mathbb{N}, i \leq n\}$ be the set of real solutions of the equation

$$\tan^{-1} \frac{1}{x} + \tan^{-1} \frac{2}{x} + \tan^{-1} \frac{3}{x} + \tan^{-1} \frac{4}{x} = \frac{\pi}{2}, \text{ Then select the CORRECT alternative.}$$

A) $n = 4$ and $x_1 x_2 x_3 x_4 = 24$

B) $x \geq \sqrt{34}, \forall x \in S$

C) $n = 4$ and $x_1 + x_2 + x_3 + x_4 = 0$

D) $x \leq \sqrt{35}, \forall x \in S$

42. Which of the following statements are INCORRECT?

A) If function $f(x) \cdot g(x)$ is continuous at $x = a$, then at least one of $f(x)$ and $g(x)$ must be continuous at $x = a$

B) The function $\{x+1\} + [x+2]$ is continuous and differentiable $\forall x \in \mathbb{R}$ (where $\{ \cdot \}$ denotes fractional part function and $[\cdot]$ denotes greatest integer function)

C) If a continuous function is non differentiable at $x = a$, then geometrically it implies that there is a sharp corner at $x = a$

D) If $f(x)$ is derivable at $x = a$, then $f'(x)$ is continuous at $x = a$

43. The tangent at point P(1,1) on the curve $C: y = x^3 + x - 1$ cuts the curve at Q. The tangent at Q on the curve cuts the curve at R. Then

A) (1,0) and centroid of ΔPQR lies on the opposite side of curve C

B) The abscissae of P, Q and R are in G.P.

C) The ordinates of P, Q and R are in G.P.

D) Length of subnormal at Q is less than 100

44. AB is a diameter of a circle of radius 2 units and centre O. CD is a chord of the circle inclined at an angle of 60° to side AB. If bisector of angle OCD meets the circle at P such that the points D, P lie on the same semicircle with diameter AB of this circle. Then select the CORRECT alternative/s.

- A) The maximum area of triangle ABP is $2\sqrt{3}$
 B) The arc traced by point P subtends an angle 30° at the centre of the circle
 C) The sum of the distances of P from the points A and B is constant
 D) The minimum area of the triangle ABP is 2

45. The expression $(\cos 3\theta + \sin 3\theta) + (2 \sin 2\theta - 3)(\sin \theta - \cos \theta)$ is positive for θ belonging to

- A) $\left(2n\pi - \frac{3\pi}{4}, 2n\pi + \frac{\pi}{4}\right), n \in \mathbb{I}$ B) $\left(2n\pi - \frac{\pi}{2}, 2n\pi + \frac{\pi}{6}\right), n \in \mathbb{I}$
 C) $\left(2n\pi - \frac{\pi}{3}, 2n\pi + \frac{\pi}{3}\right), n \in \mathbb{I}$ D) $\left(2n\pi - \frac{\pi}{4}, 2n\pi + \frac{3\pi}{4}\right), n \in \mathbb{I}$

46. Let $\vec{a} \times \vec{x} + \vec{b} \times \vec{y} = \vec{a} + \vec{b}$ and $\vec{a} \times \vec{b} = \vec{x} + \vec{y}$ where $|\vec{a}| = |\vec{b}| = 1$, then which of the following must be true (Here the RHS in options A and D are null vectors)

- A) $\vec{a} + \vec{b} = 0$ B) $|\vec{x}| = 1$ C) $[\vec{a} \ \vec{b} \ \vec{x}] = 0$ D) $\vec{b} \times \vec{x} = 0$

47. If maximum and minimum values of the determinant $\begin{vmatrix} 1 + \sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1 + \cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1 + \sin 2x \end{vmatrix} \forall x \in \mathbb{R}$ are α and β respectively then which one of the option(s) is/are CORRECT?

- A) $\alpha + \beta^{99} = 4$
 B) $\alpha^3 - \beta^{17} = 26$
 C) $(\alpha^{2n} - \beta^{2n})$ is always an even integer for $n \in \mathbb{N}$
 D) A triangle can be constructed having its sides as α, β and $5\alpha - 13\beta$

48. If $\int \frac{x}{\sqrt[2019]{(2+x^2)^{4037}} (1+x^2)} dx = \frac{n+1}{2n} (1+f(x))^{\frac{-n}{n+1}} + c$, Then

A) Range of $f(x)$ is $(0, 1]$

B) $n = 2019$

C) $\frac{1}{f(n)}$ is a prime number

D) $n = 2018$

SECTION – III (Numerical Value Answer Type)

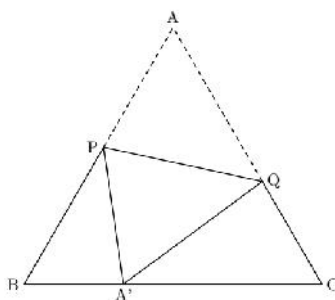
This section contains 6 questions. The answer to each question is a **Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place)**.
Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme : +3 for correct answer, 0 in all other cases.

49. Let $S = \{z = a + ib : a, b \in \mathbb{I}\}$. Let $A = \left\{z \mid \text{if } z \in S, \text{ then } \frac{4}{z} \in S\right\}$, then the number of elements in the set A is

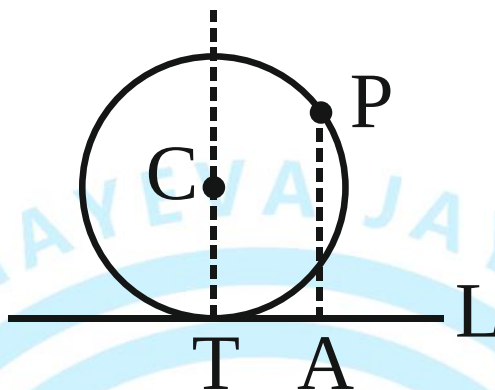
50. Area of triangle formed by common tangents to parabola $y^2 = 8x$ and hyperbola $\frac{x^2}{1} - \frac{y^2}{3} = 1$ and directrix of the given parabola is

51. A part of equilateral triangle ABC is folded as shown so that the vertex A falls on side BC such that the lengths $BA' = \frac{5}{\sqrt{21}}$, $A'C = \frac{10}{\sqrt{21}}$. Then the length PQ =

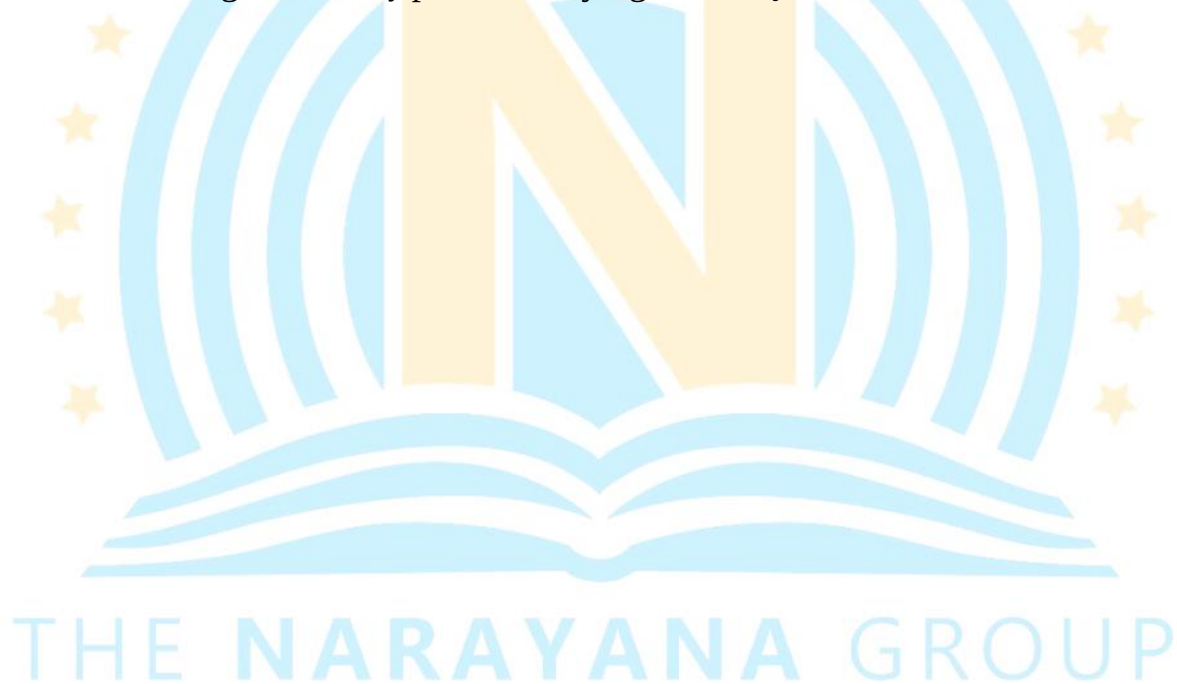


52. If $\tan^{-1} x + \operatorname{cosec}^{-1} \sqrt{10} = \tan^{-1} \left(\frac{1}{|y|} \right)$, where x is a positive integer, then the smallest absolute value of y satisfying above is

53. Let P be a variable point on a unit radius circle (as shown in figure). Let line L be tangent to circle at fixed point T and point A be projection of point P on L. If C be center of circle, then $\lim_{P \rightarrow T} \frac{(\text{Area of sector CPT})^3}{\text{Area of } \triangle PAT}$ is



54. The area of the region in $x - y$ plane satisfying $x^6 - x^2 + y^2 \leq 0$ is





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INDIA

Sec: Sr.N-Super Chaina
Time: 3hr

Date: 14-06-20
Max.Marks:186

Name of the Student: _____

H.T. NO:

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14-06-20_SR.N-SUPER CHAINA _Jee-Adv_GTA-18(P-2)_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

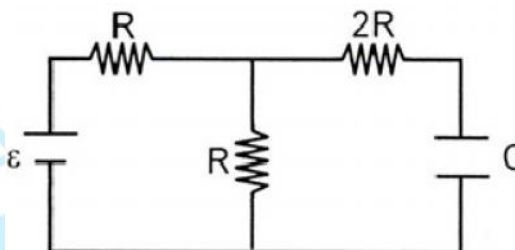
MATHEAMTICS : TOTAL SYLLABUS

SECTION – I
(MULTIPLE CORRECT CHOICE TYPE)

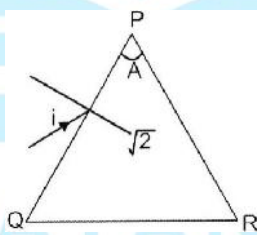
This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

1. Circuit shown in the figure is in steady state. Now the capacitor is suddenly filled with medium of dielectric constant $K=2$.

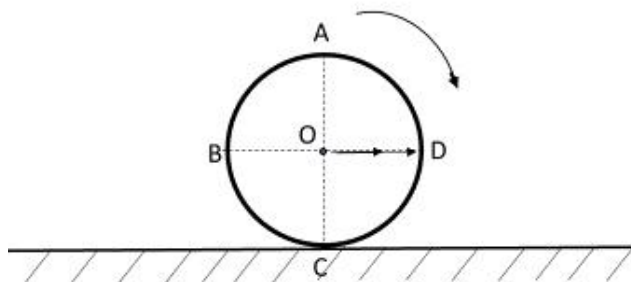


- A) current through $2R$ just after this moment is $\frac{\epsilon}{10R}$
- B) current through $2R$ just after this moment is $\frac{\epsilon}{15R}$
- C) current through battery just after this moment is $\frac{11\epsilon}{20R}$
- D) potential difference across capacitor just after this moment is $\frac{\epsilon}{4}$
2. A prism of refractive index $\sqrt{2}$ and apex angle A is in air as shown. Light is incident from PQ side at an angle of incidence i . ($0 < i < 90^\circ$). Choose the correct options

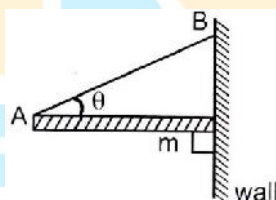


- A) if $A=40^\circ$ then light incident at all angles will be refracted at surface PR
- B) If $A=80^\circ$ then light incident at some angles will be refracted and at some other angles light will be reflected at surface PR.
- C) If $A=92^\circ$ then light incident at all angles will be reflected at surface PR
- D) whatever is the value of A , light definitely emerge from the surface PR

3. A wheel rolls on a plane surface without slipping. At a certain instant velocity and acceleration of centre 'O' are 1 m/s and 3 m/s^2 . Radius of the wheel is 0.5 m , BD is horizontal, and AC is vertical. Then at that instant

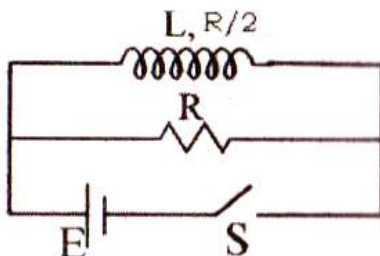


- A) accelerations of all four points i.e. A,B,C,D are equal in magnitude
 B) accelerations of points B and D are equal in magnitude
 C) acceleration of 'C' is 2 m/s^2
 D) acceleration of 'A' is $\sqrt{40}\text{ m/s}^2$
4. A uniform rod of mass m is supported by string AB and friction at the wall. Then choose the correct option(s): (g is acceleration due to gravity)

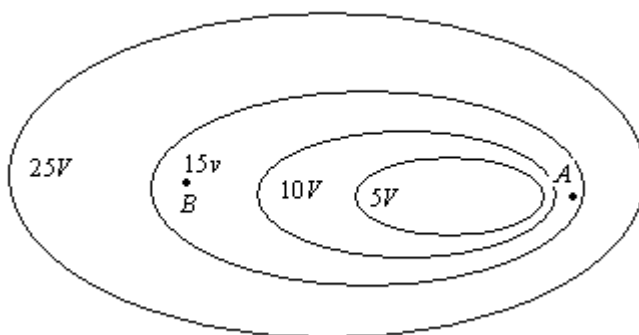


- A) tension in the string is $\frac{mg}{2} \cos\theta$
 B) friction force on the rod due to wall is mg downward
 C) friction force on the rod due to wall is $\frac{mg}{2}$ upward
 D) tension in the string is $\frac{mg}{2} \operatorname{cosec}\theta$

5. A solenoid of inductance L and resistance $\frac{R}{2}$ is connected in parallel to a resistance R . An ideal battery of emf E is connected across the parallel combination as shown in figure. Switch S is kept closed for long time and it is opened at time $t = 0$. The current through solenoid immediately after switch opened is I and total heat generated in the solenoid after switch opened is H . Then



- A) $I = \frac{E}{R}$ B) $I = \frac{2E}{R}$ C) $H = \frac{2}{3} \frac{LE^2}{R^2}$ D) $H = 2 \frac{LE^2}{R^2}$
6. The figure shows projection of few equi-potential surfaces with their potential values. These surfaces are ellipsoidal in shape and are symmetrical about the axis AB. Pick up the correct option(s) from the following statements:



- A) the field has a greater magnitude at point A compared to B and it is directed towards left
- B) the field has a greater magnitude at point A compared to B and it is directed towards right
- C) the field has a smaller magnitude at point B compared to A and it is directed towards right
- D) the field has a greater magnitude at point B compared to A and it is directed towards left

7. Two metallic solid spheres A and B are made of same material and have got identical surface finish. The mass of sphere A is four times that of B. Both the spheres are heated to the same temperature and placed in a room having lower temperature but thermally insulated from each other
- A) The ratio of heat loss of A to that of B is $2^{4/3}$
B) The ratio of heat loss of A to that of B is $2^{2/3}$
C) The ratio of initial rate of cooling of A to that of B is $2^{-2/3}$
D) The ratio of initial rate of cooling of A to that of B is $2^{-4/3}$
8. In a stationary wave,
- A) all the particles of the medium vibrate in phase
B) all the antinodes vibrate in phase
C) the alternate antinodes vibrate in phase
D) all the particles between consecutive nodes vibrate in phase

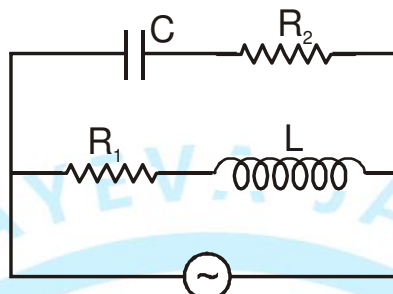
SECTION – II
(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a **Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place)**.
Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme : +3 for correct answer, 0 in all other cases.

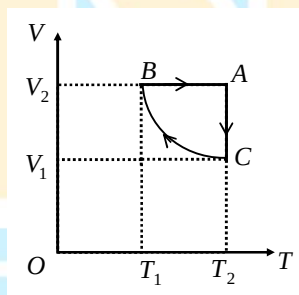
9. A particle of positive charge 1 mC and mass 1gm is projected from origin with velocity 20 m/s towards +ve x – axis in a non – uniform magnetic field $\vec{B} = -b_0 x \hat{k}$, where $b_0 = \frac{1}{10} T/m$ and x is in metre. If the maximum positive x – co – ordinate of the particle during its motion is $4x_0$ (in m) then x_0 is (Neglect gravity and air drag)
10. A neutron strikes a hydrogen atom which is initially at rest and free to move. The initial speed of the neutron is v_0 . After the collision both neutron and the H atom move making 30° angle with the initial velocity direction of the neutron. Also, the H atom is found to get excited to first excited state. If the initial KE of neutron is 'x' times first excitation energy of the H atom, find the value of 'x' (assume mass of neutron = mass of H atom)

11. In the shown circuit, $R_1 = 10\Omega$, $L = \frac{\sqrt{3}}{10}H$, $R_2 = 20\Omega$, $C = \frac{\sqrt{3}}{2}$ milli-farad and t is time in seconds. Then at the instant current through R_1 is $10\sqrt{2}A$; find the current through resistor R_2 in amperes

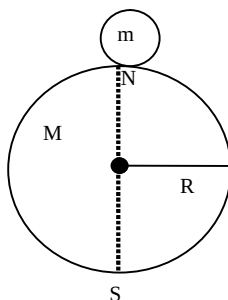


$$V = 200\sqrt{2} \sin(100t) \text{ volt}$$

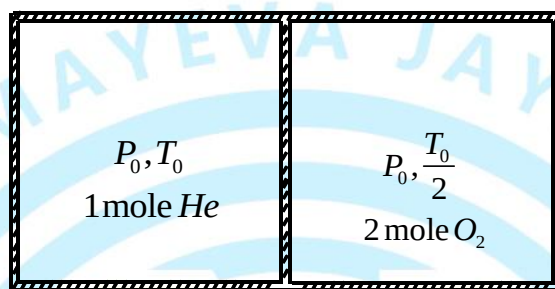
12. Consider a cyclic process ACBA consisting of $BA \rightarrow$ isochoric, $AC \rightarrow$ isothermal, $CB \rightarrow VT = \text{constant}$. Working substance is 2 moles of monoatomic ideal gas. If net heat exchange in the process is $xR(T_2 - T_1) + yRT_2 \ln \frac{V_2}{V_1}$, Find $(x+y)$.



13. A satellite of mass m and radius r is projected with a velocity V_0 tangential to the surface of the earth (radius R and mass M) at the North Pole. Given $r \ll R$. If that the satellite will go to infinity, for $RV_0^2 = xGM$. Find x .



14. An adiabatic piston divides a cylindrical container with adiabatic walls into two equal parts. The left part contains 1 mole of helium at temperature T_0 & pressure P_0 . The right part contains 2 moles of oxygen at temperature $\frac{T_0}{2}$ & pressure P_0 . Now the piston is removed and gases mix homogeneously and final pressure becomes $P = \frac{4n}{13} P_0$. Find the value of n



SECTION-III (MATCHING LIST TYPE)

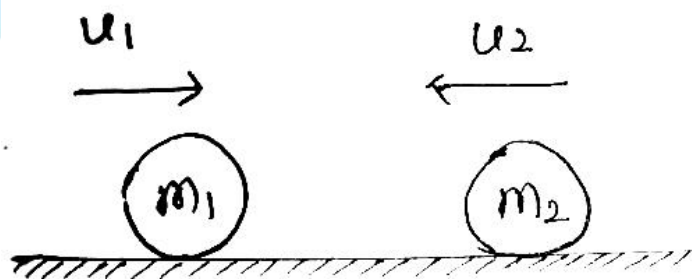
This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.15 and Q.16 by appropriately matching the lists based on the information given in the paragraph.

Two balls of masses m_1 and m_2 moving towards each other with speeds u_1 and u_2 , respectively. They collide head on and their speeds are v_1 and v_2 after collision

($m_1 = 8 \text{ kg}$, $m_2 = 2 \text{ kg}$, $u_2 = 3 \text{ m/s}$)



LIST -I		LIST -II(SI units)	
I)	The speed u_1 (in m/s) so that both balls move in same direction if coefficient of restitution is $e = 0.5$	P)	$\frac{1}{14}$
II)	The speed u_1 (in m/s) so that the maximum fraction of energy is transmitted to m_2 assume elastic collision	Q)	$\frac{1}{8}$
III)	Coefficient of restitution if m_2 stops after collision and $u_1 = 0.5$ m/s	R)	2
IV)	If collision is inelastic and $u_1 = 3$ m/s the loss of kinetic energy in (J) after collision may be	S)	4
		T)	3
		U)	$\frac{1}{6}$

15. Choose the correct option

A) I-R, II-Q, III-P, IV-P

B) I-P, II-S, III-P, IV-Q

C) I-T, II-T, III-U, IV-R

D) I-R, II-R, III-P, IV-R

16. Choose the incorrect option

A) I-T, II-R, III-P, IV-U

B) I-S, II-R, III-U, IV-P

C) I-R, II-R, III-P, IV-S

D) I-S, II-R, III-P, IV-P

Answer Q.17 and Q.18 by appropriately matching the lists based on the information given in the paragraph

Some physical quantities are given in List I and some possible SI units in which these quantities may be expressed are given in List II. Match the physical quantities in List I with the units in List II

LIST -I		LIST -II	
I)	$G \text{ Me Ms}$	P)	(volt) (columb) (meter)
II)	$\frac{3RT}{M}$	Q)	(kilogram) (meter) ³ (second) ⁻²
III)	$\frac{F^2}{Q^2 B^2}$	R)	(meter) ² (second) ⁻²
IV)	$\frac{GMe}{Re}$	S)	(farad) (volt) ² (kg) ⁻¹
		T)	(coulomb) (volt) ⁻¹
		U)	(meter) (second)

where G is universal gravitational constant; Me -mass of the earth, Ms -mass of the sun, Re -radius of earth, R -universal gas constant, T -absolute temperature, M -molar mass, F -force, Q -charge, B -magnetic field.

17. Choose the correct option

- A) I-T, II-P, III-U, IV-P B) I-P, II-Q, III-R, IV-S
C) I-P, II-R, III-S, IV-R D) I-R, II-R, III-S, IV-U

18. Choose the incorrect option

- A) I-P, II-S, III-R, IV-S B) I-P, II-R, III-S, IV-S
C) I-Q, II-P, III-R, IV-T D) I-Q, II-R, III-R, IV-R

SECTION – I
(MULTIPLE CORRECT CHOICE TYPE)

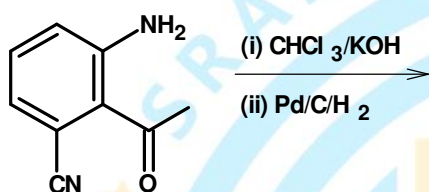
This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

Marking scheme +4 for correct answer, 0 if not attempted and -1 in all other cases.

19. The metals that can be reduced from their sulphides without any external reducing agent is (are)

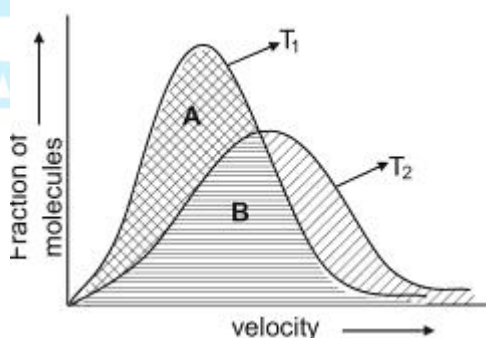
- A) Copper B) Lead C) Silver D) Gold

20. The major product obtained in the following reaction is :



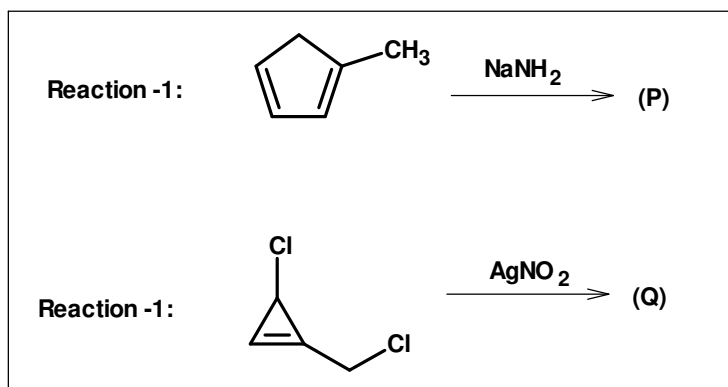
- A) B) C) D)

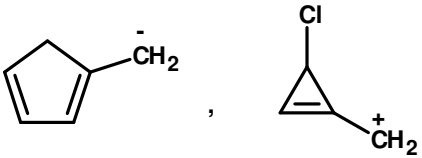
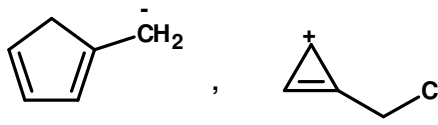
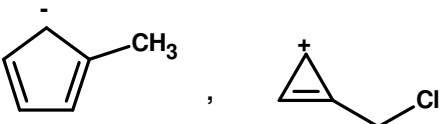
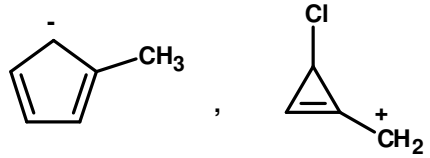
21. Let A and B be the area under the curve for a given sample of gas. If $T_2 > T_1$ then



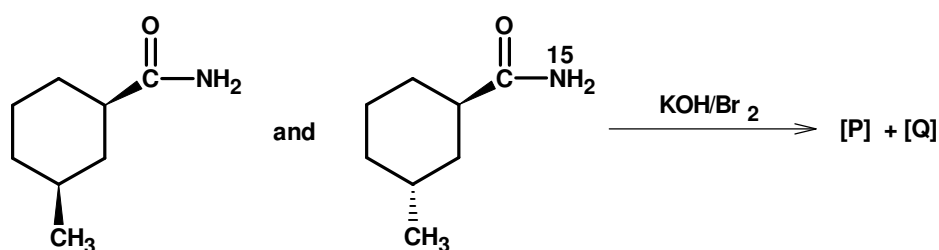
- A) $A = B$ B) $A > B$ C) $A < B$ D) cannot be predicted

22. The products P and Q in the following reaction respectively are

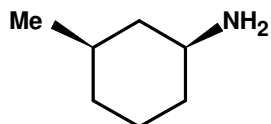
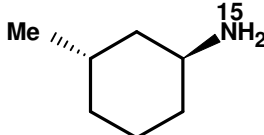
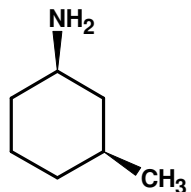
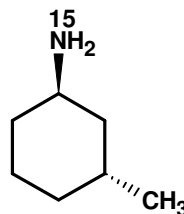


- a)  (P) (Q) B)  (P) (Q)
- c)  (P) (Q) D)  (P) (Q)

23.



[P] and [Q] are

- A) [P] =  [Q] =  B)
- C) [P] =  [Q] =  D)

24. Select the **incorrect** statement(s) from the following.

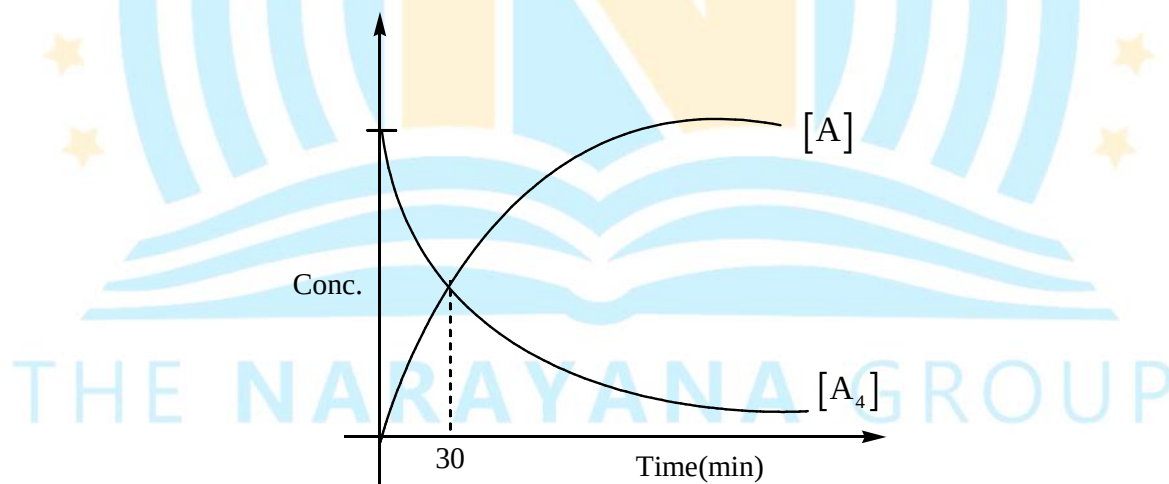
- A) If the critical micelle concentration (CMC) of a soap is 10^{-3} M, then 10^{-4} M solution of this soap will have colloidal nature.
- B) Charge on the colloidal particles may be determined with the help of dialysis.
- C) Peptization is the process of conversion of a colloidal sol into precipitate by adding some electrolyte.
- D) When some oil-soluble dye is added in the water in oil emulsion, the background becomes coloured.

25. Among the following, the alum(s) is/are:

- A) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ B) $\text{FeSO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
- C) $\text{K}_2\text{SO}_4 \cdot \text{Fe}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ D) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

26. Consider the following first-order decomposition reaction $\text{A}_4(g) \rightarrow 4\text{A}(g)$.

Which of the following statement(s) is/are correct regarding the reaction? ($\log 2 = 0.3$)



- A) At 30 min, only 20% reaction is complete.
- B) $t_{1/2}$ of reaction is 90_min.
- C) Rate of reaction decreases linearly with time.
- D) The time for intersection of two curves is independent of initial concentration of A_4 .

SECTION – II

(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place). Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

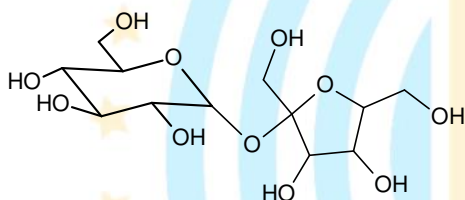
Marking scheme : +3 for correct answer, 0 in all other cases.

27. 0.5 molar HA (a weak acid) solution and 0.8 molar urea solution are found to be isotonic at 27°C . Under these conditions degree of dissociation of NaCl will be 10x%.

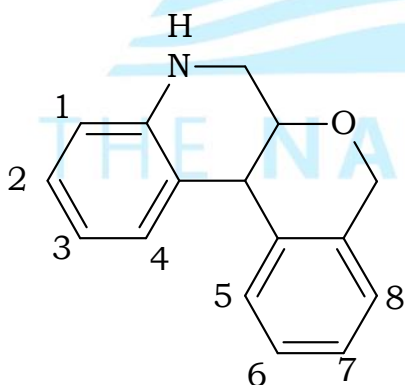
The value of x is__

28. A certain mass of ideal gas absorbs 80kJ heat and gas expanded from 2L to 10L against a constant pressure of 25 bar then ΔU for this process in kJ/mole is x then value of $\frac{x}{10}$ is (1bar-L=100J)

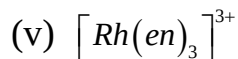
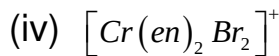
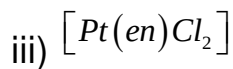
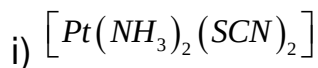
29. The number of acetal (or ketal) groups in the following saccharide is:



30. From which position does NO_2^+ replace a H^+ from the following compound predominantly?



31. Consider the following complex compounds.



Then calculate sum of total number of geometrical isomers in all above complex compounds.

32. In the following reaction $SO_2 + PCl_5 \rightarrow A + B$, the total number of π - bonds in the products A and B is

SECTION-III (MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.33 and Q.34 by appropriately matching the lists based on the information given in the paragraph.

Match the ions of column I with the reagent in column II.

Which responds to a positive test?

Column I (Radicals)		Column II (Reagents)	
I)	Cl^-	P)	$K_2Cr_2O_7, H^+$
II)	$C_2O_4^{2-}$	Q)	H_2SO_4 (Conc.)
III)	S^{2-}	R)	H_2SO_4 (dil.)
IV)	NO_3^-	S)	AgNO ₃ solution

33. Which of the following is correct combination for the above?

A) I, Q, R B) II, P, R C) I, P, S D) III, P, R

34. Which of the following is a correct combination?

A) I, R, S B) IV, Q, R C) III, P, Q D) IV, R

Answer Q.35 and Q.36 by appropriately matching the lists based on the information given in the paragraph.

Column I		Column II	
(I)		(P)	Diastereomers
(II)		(Q)	Recemic mixture
(III)		(R)	Meso compound
(IV)		(S)	CO ₂ gas is evolved

35. The correct combination in the following is

- A) I, P, S B) I, Q, S C) II, Q, R D) II, P, R

36. The **incorrect** combination in the following is

- A) I, P, S B) II, P, R C) III, S D) IV, P, R

MATHEMATICS

Max.Marks:62

SECTION – I
(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/ are correct.

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

37. If $\sin^{-1}(a^2x^2 + b^2y^2) + \cos^{-1}|ax + by| = \pi$, where $ab < 0$, then which of the following is possible?

A) $xy < 0$

B) $ax = \frac{1}{\sqrt{2}}, by = -\frac{1}{\sqrt{2}}$

C) $a^2 + b^2 = 1$

D) $abxy = \frac{1}{2}$

38. A solution of the equation: $\cos^2\left(\frac{\pi}{4}(\sin x + \sqrt{2}\cos^2 x)\right) - \tan^2\left(x + \frac{\pi}{4}\tan^2 x\right) = 1$ is

A) $x = -\frac{\pi}{4} + 2n\pi ; n \in \mathbb{Z}$

B) $x = \frac{5\pi}{4} + 2n\pi ; n \in \mathbb{Z}$

C) $x = 2n\pi ; n \in \mathbb{Z}$

D) $x = n\pi ; n \in \mathbb{Z}$

39. Let $f(x) = x^{13} + 2x^{12} + 3x^{11} + \dots + 13x + 14$ and $\alpha = \cos\frac{2\pi}{15} + i\sin\frac{2\pi}{15}$

If $N = f(\alpha)f(\alpha^2)f(\alpha^3)\dots f(\alpha^{14})$, then

A) Number of divisor of N is 196

B) Number of divisor of N is 256

C) Number of divisor of N which are perfect squares is 49

D) Number of divisor of N which are perfect cubes is 16

40. If the number of solutions satisfying the equation

$$(\sin\theta - 1)(2\sin\theta - 1)(3\sin\theta - 1)\dots(n\sin\theta - 1) = 0, \quad n \in \mathbb{N} \text{ in } [0, \pi] \text{ is } 1979 \text{ and ordered pair}$$

(x, y) satisfy the equation $1978 + \operatorname{cosec}^2 x + 1979^{\sin^2 y} = 2n$ where $0 \leq x, y \leq 2\pi$, then

A) No. of odd positive integral divisors of n are 12

B) Sum of possible values of x is 2π

C) No. of ordered pair of (x, y) are 6

D) Product of non-zero possible values of y is $2\pi^2$

41. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a bijective function and the graph of $f(x)$ is symmetrical about point $(1010, 0)$, then which of the following is **INCORRECT**

A) $f(1000) + f(1020) = 20$

B) $f^{-1}(2020) + f^{-1}(-2020) = 1010$

C) $f(990) + f(1030) = 2020$

D) $f^{-1}(4040) + f^{-1}(-4040) = 2020$

42. Let $Sai_n = \sum_{r=1}^n \left(\frac{r^4 + r^3n + r^2n^2 + 2n^4}{n^5} \right)$ and $Teja_n = \sum_{r=0}^{n-1} \left(\frac{r^4 + r^3n + r^2n^2 + 2n^4}{n^5} \right)$

$n = 1, 2, 3, \dots$ then in the list $\left\{ Sai_n, Teja_n, \frac{Sai_n + Teja_n}{2}, \frac{167}{60} \right\}$, the biggest and the second biggest numbers are

A) Sai_n

B) $Teja_n$

C) $\frac{Sai_n + Teja_n}{2}$

D) $\frac{167}{60}$

43. For the equation

$$\frac{x^2}{x-1} + \sqrt{x-1} + \frac{\sqrt{x-1}}{x^2} = \frac{x-1}{x^2} + \frac{1}{\sqrt{x-1}} + \frac{x^2}{\sqrt{x-1}}$$

which of the following is **CORRECT**

A) No. of real root of the equation is 1

B) Sum of the real roots of the equation is 4

C) Product of the real roots of the equation is 2

D) The real root/s of the equation is/are greater than 1

44. If $f_1(x) = x$, $f_2(x) = 1 - x$, $f_3(x) = \frac{1}{x}$, $f_4(x) = \frac{1}{1-x}$, $f_5(x) = \frac{x}{x-1}$, $f_6(x) = \frac{x-1}{x}$ suppose that

$f_6(f_m(x)) = f_4(x)$ and $f_n(f_4(x)) = f_3(x)$, then the value of $m + n$ is

A) Less than 8

B) Prime Number

C) Greater than 5

D) Odd Number

SECTION – II
(Numerical Value Answer Type)

This section contains 6 questions. Each question is numerical value. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place.

(e.g. 6.25, 7.00, -0.33, -30, 30.27, -127.30).

Marking scheme: +3 for correct answer , 0 if not attempted and 0 in all other cases.

45. If x, y, z, t , and u are positive real number such that $x + y + z + t + u = 8$, then

$$x^2 + y^2 + z^2 + t^2 + u^2 = 16, \text{ then the minimum value of } \frac{4}{u} \text{ is}$$

46. Let ABCD be a square. P and Q are any two points on BC and CD respectively. Such that $AP = 4$ cm, $PQ = 3$ cm, $AQ = 5$ cm. The area of the square is given by Δ , then $[\Delta]$ (where $[\cdot]$ is G.I.F.)

47. Let P be a fixed point on the parabola $y^2 = 5x$ which is not the vertex. Point Q on the parabola is such that the acute angle between the tangents to the parabola at P and Q is equal to the acute angle between x – axis and the normal to the parabola at Q. It is known that there are two such possible locations for point Q, namely points A and B on the parabola. If AB always passes through the fixed point (a, b) for any position of point P on the parabola, then $a + b =$

48. The value of $\frac{2022!}{2^{2020}} - \sum_{k=1}^{2020} \frac{k \cdot k!}{2^k} - \sum_{k=1}^{2020} \frac{k^2 \cdot k!}{2^k}$ is

49. Let $P_n = a^{P_{n-1}} - 1, \forall n = 2, 3, \dots$, and Let $P_1 = a^x - 1$ where $a \in \mathbb{R}^+$

$$\lim_{x \rightarrow 0} \frac{P_{2021}}{x} = k(\ln a)^n, \text{ then } n - k \text{ is equal to}$$

50. The projection length of a variable vector $x\hat{i} + y\hat{j} + z\hat{k}$ on the vector $\vec{P} = \hat{i} + 2\hat{j} + 3\hat{k}$ is 6. Let c be the minimum projection length of the vector $x^2\hat{i} + y^2\hat{j} + z^2\hat{k}$ on the vector $\vec{P}$ then the value of $\sqrt[3]{c^2 + 15^2}$ is

SECTION-III (MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q.51 and Q.52 by appropriately matching the lists based on the information given in the paragraph.

$a_1a_2a_3a_4a_5a_6$ is a six digit number formed using the digits 1, 2, 3, 4, 5, 6 once each such that the number $a_1a_2...a_k$ is divisible by k for $1 \leq k \leq 6$.

LIST -I		LIST -II	
I)	$a_1 + a_3 + a_6$	P)	12
II)	$a_1a_3 + a_5a_6$	Q)	9
III)	$a_1 + a_2 + a_3 + a_4$	R)	8
IV)	$(a_5^2 - a_6^2)(a_2 + a_4)$	S)	23
		T)	72

51. Which of the following is a **CORRECT** combination?

- A) (I), (P) B) (II), (S) C) (II), (T) D) (I), (Q)

52. Which of the following is an **INCORRECT** combination

- A) (I), (R) B) (IV), (T) C) (III), (P) D) (II), (P)

Answer Q.53 and Q.54 by appropriately matching the lists based on the information given in the paragraph.

Let $f(x) = 4\left(\sqrt{4 + \sqrt{16x^2 - 8x^3 + x^4}} - 2\right)$ be a real valued function defined on $\mathbb{R}$.

Let $A = \{a : f(x) \text{ is not differentiable at } x = a\}$

LIST –I		LIST –II	
I)	Sum of elements of A is	P)	6
II)	$f(x)$ has local max at $x =$	Q)	1
III)	No. of points where $f(x)$ has local min. is/are	R)	3
IV)	$f(x) = K$ has four different solution then K can be	S)	4
		T)	2

53. Which of the following is a **CORRECT** combination?

- A) (I). (P) B) (II). (T) C) (II). (S) D) (I). (Q)

54. Which of the following is an **INCORRECT** combination?

- A) (IV). (S) B) (I). (S) C) (III). (T) D) (II). (T)



PHASE-III

JEE ADVANCED

GRAND TEST SERIES

GRAND TEST-5



Sri Chaitanya IIT Academy, India

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON CENTRAL OFFICE, MADHAPUR-HYD

Sec: Sr.IPLCO/IC/ISB/IIIT

Time: 09:00 AM to 12:00 Noon

Date: 21-04-16

Max. Marks:240

Name of the Student: _____

H.T. NO:

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PAPER-I

21-04-16_ Sr.IPLCO/IC/ISB/IIIT_GTA-5_Weekend Syllabus

Mathematics: Total Syllabus

Physics : Total Syllabus

Chemistry : Total Syllabus

MATHS**Max Marks : 80****SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and --1 in all other cases.

- The area of the region bounded by the curves $y = e^{|x|} \ln|x|$, $x^2 + y^2 - 2(|x| + |y|) + 1 = 0$ and x-axis where $|x| \leq 1$, if α is the x-co-ordinate of the point of intersection of curves in 1st quadrant is

A) $4 \left[\int_0^\alpha e^x \ln x \, dx + \int_\alpha^1 \left(1 - \sqrt{1 - (x-1)^2} \right) dx \right]$

C) $2 \left[-\int_0^\alpha e^x \ln x \, dx + \int_\alpha^1 \left(1 - \sqrt{1 - (x-1)^2} \right) dx \right]$

B) $4 \left[\int_0^\alpha e^x \ln x \, dx - \int_1^\alpha \left(1 - \sqrt{1 - (x-1)^2} \right) dx \right]$

D) $2 \left[-\int_0^\alpha e^x \ln x \, dx - \int_\alpha^1 \left(1 - \sqrt{1 - (x-1)^2} \right) dx \right]$
- $\int_{-1}^1 \frac{dx}{x^2 + x + 1 + \sqrt{x^4 + 3x^2 + 1}}$ is equal to

A) $\frac{\pi}{2}$

B) $\frac{\pi}{4}$

C) $\frac{\pi}{3}$

D) $\frac{\pi}{8}$
- There are twenty points $A_1, A_2, \dots, A_{20}$ on a circle. The number of ways of selecting 4 points such that there are atleast two points in between any two selected points is

A) ${}^{20}C_1 \cdot {}^{11}C_3$

B) $5 \cdot {}^{11}C_3$

C) $5 \cdot {}^{10}C_3$

D) none of these

4. The perpendicular distance from origin to the line $\bar{r} \cdot \bar{a} = 1, \bar{r} \cdot \bar{b} = 1$,

A) $\left| \frac{\bar{a} + \bar{b}}{\bar{a} \times \bar{b}} \right|$

B) $\frac{|\bar{a} - \bar{b}|}{|\bar{a} \times \bar{b}|^2}$

C) $\sqrt{\frac{|\bar{a} - \bar{b}|}{|\bar{a} \times \bar{b}|}}$

D) $\frac{|\bar{a} - \bar{b}|}{|\bar{a} \times \bar{b}|}$

5. If a and b are distinct positive integers and the quadratic equations

$(a-1)x^2 - (a^2+2)x + (a^2+2a) = 0$ and $(b-1)x^2 - (b^2+2)x + (b^2+2b) = 0$ have a common root. Then which of the following can be true.

A) $a^2 + b^2 = 45$

B) $a = 2b$

C) $b = 3a$

D) $ab = 18$

SECTION - II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which MORE than ONE option can be correct.

Marking scheme: +5 for correct answer, 0 if not attempted and -2 in all other cases.

6. Let $f(x) = x^2 e^x$ and $g(x) = c$. Let $x_0 \in \mathbb{R}$ such that $f(x_0) = g(x_0)$ have exactly two solutions. Then which of the following is/are correct.

A) $g(x_0) = 4e^{-2}$

B) $f(x) + g(x)$ is a monotonic function.

C) $f(x)$ is decreasing in $(-1, -1/2)$

D) area bounded by $y = f(x)$, $y = f(x_0)$ and $x = 0$ is more than $1(\text{unit})^2$

7. A fair die is rolled and the random variable Z takes 2 if the die shows even number other wise Z takes 3. Now a coin is tossed Z times
- A) If the coin tossed is fair and shows exactly 1 head then the probability that “ Z ” takes 2 is $\frac{4}{7}$
- B) If the coin tossed is fair and shows exactly 1 head then the probability that “ Z ” takes 2 is $\frac{3}{7}$
- C) If the tossed coin is biased with a probability of $\frac{3}{4}$ for showing head and shows exactly 1 head then the probability that “ Z ” takes 2, is $\frac{8}{11}$
- D) If the tossed coin is biased with a probability of $\frac{3}{4}$ for showing head and shows exactly 1 head then the probability that “ Z ” takes 2, is $\frac{24}{31}$
8. Normal at any point P on a rectangular hyperbola, meets its axes in “ G ” and “ g ” and C be the centre of the rectangular hyperbola. Then which of the following statement(s) is / are correct
- A) $PG = Pg$ B) $PG = PC$ C) $Pg = PC$ D) $PG = 2PC$

9. Consider $L = 3\sqrt{12} + 3\sqrt{13} + 3\sqrt{14} + \dots + 3\sqrt{1011}$

$$R = 3\sqrt{13} + 3\sqrt{14} + 3\sqrt{15} + \dots + 3\sqrt{1012}$$

$I = \int_{12}^{1012} 3\sqrt{x} \, dx$ then which of the following is/are correct?

A) $L + R < 2I$

B) $R > I$

C) $L < I$

D) $L + R > 2I$

10. Consider $f(x) = a\sqrt{x-1} + b\sqrt{2x-1} - \sqrt{2x^2 - 3x + 1}$ and $a, b \in R$, $a + 2b = 2$, then

A) for $b > 0$, there exists some $C \in (1, 5)$ such that $f(C) = 0$

B) for $b < 0$, there exists some $C \in (1, 2)$ such that $f(C) = 0$

C) for $b < -2$ there exists some $C \in (1, 5)$ such that $f(C) = 0$

D) for $b = 0$ & $a \in R$ there exists some "c" such that $f'(c) = 0$

SECTION - III (PARAGRAPH TYPE)

This section contains **2 Paragraph of questions**. Each paragraph has 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONE OR MORE** is/ are correct. **Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.**

Passage: 1

Consider the curve $xy = 15!$ and the point (α, β) lies on it $(\alpha, \beta \in I^+)$

11. Which of the following is/are correct?
- A) number of distinct ordered pairs (α, β) such that HCF of (α, β) is 1, is 64
 - B) number of ways of writing $N = 15!$ as a product of two positive co-prime numbers, is 32
 - C) number of ways of writing $N = 15!$ as product of positive numbers is 16
 - D) number of ordered pairs (α, β) such that $\alpha\beta = 15!$ (α, β even integers) is 3360
12. Which of the following is/are correct?
- A) Product of all divisors of $N = 15!$ is $(15!)^{2016}$
 - B) Sum of all odd divisors of $N = 15!$ is $(3^7 - 1)(5^4 - 1)(7^3 - 1)\frac{7}{2}$
 - C) Product of all divisors of $N = 15!$ which are not divisible of 5, is $(15!)^{1008}$
 - D) Sum of all odd divisors of $15!$ is $(3^7 - 1)(5^4 - 1)(7^3 - 1)7$
13. Which of the following is/are correct.
- A) Number of ordered pairs (α, β) such that α divides β is 96
 - B) number of divisors of $N = 15!$ is 4032
 - C) Number of ordered pairs (α, β) such that α divides β is 192
 - D) If 12^x divides $15!$, then the maximum value of x is 5

Passage: 2

Consider a parabola $y^2 = 4x$ & an ellipse $\frac{x^2}{50} + \frac{y^2}{8} = 1$. Consider a point $P(h, k)$, where $h, k \in \mathbb{N} \cup \{0\}$ from which normals, are drawn to the parabola.

14. Which of the following is/ are correct

A) Number of points $P(h, k)$ on the ellipse from which exactly two **distinct** normals can be drawn, is 2

B) Number of points $P(h, k)$ on the ellipse from which exactly two **distinct** normals can be drawn, is 4

C) If $k=0$, then number of points $P(h, k)$ inside the ellipse from which exactly three normals can be drawn, 5

D) If $k=0$, then number of points $P(h, k)$ inside the ellipse from which exactly three normals can be drawn, 6

15. Which of the following is/ are correct

A) If $h=6$, then number of points $P(h, k)$ outside the ellipse from which exactly three normals can be drawn, is **2**

B) If $|h| > 2$, then number of points $P(h, k)$ inside the ellipse from which exactly one normal can be drawn, is **6**

C) If $h=6$, then number of points $P(h, k)$ outside the ellipse from which exactly three normals can be drawn, is 6

D) If $|h| > 2$, then number of points $P(h, k)$ inside the ellipse from which exactly one normal can be drawn, is 8

16. Which of the following is/ are correct

- A) If $h < 6$, then number of points $P(h, k)$ inside the ellipse from which exactly three normals can be drawn, is **5**
- B) If $h < 6$, then number of points $P(h, k)$ inside the ellipse and outside the parabola from which exactly three normals can be drawn, is **0**
- C) If $h = 7$, then number of points $P(h, k)$ outside the ellipse from which exactly three normals can be drawn, is **5**
- D) If $6 < h < 9$, then number of points $P(h, k)$ inside the ellipse from which exactly three normal can be drawn, is **1**

SECTION – IV
(INTEGER ANSWER TYPE)

This section contains 4 questions. The answer is a Three digit integer ranging from 000 to 999 (both inclusive-Ex: If answer is 1 you should mark as 001).

Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

17. Let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$ be a set

Let $f : A \rightarrow A$ be a function such that

i) $f(x) \neq x \quad \forall x \in A$,

ii) $f(x)$ is even whenever x -is even and

iii) $f(x)$ is odd whenever x -is odd,

then the number of onto functions satisfying these conditions, will be

18. The triangle ABC right angled at C, has medians AD, BE, CF. AD lies along the line $y = x + 3$, BE lies along the line $y = 2x + 4$. If the length of the hypotenuse is 60, then area of the triangle ABC in square units is.....
19. A quadratic equation with roots α, β where $\alpha\beta = 4$, satisfies the equation $\frac{\alpha}{\alpha-1} + \frac{\beta}{\beta-1} = \frac{a^2-7}{a^2-4}$, then number of integral values of "a" for which $\alpha, \beta \in (1, 4)$, is/are _____
20. If $C = \{\cos \alpha, \cos 2\alpha, \cos 3\alpha\}$ and $S = \{\sin \alpha, \sin 2\alpha, \sin 3\alpha\}$ are equal sets then number of values α in $[0, 2\pi]$

PHYSICS

Max Marks : 80

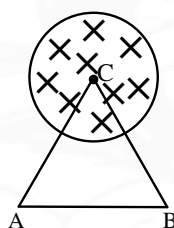
SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

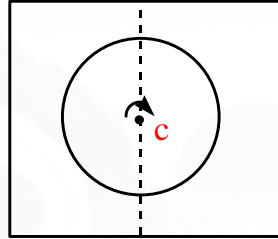
Marking scheme: + 3 for correct answer, 0 if not attempted and -- in all other cases.

21. A uniform magnetic field of intensity $B = C_1 t + C_2$ is directed into the plane of the paper exists in the cylindrical region of radius r . Here C_1 and C_2 are positive constant and t is time in seconds. A loop in form of an equilateral triangle ABC of side length $2r$ is placed such that vertex C lies on axis of cylindrical region and plane of loop is perpendicular to axis. Resistance of unit length of wire of loop = $\lambda \Omega/m$. Find the potential difference $V_A - V_B$ at any time t

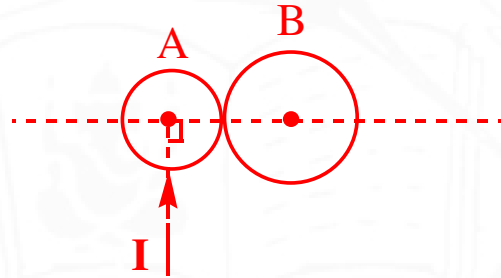


- A) $+\frac{C_1 \pi r^2}{6}$ B) $-\frac{C_1 \pi r^2}{6}$ C) $+\frac{C_1 \pi r^2}{9}$ D) $-\frac{C_1 \pi r^2}{9}$
22. One digital watch 'A' was showing 1 : 12 : 32 and an old watch 'B' not having seconds hand was showing 1 : 10 ; at the moment actual time was 1 : 10 : 15
- A) 'A' is more precise but less accurate than 'B'
- B) 'A' is less precise but more accurate than 'B'
- C) 'A' is both more accurate and precise
- D) 'B' is both more accurate and precise

23. In the figure shown a uniform ring of mass m is placed on a rough horizontal platform. The coefficient of friction between left half of ring and table is μ_1 whereas between right half and table is μ_2 . The ring has angular velocity in clockwise sense. Find the magnitude of instantaneous acceleration of centre C of ring. Given $\mu_1 = 0.6\pi$, $\mu_2 = 0.2\pi$ and $g = 10 \text{ m/s}^2$.



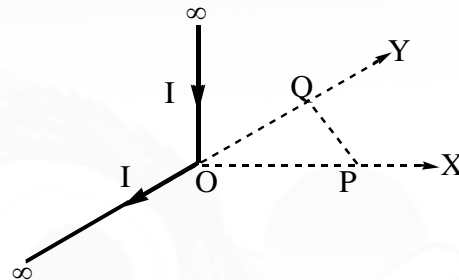
- A) $4m/s^2$ B) $2m/s^2$ C) $1m/s^2$ D) Zero
24. Figure shows an isolated system of two smooth spheres 'A' and 'B' at rest. $m_A = m_B = m$; $2R_A = R_B = R$. Now an impulse is imparted to sphere A. Find the maximum value of impulse such that sphere 'A' does not lose contact with 'B'.



- A) $\sqrt{\frac{Gm^3}{3R}}$ B) $\sqrt{\frac{2Gm^3}{3R}}$ C) $\sqrt{\frac{4Gm^3}{3R}}$ D) Zero

25. Figure shows combination of two semi infinite long wires kept on Z and Y-axis. It carries constant current I as shown in figure. Calculate $\int_Q^P \vec{B} \cdot d\vec{l}$ on line QP.

Co-ordinate of point P and Q are $(a, 0, 0)$ and $(0, a, 0)$ respectively.



- A) $\mu_0 I$ B) $\frac{\mu_0 I}{2}$ C) $\frac{\mu_0 I}{4}$ D) $\frac{\mu_0 I}{8}$

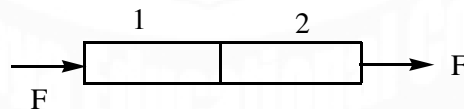
SECTION – II

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which MORE than ONE option can be correct.

Marking scheme: + 5 for correct answer, 0 if not attempted and -2 in all other cases.

26. Figure shows two identical rods 1 and 2 having same length (l), same area of cross-section (A) and made up of same material. Young's modulus of material of rods is 'Y'. Constant and equal forces each of magnitude 'F' is applied at both ends as shown below:



A) Net deformation of two rod system is $\frac{2Fl}{AY}$

B) Net deformation of two rod system is $\frac{Fl}{AY}$

C) Elastic potential energy stored in two rod system is $\frac{2F^2l}{3AY}$

D) Elastic potential energy stored in two rod system is $\frac{F^2l}{3AY}$

27. Suppose the coloumbs law is defined as $\vec{F}_{12} = \frac{Kq_1q_2}{r_{12}^3} \hat{r}_{21}$. Then which of the following statements will be correct?

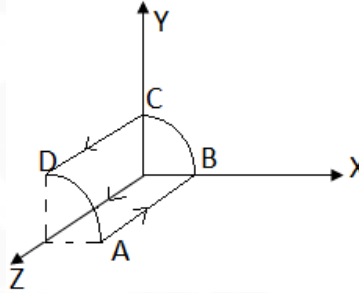
A) Gauss theorem is still valid

B) The electric potential due to a positive charge increases with distance from that positive charge

C) Like charges would repel

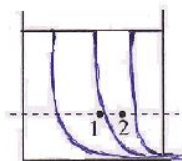
D) Like charges would attract

28. A long thin wire is carrying current along positive Z-axis. Another current carrying loop made of uniform wire is arranged with its vertices at $A(a,0,a)$ $B(a,0,0)$ $C(0,a,0)$ $D(0,a,a)$ as shown in figure. BC and DA are in the shape of a quarter circle centered on Z-axis. The loop is constrained to rotate about Z-axis without friction.

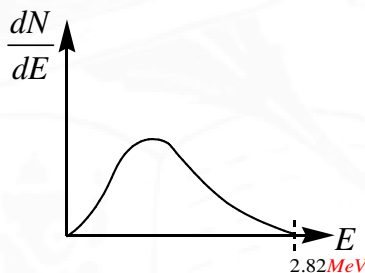


- A) Mutual inductance between the long wire and the loop is zero.
 B) Net magnetic force acting on the loop due to the long wire is zero
 C) Net torque due to magnetic force on the loop about Z-axis is zero.
 D) Consider an ampere loop in the shape of a circle of radius 'a'. Plane of the loop is parallel to XZ-plane with its centre at $(0, a/2, 0)$. Let $\vec{B}$ be the net magnetic field in space. Then $\oint \vec{B} \cdot d\vec{l}$ around the Ampere loop is zero.

29. Water is filled in a vessel. On opening the tap, water starts flowing out. Figure shows streamline pattern in the vessel at an instant. Points '1' and '2' lie on same horizontal level, 'P' and 'V' represent pressure and speed of fluid element passing through a point in vessel. Then,



- A) $P_1 = P_2$ B) $P_1 > P_2$ C) $V_1 = V_2$ D) $V_1 < V_2$
30. One mole of a nuclide undergoes β^+ decay and turns completely into a stable substance. Let dN be the number of β^+ particles that possess energies between E and $E+dE$. The plot of $\frac{dN}{dE}$ versus E is given below.



- A) All the β^+ particles must have nearly uniform linear momentum
 B) Conservation of linear momentum is not valid in this decay process
 C) Total mass defect in the process is 0.003 g
 D) Area of the region bounded by the graph with E axis is equal to Avogadro number

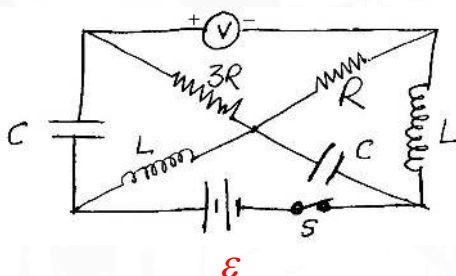
SECTION - III

(PARAGRAPH TYPE)

This section contains **2 Paragraph of questions**. Each paragraph has 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONE OR MORE** is/ are correct. **Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.**

PASSAGE-1

The circuit consists of two resistances of values R and $3R$, two identical capacitors of capacitance C , two identical inductors each of inductance L and an ideal battery of emf ' ε ' and a switch ' S '. Initially the switch is closed and the circuit is in steady state. V is an ideal volt meter with its terminals as indicated in the diagram



31. select correct option(s)
- A) The reading of volt meter is ε
 - B) The reading of volt meter is zero
 - C) Current through two inductors is same
 - D) Both capacitors are uncharged

32. Now, the switch 'S' is opened.
- A) the reading of voltmeter immediately after 'S' is opened is ε
 - B) the reading of voltmeter immediately after 'S' is opened is zero
 - C) Long time after the opening of the switch S, the reading of volt meter is zero
 - D) Long time after opening the switch S, reading of volt meter is ε
33. Find the total amount of heat dissipated in the resistors after 'S' is opened.
- A) $\frac{L\varepsilon^2}{R^2} + c\varepsilon^2$
 - B) $2\frac{L\varepsilon^2}{R^2} + \frac{1}{2}c\varepsilon^2$
 - C) $\frac{1}{2}c\varepsilon^2 + \frac{L\varepsilon^2}{R^2}$
 - D) $\frac{1}{2}c\varepsilon^2 + \frac{1}{2}\frac{L\varepsilon^2}{R^2}$

PASSAGE -2

Light from discharge tube containing hydrogen atoms falls on surface of a piece of sodium. The kinetic energy of fastest photo electron emitted from sodium is 0.73 eV. The work function of sodium is 1.82 eV. Ionization potential of hydrogen is 13.6 eV. Assume initially hydrogen atoms emitting electrons, are at rest.

34. SELECT CORRECT OPTIONS
- A) The energy of photon causing photoelectric emission is 1.09 eV
 - B) The energy of photon causing photoelectric emission is 2.55 eV
 - C) The quantum number of two levels involved in emission of these photons is. $4 \rightarrow 2$
 - D) The quantum number of two levels involved in emission of these photons is $2 \rightarrow 1$.

35. change in angular momentum of the electron in hydrogen atom in the above transition is

- A) $2h/\pi$ B) $-h/\pi$ C) $+h/\pi$ D) $-2h/\pi$

36. The recoil speed of the emitting atom is

- A) 0.426 m/s B) 0.814 m/s C) 0.213 m/s D) 1.628 m/s

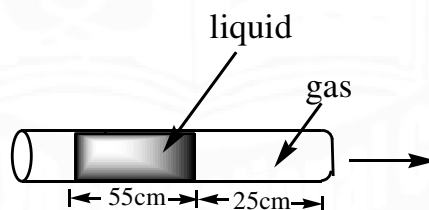
SECTION – IV

(INTEGER ANSWER TYPE)

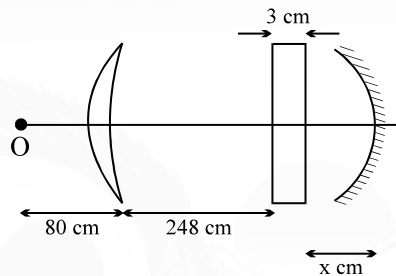
This section contains 4 questions. The answer is a Three digit integer ranging from 000 to 999 (both inclusive-Ex: If answer is 1 you should mark as 001).

Marking scheme + 4 for correct answer , 0 if not attempted and 0 in all other cases.

37. A 1m long tube is open at one end and closed at the other end. Initially it contains a liquid column of 55 cm which traps 25 cm column of air when held horizontal. Now the tube is held vertical with open end at bottom. Now what will be the equilibrium height of liquid column (in cm), after fall of some liquid, if some liquid falls? (Neglect surface tension, assume temperature of the gas remains constant). $P_{\text{atm}} = 10^5 \text{ Pa}$, $\rho_{\text{liquid}} = 10^2 \text{ kg/m}^3$, $g = 10 \text{ m/s}^2$

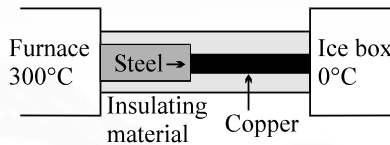


38. A concavo-convex lens made of glass of refractive index 1.5 has surfaces of radii 20 cm and 60 cm. A concave mirror of radius of curvature 20 cm is placed co-axially to the lens. A glass slab of thickness 3 cm and refractive index 1.5 is

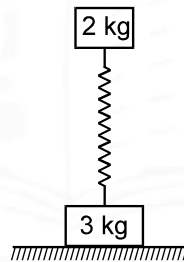


- placed close to the mirror in the space between the mirror and lens as shown in figure. The distance of the nearest surface of the slab from lens is 248 cm. An object is placed 80 cm to the left of the lens. If the final position of the image formed after the refraction from lens, refraction from slab, reflection from mirror, refraction from the slab and lens is at the object O, find the distance x (in cm) of the mirror from nearer surface of slab.
39. What is the temperature in $^{\circ}\text{C}$ of the steel-copper junction in the steady state of the system shown in the figure. Length of the steel rod = 25 cm, length of the copper rod = 50 cm, temperature of the furnace = 300°C ,

temperature of the other end = 0°C . The area of cross section of the steel rod is twice that of the copper rod. (Thermal conductivity of steel = $50\text{Js}^{-1}\text{m}^{-1}\text{K}^{-1}$ and of copper = $400\text{Js}^{-1}\text{m}^{-1}\text{K}^{-1}$)



40. The ends of a spring are attached to blocks of masses 3 kg and 2 kg. The 3 kg block rests on a horizontal surface and the 2 kg block which is vertically above it is in equilibrium producing a compression of 1 cm of the spring initially. The **minimum** value of the distance to which the 2 kg mass must be displaced **down** (in mm), so that when it is released, the 3 kg block may be lifted off the ground is :
 ($g = 10\text{m/s}^2$)



CHEMISTRY**Max Marks : 80****SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: + 3 for correct answer, 0 if not attempted and -- in all other cases.

41. During the ring test for nitrate; when freshly prepared FeSO_4 solution is added to aqueous solution of NO_3^- , followed by addition of conc. H_2SO_4

A) The complex is formed by Fe^+ and NO

B) During the formation of brown complex charge transfer takes place as

$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}; \text{NO} \rightarrow \text{NO}^+$

C) During the formation of brown complex charge transfer takes place as

$\text{Fe}^{2+} \rightarrow \text{Fe}^+; \text{NO} \rightarrow \text{NO}^+$

D) The complex is formed by Fe^{3+} and NO

42. For the following equilibrium in a closed container



At a constant temperature, the volume of the container is halved suddenly, which of the following is correct for K_p and α ?

A) Both K_p and α changes

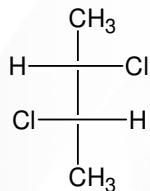
B) None of the K_p and α change

C) K_p changes but α does not

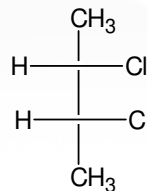
D) α - changes but K_p does not

43. If the specific rotation produced by A is $+36^\circ$, then the specific rotation produced by B will be

A is



B is



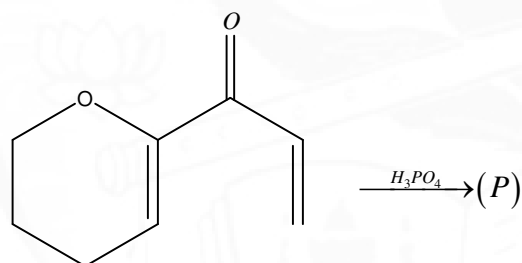
A) -36°

B) $+36^\circ$

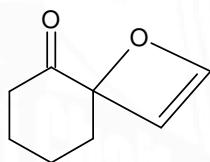
C) 0°

D) it is experimental, can't be predicted

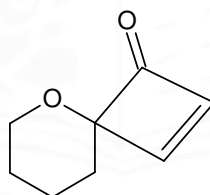
44.



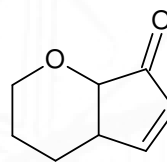
A)



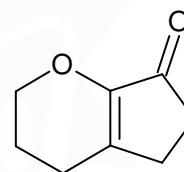
B)



C)



D)



45. If the energy of H-atom in the ground state is $-E$, the velocity of electron emitted when a photon having energy E_p strikes a stationary Li^{2+} ion in ground state, is given by:

A) $v = \sqrt{\frac{2(E_p - E)}{m}}$ B) $v = \sqrt{\frac{2(E_p + 9E)}{m}}$ C) $v = \sqrt{\frac{2(E_p - 9E)}{m}}$ D) $v = \sqrt{\frac{2(E_p - 3E)}{m}}$

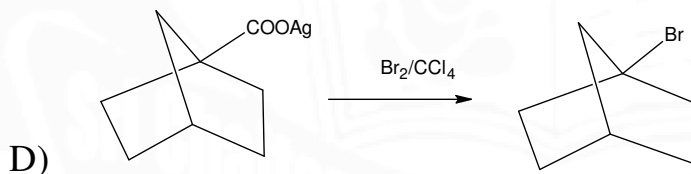
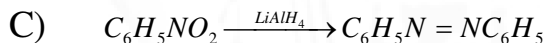
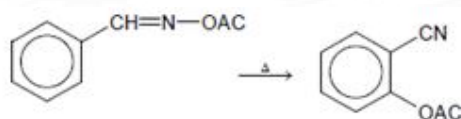
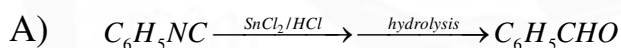
SECTION - II

(MULTIPLE CORRECT ANSWER TYPE)

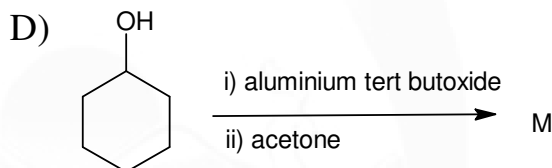
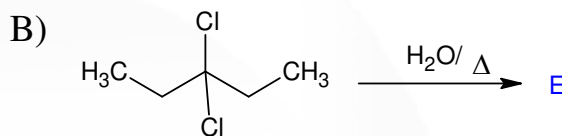
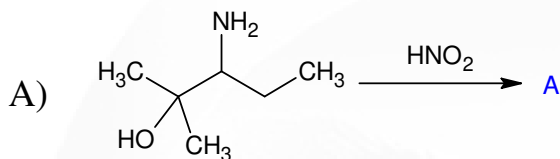
This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which MORE than ONE option can be correct.

Marking scheme: + 5 for correct answer, 0 if not attempted and -2 in all other cases.

46. Which of the following reaction(s) are incorrect?



47. In which of the following reaction(s), one of the possible product can give haloform test?



48. How many of the following are correct statement(s)?

- A) CaSO_4 is insoluble but it is not precipitated when excess of $(\text{NH}_4)_2\text{SO}_4$ is added to CaCl_2
- B) H_2O_2 is obtained in better yield by treating BaO_2 with H_3PO_4 rather with H_2SO_4
- C) XeF_6 is a poor conductor of electricity but once dissolved in HF , it becomes good conductor
- D) $\text{Ni}(\text{CO})_4$ exists but similar stable complex is not formed by Pd or Pt

49. 34.1 gm Pb_3O_4 is dissolved in 500 ml of 4M HNO_3 leaving some residue; then (given $M_{\text{Pb}} = 206$ gm/mole)
- A) Lead oxide is limiting reagent
 - B) The weight of residue is 19.9 gm
 - C) 300 ml of 6 M NaOH is required to neutralise excess of HNO_3
 - D) Total 2.3 moles of NaOH will be used up by the contents remaining after reaction of Pb_3O_4 with HNO_3
50. Which of the following statement(s) is/are true?
- A) The equivalent conductances of both strong and weak electrolytes increase on dilution.
 - B) For dilute solutions, the increase in equivalent conductance for both strong and weak electrolytes on further dilution is due to increase in no of ions.
 - C) If $\text{CH}_3\text{COOH}(\text{aq})$ is separated from pure water by semi permeable membrane, then equivalent conductance of solution increases gradually and ultimately becomes constant.
 - D) *for the cell, $\text{Ti}(\text{s}) | \text{Ti}^+(0.001\text{M}) || \text{Cu}^{+2}(0.1\text{M}) | \text{Cu}(\text{s})$; $E_{\text{cell}} = 0.83\text{V}$ at 298K , The cell potential doesn't change by increasing or decreasing $[\text{Ti}^+]$*

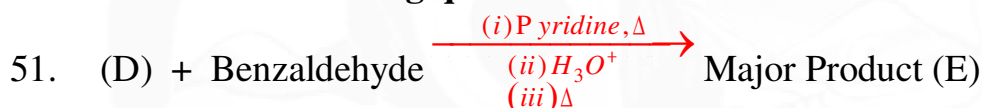
SECTION - III

(PARAGRAPH TYPE)

This section contains **2 Paragraph of questions**. Each paragraph has 3 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONE OR MORE** is/ are correct. **Marking scheme: +4 for correct answer, 0 if not attempted and -2 in all other cases.**

Passage: 1

An organic compound (A) $C_3H_6Cl_2$ on reduction with red P and HI gives propane. (A) on hydrolysis by an alkali followed by oxidation gives (B) $C_3H_4O_4$, which on heating gives (C). Both (B) and (C) give effervescence with $NaHCO_3$. (B) on reacting with ethyl alcohol gives (D) $C_7H_{12}O_4$, a well known synthetic reagent.

Now answer the following questions

The major product (E) is

A) crotonic acid B) cinnamic acid C) benzoic acid D) Mandelic acid

52. Choose the correct statement(s)

A) Out of A, B, C, D and E only B, C and E give effervescence with sodium bicarbonate

B) Product formed when C is treated with P_2O_5 is a lactone

C) Out of A, B, C, D and E; only E can show any type of stereoisomerism considered

D) The lower homologue of C reduces mercuric chloride to mercurous chloride

53. Conversion of benzaldehyde to E can also be done by
- A) Treating benzaldehyde with anhydride of C in the presence of sodium salt of C
 - B) Oxidation of crotonaldehyde(but-2-enal) with potassium dichromate
 - C) Oxidation of 3-phenyl-2-propenal with ammonical AgNO_3 followed by acidification.
 - D) Treating benzaldehyde with malonic anhydride in the presence of sodium salt of malonic acid followed by acidic hydrolysis and heating

Passage: 2

The statement of the first law of thermodynamics given by Clausius, is: 'Die Energie der welt ist konstant'. That is, the energy of the universe is constant. Mathematically it can be written as: $\Sigma \Delta U = 0$; (universe)

The first law predicts that energy added to or removed from a system must be accounted for, by a change in the internal energy ΔU . Energy is added to or removed from a system as work (w) or heat (q), in which case $\Delta U = q + w$; for an infinitesimal process, this equation becomes $\delta U = \delta q + \delta w$.

Heat and work represent ways that energy can be transferred from one system to another. q and w, both are path functions and both are interconvertible. Extent of work done can be calculated by

$$w = - \int_{v_1}^{v_2} P_{ext} dV .$$

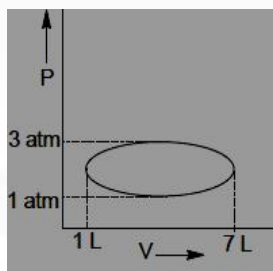
54. Which of the following is correct statement?

- A) The sum of the two path dependent properties, is always path dependent.
- B) Work appears at the boundary of the system as well as inside the system.
- C) Work and heat are completely inter convertible in real life systems.
- D) For a gas enclosed in a cylinder, magnitude of work done during isothermal irreversible expansion is less than isothermal irreversible compression for a given pair of initial and final states.

55. Which of the following is/are correct?

- A) Work done in a cyclic process is always zero.
- B) A two litre container is filled with the gas at 10^{-3} atm pressure and the gas is allowed to diffuse into an empty and closed container of volume 4 litre. Then work done in the total process is zero due to no opposing force.
- C) Work is a path function
- D) Work & heat are modes of transfer of energy not forms of energy.

56. PV work done in a cyclic process (ellipse), as shown in diagram next (for 1 mole ideal gas) is approximately equal to



A) 955 J

B) 801 J

C) 726 J

D) 540 J

SECTION – IV
(INTEGER ANSWER TYPE)

This section contains 4 questions. The answer is a Three digit integer ranging from 000 to 999 (both inclusive-Ex: If answer is 1 you should mark as 001).

Marking scheme +4 for correct answer , 0 if not attempted and 0 in all other cases.

57. From the following choose the appropriate number as mentioned and report your answer as a three digit number

	0	1	2
More basic:			
HIGHEST RESONANCE ENERGY PER RING	BENZENE	PYRIDINE	NAPHTHALENE
Compound used in Holme's signal	Calcium cyanamide	Calcium nitride	Calcium phosphide

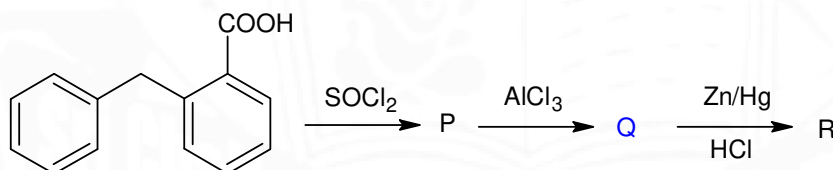
58. From the following choose the appropriate number as mentioned and report your answer as a three digit number

	1	2	3	4
MOST BASIC	P ₂ O ₅	SnO	B ₂ O ₃	MgO
Highest MP	AlCl ₃	AlF ₃	AlBr ₃	AlI ₃
Least solubility in water	NaHCO ₃	KHCO ₃	RbHCO ₃	LiHCO ₃

59. From the following choose the appropriate number as mentioned and report your answer as a three digit number

	1	2	3	4
Aromatic amino acid	Tyrosine	Threonine	Methionine	Asparagine
Most stable	SiH ₄	GeH ₄	SnH ₄	PbH ₄
Smallest K _{SP}	NiS	CuS	ZnS	HgS

60.



Molecular mass of R is?

TG ~ @bohring_bot

JEE ADVANCED UNIT TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

Sec: Sr.N-Super Chaina
Time: 3hr

Date: 30-04-20
Max.Marks:186

Name of the Student: _____

H.T. NO:

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30-04-20_SR.N-SUPER CHAINA _Jee-Adv_GUTA-1(P-2)_SYLLABUS

PHYSICS: Gravitation, Electrostatics, Current Electricity, Capacitors, Magnetism, Electromagnetic Induction, Alternate Current, Heat & Thermodynamics, Modern Physics

CHEMISTRY: Stoichiometry, Thermo, Gaseous state, Solution, Solid state., Atomic Structure
Chemical bonding, d-Block, Metallurgy, s-Block, Group-14, Group-15 & Group-18
Aldehydes & Ketones, Carboxylic acids & derivatives, Phenol, Ethers, amines, Alcohol, Biomolecules.

MATHEMATICS : Total calculus, Total Co-ordinate Geometry, Vector-3D

PHYSICS

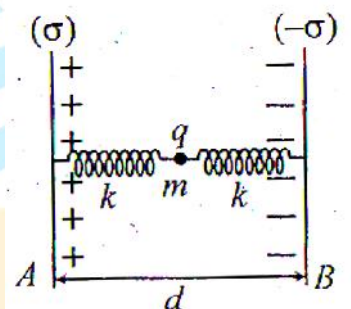
Max.Marks:62

SECTION – I
(MULTIPLE CORRECT CHOICE TYPE)

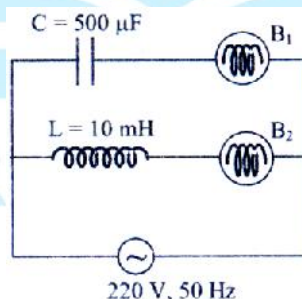
This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

1. Two large non conducting plates having surface charge densities $+\sigma$ and $-\sigma$, respectively, are fixed 'd' distance apart. A small test charge q of mass m is attached to two non – conducting identical springs of spring constant k as shown in the adjacent fig. The charge q is now released from rest with springs in natural length. Then q will [neglect gravity]



- A) Perform SHM with angular frequency $\sqrt{\frac{2k}{m}}$
 B) Perform SHM with amplitude $\frac{\sigma q}{2k\epsilon_0}$
 C) Not perform SHM, but will have a periodic motion if charges are removed on plates as well as on m
 D) Remain stationary
2. In the circuit shown in fig, if both the bulbs B_1 and B_2 are identical

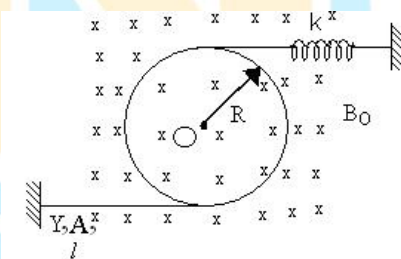


- A) their brightness will be the same
 B) B_2 will be brighter than B_1
 C) as frequency of supply voltage is increased, brightness of B_1 will increase and that of B_2 will decrease
 D) Only B_2 will glow because the capacitor has infinite impedance

3. A cylindrical shell of length l and inner, outer radii are a and b respectively is made of material of resistivity r . Then

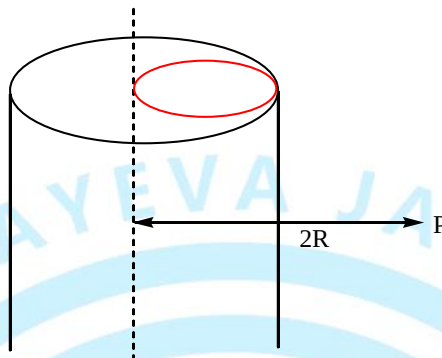
- A) Its resistance between the ends is $\rho \frac{l}{\pi(b^2 - a^2)}$
- B) Its resistance between inner and outer surface is $\rho \frac{(b-a)}{\ell(a+b)2\pi}$
- C) Its resistance between inner and outer surface is $\frac{\rho}{2\pi l} \ln\left(\frac{b}{a}\right)$
- D) Its resistance between the ends is $\rho \frac{l}{\pi(b^2 + a^2)}$

4. A uniform non-conducting circular disc of mass M and radius R charged uniformly with charge Q is free to rotate in horizontal plane about a vertical axis passing through its fixed centre O . A mass less spring of constant k and an elastic cord of length l , cross-sectional area A and young's modulus Y are attached to diametrically opposite points of the disc such that their lengths are parallel as shown in the figure. Initially the spring is unstretched, the cord is just taut, and a uniform magnetic field B_0 exists in vertical direction. Now B_0 is switched off. Assuming B_0 is small enough for the oscillations to be small and simple harmonic, choose the correct options

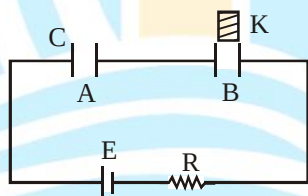


- A) The time period of oscillations is $T = 2\pi \sqrt{\frac{Ml}{2(Kl + YA)}}$
- B) The time period of oscillations is $T = \pi \sqrt{\frac{Ml}{2(Kl + YA)}} + \pi \sqrt{\frac{M}{2K}}$
- C) During the oscillations the maximum potential energy stored in the spring is $\frac{Q^2 B_0^2 R^2}{16M}$
- D) During the oscillations the maximum elastic potential energy stored in the cord is $\frac{Q^2 B_0^2 R^2}{16M}$

5. An infinitely long current carrying thick cylindrical conductor has a cylindrical cavity of radius $\frac{R}{2}$. The axis of the cylindrical cavity is at a distance $\frac{R}{2}$ from the axis of the cylindrical conductor. A current with current density J flows along the length of the conductor. The magnetic field at a point 'P' at a distance $2R$ from the axis of the cylinder is (point 'P' is on the plane containing both the axes)

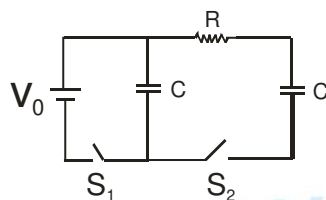


- A) Magnetic field at P is $\frac{\mu_0 JR}{12}$
 B) Magnetic field at P is $\frac{\mu_0 JR}{6}$
 C) Magnetic field on the axis of the cavity is non-zero
 D) Magnetic field on the axis of the conductor is non-zero
6. In the figure shown capacitors A and B of capacitance C each are in steady state. A dielectric slab of dielectric constant $K = 2$ and dimensions equal to the inner dimensions of the capacitor is inserted in the space between the plates of the capacitor B. Choose the correct option(s) : (in steady state)

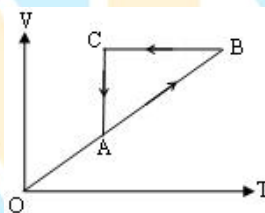


- A) Charge on each capacitor will increase by $\frac{CE}{6}$.
 B) Due to the insertion of dielectric, energy of the cell decreases by an amount of $\frac{CE^2}{6}$.
 C) Due to the insertion of dielectric, energy of the cell increases by an amount of $\frac{CE^2}{6}$.
 D) Due to the insertion of dielectric, the energy in the capacitor A increases by an amount of $\frac{7CE^2}{72}$

7. In a circuit shown in figure, the capacitance of each capacitor is equal to C and the resistance equal to R . One of the capacitance was connected to a voltage V_0 by closing switch S_1 . Now at $t = 0$, the switch S_1 is opened and S_2 is closed



- A) The current in the circuit as a function of time t is $I = \frac{V_0}{R} e^{-2t/RC}$
- B) Charge on the uncharged capacitor as a function of time is $q = \frac{CV_0}{2} (1 - e^{-2t/RC})$
- C) Heat produced in the circuit is $\frac{CV_0^2}{2}$ D) Heat produced in the circuit is $\frac{CV_0^2}{4}$
8. An ideal gas undergoes a thermodynamic cycle as shown in figure:



Which of the following graphs represents the same cycle?

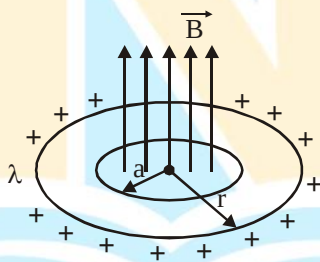
A)		B)	
C)		D)	

SECTION – II
(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a **Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place)**.
 Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme : +3 for correct answer, 0 in all other cases.

9. Heat is generated uniformly per unit volume inside a spherical volume of radius 1m at rate of 20 W/m^3 . The thermal conductivity of the spherical volume is $\frac{1}{6} \text{ W/mK}$. If the outer surface temperature of the sphere is 20°C . the temperature of the centre of the sphere is (in $^\circ\text{C}$)
10. A satellite is launched in the equatorial plane in such a way that it can transmit signals up to 60° latitude on the earth. The angular velocity of the satellite is $\sqrt{\frac{GM}{nR^3}}$. Find n?
11. A ring of mass 4 kg is uniformly charged with $\lambda = 4 \text{ C/m}$ and kept on rough horizontal surface with friction coefficient $\mu = \frac{\pi}{4}$. A time varying magnetic field $B = B_0 t^2$ is applied in a circular region of radius a ($a < r$) perpendicular to the plane of ring as shown in figure. Find out the time in sec when ring just starts to rotate on surface. (Take $a = 5 \text{ cm}$ $g = 10 \text{ m/s}^2$ $B_0 = 125 \text{ SI Unit}$).



12. A neutron having kinetic energy 12.5 eV collides headon with a hydrogen (in ground state) atom at rest. Neglect the difference in mass between the neutron and the hydrogen atom. Find the kinetic energy of the neutron after the event.
13. A stationary He^+ ion emitted a photon corresponding to the first line of the Lyman series. The photon liberates electron from a stationary hydrogen atom in the ground state. The velocity of the liberated electron is $N \times 10^6 \text{ m/s}$. Find N (You can make necessary approximations).
14. A current of 4A flows in a coil when connected to a 12V. dc source. If the same coil is connected to a 12V, $\frac{50}{\pi}$ Hz ac source a current 2.4A flows in the circuit. The inductance of the coil isH

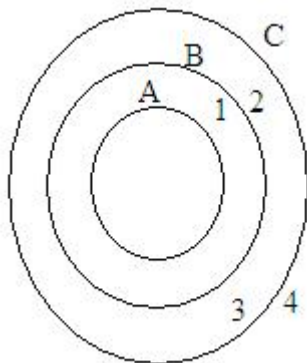
SECTION-III (MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Matching 15 & 16.

Fig. shows three concentric thin spherical shells A, B and C of radii R , $2R$ and $3R$. The shells A and C are given charges q and $2q$ respectively.



- 1 → Inner surface of B
 2 → Outer surface of B
 3 → Inner surface of C
 4 → Outer surface of C

List – I		List – II	
I)	q_1	P)	q
II)	q_2	Q)	$-q$
III)	q_3	R)	$3q$
IV)	q_4	S)	$\frac{2q}{3}$
		T)	$\frac{4q}{3}$
		U)	$\frac{-4q}{3}$

15. The charges appearing on surfaces 1, 2, 3, and 4 are q_1, q_2, q_3 and q_4 respectively
- A) I-Q, II-P, III-Q, IV-R B) I-Q, II-P, III-P, IV-P
 C) I-P, II-Q, III-P, IV-R D) I-P, II-Q, III-Q, IV-Q
16. Now shell B is earthed and the charges appearing on surfaces 1, 2, 3, and 4 are q_1, q_2, q_3 and q_4 respectively
- A) I-Q, II-T, III-T, IV-S B) I-Q, II-U, III-T, IV-S
 C) I-P, II-T, III-U, IV-T D) I-P, II-U, III-T, IV-S

Matching 17 & 18.

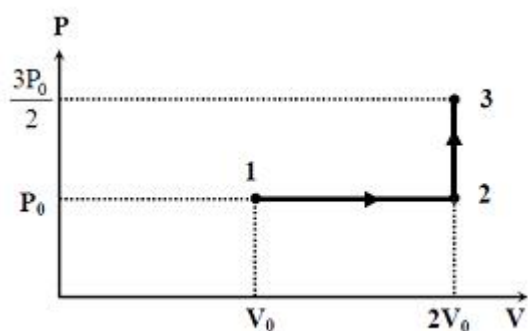
Answer the following by appropriately matching the list based on the information given in the paragraph.

In a thermodynamic process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by $T (\Delta X)$, where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas $X = \frac{3}{2} R \ln\left(\frac{T}{T_A}\right) + R \ln\left(\frac{V}{V_A}\right)$. Here, R is gas constant, V is volume of gas T_A and V_A are constants.

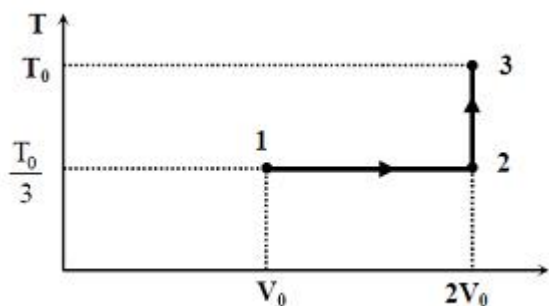
The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

List – I		List – II	
I)	Work done by the system in process $1 \rightarrow 2 \rightarrow 3$	P)	$\frac{1}{3} RT_0 \ln 2$
II)	Change in internal energy in process $1 \rightarrow 2 \rightarrow 3$	Q)	$\frac{1}{3} RT_0$
III)	Heat absorbed by the system in process $1 \rightarrow 2 \rightarrow 3$	R)	RT_0
IV)	Heat absorbed by the system in process $1 \rightarrow 2$	S)	$\frac{4}{3} RT_0$
		T)	$\frac{1}{3} RT_0 (3 + \ln 2)$
		U)	$\frac{5}{6} RT_0$

17. If the process carried out on one mole of monatomic ideal gas is as shown in figure in the PV-diagram with $P_0V_0 = \frac{1}{3}RT_0$, the correct match is



- A) I-Q, II-R, III-S, IV-U B) I-Q, II-S, III-R, IV-U
C) I-Q, II-R, III-P, IV-U D) I-S, II-R, III-Q, IV-T
18. If the process is carried out on one mole of monatomic ideal gas is as shown in the TV-diagram with $P_0V_0 = \frac{1}{3}RT_0$, the correct match is



- A) I-P, II-T, III-Q, IV-T B) I-S, II-T, III-Q, IV-U
C) I-P, II-R, III-T, IV-S D) I-P, II-R, III-T, IV-P

SECTION – I
(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

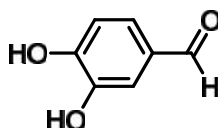
19. Correct statement(s) among the following is/are

- A) The pKa of $-\text{COOH}$ group in an α -amino acid is lower than the pKa of acetic acid.
- B) Proline is an example of aromatic heterocyclic ring containing amino acids
- C) The pentaacetate of glucose does not react with hydroxylamine
- D) $\alpha - D - (+) - \text{Glucopyranose}$ and $\beta - D - (+) - \text{Glucopyranose}$ are differ in their configuration at C2 position

20. Which of the following is/are correct ?

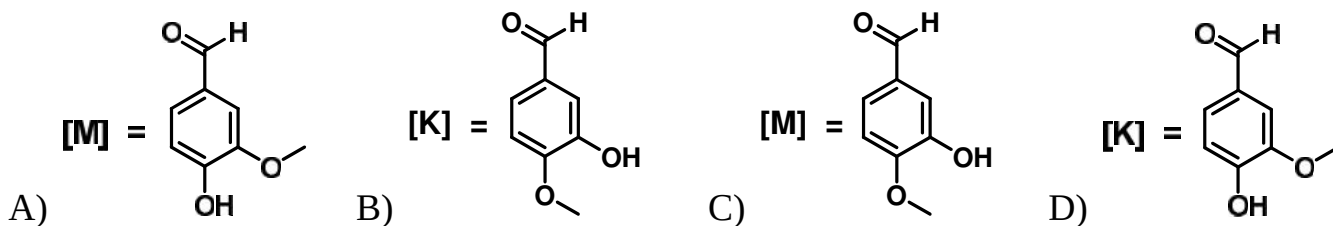
- A) 1g of dry green algae absorbs 4.7×10^{-3} mol. CO_2 per hour by photosynthesis. Algae will take 7.88 hours to double its own weight if all carbon atoms of absorbed CO_2 is stored in the form of starch after photosynthesis.
- B) If 3g of carbon reacts completely with 5g oxygen, volume percentage of 'CO' in the product will be 75% (No reactant is remained after reaction)
- C) If 3g of mixture of Na_2CO_3 and NaHCO_3 loses 0.248 g on heating to 300°C then percentage of Na_2CO_3 in the original sample is 77.6% (approximately)(Considering only decomposition of NaHCO_3)
- D) If 9g of H_2O are added to 100g of oleum sample labeled as 112% volume of free SO_3 remaining in the solution will be 0.373 lit at STP.

21. Protocatechualdehyde with dimethyl sulphate in the presence of NaOH gives K along with L and M. In this reaction product ratios can be controlled by the reaction conditions. With excess of NaOH and excess of dimethyl sulphate, product L was formed exclusively. When one equivalent of NaOH and one equivalent of dimethyl sulfate were used, K was formed in 87% yield. The same reaction when carried out with excess of NaOH and only one equivalent of dimethyl sulphate M was formed in 95% yield.

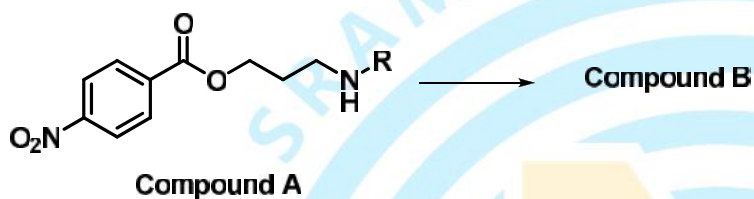


Protocatechualdehyde

Identify the correct options



22. When the following compound is allowed to stand in pentane, converted into a constitutional isomer. Which of the following is/are correct about the constitutional isomers (A) and (B)?

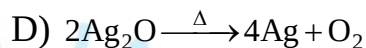
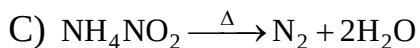
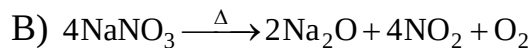
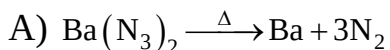


- A) (A) & (B) are functional isomers
 B) Both give same products on hydrolysis
 C) **Hydrolysed** products can be separated by **aqueous. HCl / Ether in separating funnel**
 D) **Hydrolysed** product can be separated by **aqueous. NaOH / Ether in separating funnel**
23. Which of the following is/are correct ?
- A) Ratio of moles of SO_2 to CH_4 in 6th diffusion chamber will be 1 : 4 if their initial moles ratio is 8 : 1
 B) $\text{N}_2\text{O}_5 \rightarrow \text{N}_2\text{O}_4 + \frac{1}{2}\text{O}_2$; $k = 0.2303\text{min}^{-1}$ average gram molar mass of the system after 10 minutes of decomposition would be 74.48g (approximately) if 1 mole of N_2O_5 is present initially
 C) In BCC, each face has 1 distorted octahedral void and 4 distorted tetrahedral voids.
 D) Equation of state for **a** real gas given by $P = \frac{RT}{V_m - b} - \frac{a}{V_m}$ (a, b are non zero positive constants, V_m is molar volume) can be liquefied.

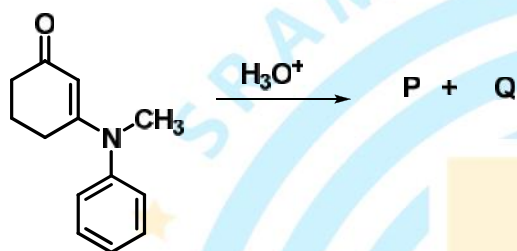
24. The cyanide process of gold extraction involves leaching out gold from its ore with CN^- in presence of 'Q' in water to form 'R' subsequently, 'R' is treated with 'T' to obtain 'Au' & 'Z'. Choose the correct options.

A) Z is $[\text{Zn}(\text{CN})_4]^{-2}$ B) 'T' is 'Zn' C) R is $[\text{Au}(\text{CN})_2]^{-1}$ D) 'Q' is O_2

25. Which of the following reactions are correct (' Δ ' indicates heating)



26.



Choose the correct statements regarding the product mixture

A) One of the product gives positive DNP test

B) Both products gives the positive tollens test

C) One of the product gives idoform test

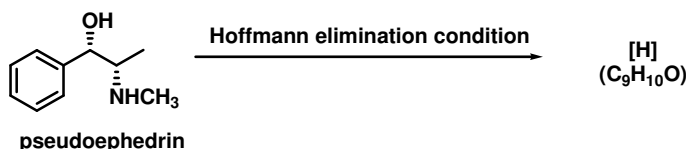
D) Number of lone pair of e^- carbon atoms in both P and Q together is 5

SECTION – II (Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place). Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme : +3 for correct answer, 0 in all other cases.

27. Pseudoephedrin, a natural alkaloid used as nasal decongestant. When we treat the pseudoephedrin with Hoffman elimination condition gives [H] ($\text{C}_9\text{H}_{10}\text{O}$). However H does not decolorize Br_2 water and does not form an oxime derivative. Number of stereoisomers (including H) possible for [H] is?



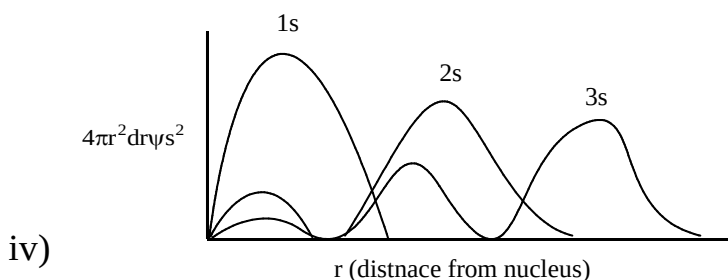
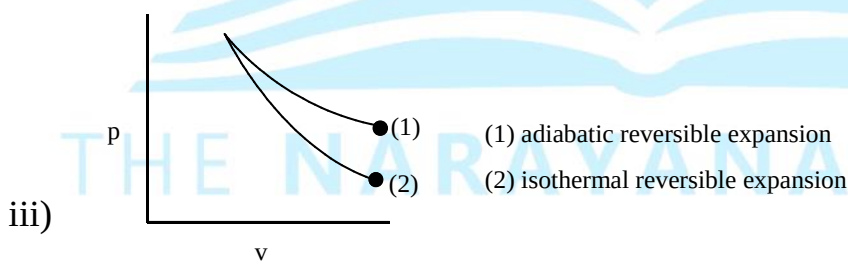
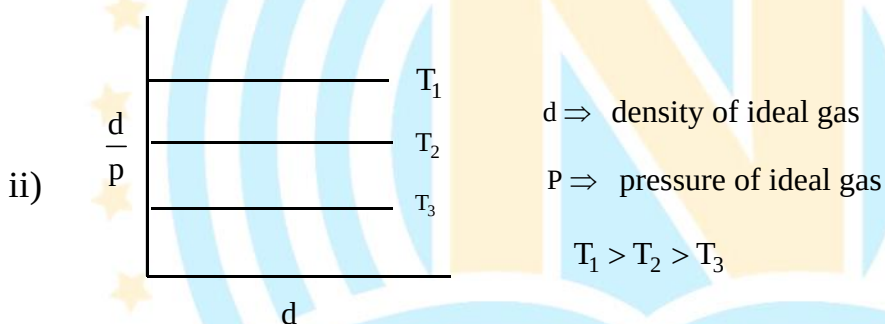
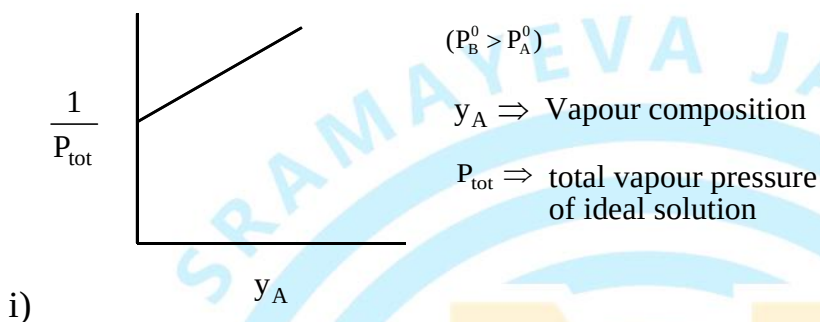
28. Find the atomic mass of 'Ca' in new scale from the following data.

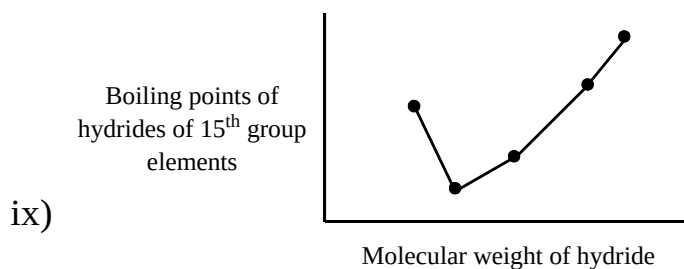
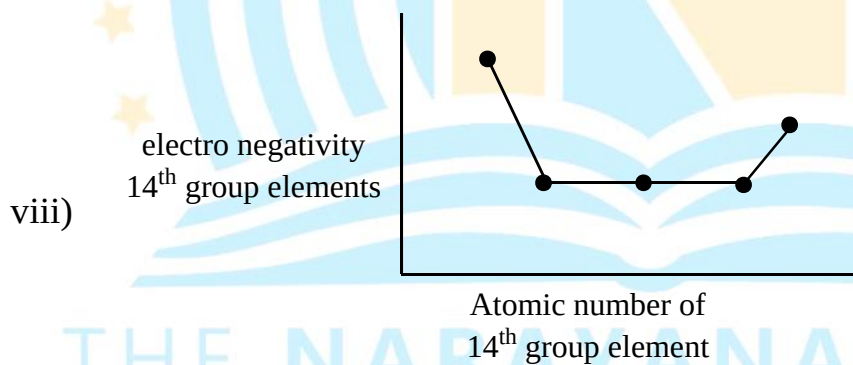
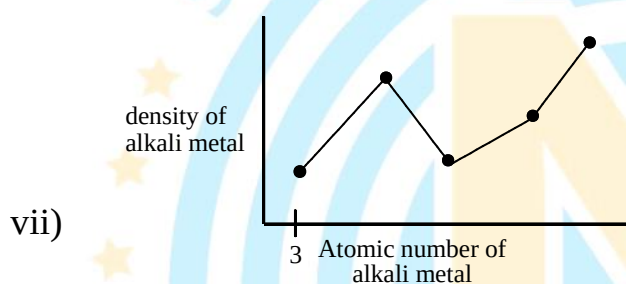
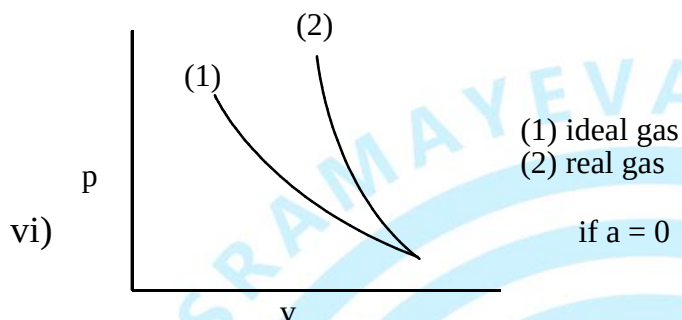
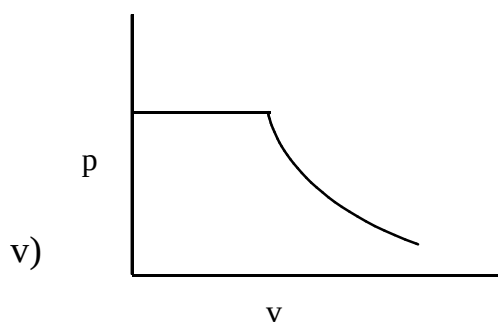
Atomic mass of 'Al' in carbon scale is 27

Atomic mass of 'Ca' in carbon scale is 40

Atomic mass of 'Al' in new scale is 36

29. How many of the following graphs are correct





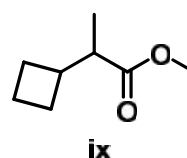
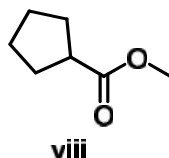
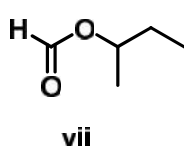
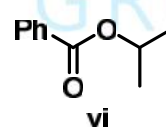
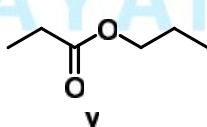
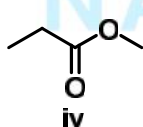
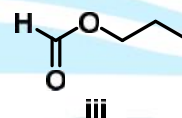
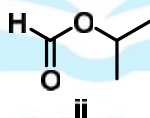
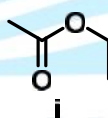
30. How many of the following are correct.

- Bond lengths order $\text{CO} > \text{CO}^+$
- If there is no 'SP' mixing, B_2 will be diamagnetic
- There are 12 five membered rings and 25 six membered rings in C_{70}
- In double chain silicates some tetrahedra share two corners, some tetrahedra share three corners in such a way that on average 2.5 corners of one tetrahedral are shared.
- In the reaction of NH_3 with HNO_2 , n – factor values of NH_3 and HNO_2 are not same.
- dil. HNO_3 on reaction with 'Cu' produces a gas which can form brown ring complex with suitable reagents.
- order of 'Xe – F' bond lengths $\text{XeF}_2 > \text{XeF}_4 > \text{XeF}_6$

31. An organic compound contains C, H & N. When 9 volumes of a gaseous mixture containing organic compound and just sufficient quantity of O_2 was exploded, 2 volumes of N_2 , 6 volumes of water vapour and 4 volumes of CO_2 were produced. If all measurements are done under similar conditions of temperature and pressure, the molecular formula of organic compound will be $\text{C}_x\text{H}_y\text{N}_z$.

Value of $\frac{(x+y+z)2}{7}$

32. An organic compound reacts with excess MeMgBr followed by H_2O to give two organic products of which one reacts with NaOI to give yellow precipitate and **OTHER DOES NOT**. How many of the following compounds can satisfy this condition?



SECTION-III (MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

33 & 34: MATCHING

Consider the Bohr's model of a one-electron atom where the electron moves around the nucleus. In the following List-I contains some quantities for the n^{th} orbit of the atom and List-II contains options showing how they depend on n .

COLUMN – I		COLUMN – II	
I)	Radius of the n^{th} orbit	P)	$\propto n^{-2}$
II)	Angular momentum of the electron in the n^{th} orbit	Q)	$\propto n^{-1}$
III)	Kinetic energy of the electron in the n^{th} orbit	R)	$\propto n^0$
IV)	Potential energy of the electron in the n^{th} orbit	S)	$\propto n^1$
		T)	$\propto n^2$
		U)	$\propto n^{1/2}$

33. Which of the following options has the correct combination considering List – I and List – II ?

- A) (III), (P) B) (IV), (U) C) (III), (S) D) (IV), (Q)

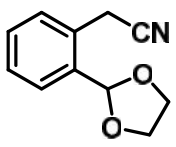
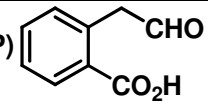
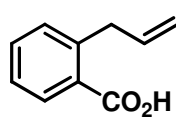
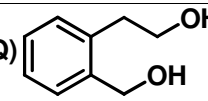
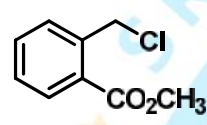
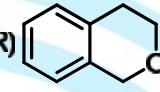
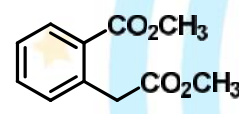
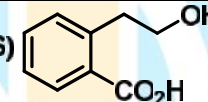
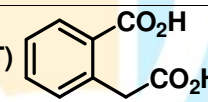
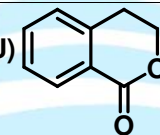
34. Which of the following options has the correct combination considering List – I and List – II ?

- A) (I), (T) B) (II), (R) C) (I), (P) D) (II), (Q)

35 & 36: MATCHING:

Answer the following by appropriately matching the lists based on the information given in the paragraph

List-I Includes starting materials and reagents of selected chemical reactions. List-II structure of the compounds that may be formed as intermediate products and/or final products from the reaction of the list-I

COLUMN – I	COLUMN – II
(I)  $\xrightarrow[\text{(iv) Con H}_2\text{SO}_4]{\begin{matrix} \text{(i) DIBAL-H} \\ \text{(ii) dil HCl} \\ \text{(iii) NaBH}_4 \end{matrix}}$	(P) 
(II)  $\xrightarrow[\text{(iv) Con H}_2\text{SO}_4]{\begin{matrix} \text{(i) O}_3 \\ \text{(ii) Zn, H}_2\text{O} \\ \text{(iii) NaBH}_4 \end{matrix}}$	(Q) 
(III)  $\xrightarrow[\text{(iv) Con H}_2\text{SO}_4]{\begin{matrix} \text{(i) KCN} \\ \text{(ii) H}_3\text{O}^+, \Delta \\ \text{(iii) LiAlH}_4 \end{matrix}}$	(R) 
(IV)  $\xrightarrow[\text{(ii) Conc H}_2\text{SO}_4]{\text{(i) LiAlH}_4}$	(S) 
	(T) 
	(U) 

35. Which of the following options has correct combination considering List-I and List-I
- A) (II), (P), (S), (U) B) (I), (S), (Q), (R) C) (II), (P), (S), (T) D) (I), (Q), (T), (U)
36. Which of the following options has a correct combination considering List-I and List-II
- A) (III), (S), (R) B) (IV), (Q), (U) C) (IV), (Q), (R) D) (III), (T), (U)

MATHEMATICS

Max.Marks:62

SECTION – I
(MULTIPLE CORRECT CHOICE TYPE)

This section contains **8 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE is/ are correct**.

Marking scheme +4 for correct answer , 0 if not attempted and -1 in all other cases.

37. A Normal at a point P on a parabola meets the curve again at Q. Tangent to the parabola at P and Q intersect in T and directrix of the parabola meets TP and TQ in M and N respectively, S being the focus of the parabola then
- A) M bisects TP B) N bisect TQ C) $\frac{PSM}{2} = \frac{\pi}{2}$ D) $\frac{QSN}{2} = \frac{\pi}{2}$
38. Let h be a twice continuously differentiable positive function on an open interval J and $g(x) = e^{-x} f(x)$ such that $f''(x) - 2f'(x) + f(x) \geq e^x$ for each $x \in J$. Then which of the following is/are correct?
- A) $g'(x) \geq 0 \forall x \in J$ B) $g'(x) \leq 0 \forall x \in J$
 C) $g''(x) \geq 0 \forall x \in J$ D) $g''(x) \leq 0 \forall x \in J$
39. Solution curve of the differential equation $\left[(2x-1)^2 + 2y^3(2x-1)\right] dy = 2y^4 dx$, passes through the point (1, 1) then solution curve
- A) Intersects $y = x$ exactly at three points
 B) Intersects $y = x$ exactly at one point
 C) Intersects $y = \frac{x^3+1}{2}$ exactly at three points
 D) Intersects $y = \frac{x^3+1}{2}$ exactly at one point
40. Let $f(x) = \begin{cases} -1; & -2 \leq x \leq 0 \\ |x-1|; & 0 < x \leq 2 \end{cases}$, $g(x) = \int_{-2}^x f(t) dt$.
- A) $g(x)$ is continuous at $x=0$ B) $g(x)$ is discontinuous at $x=1$
 C) $g(x)$ is derivable at $x=0$ D) $g(x)$ is derivable at $x=1$
41. Let $f: A \rightarrow B$ be onto function defined as $f(x) = \frac{\sin^{-1} x + \tan^{-1} x}{\cos^{-1} x + \cot^{-1} x}$
- And $g: B \rightarrow A$ be a function such that $g(f(x)) = x \forall x \in A$ & $f(g(x)) = x \forall x \in B$.
- Then which of the following is/are true?

A) $g(3)=1$

B) $g(3)=-1$

C) Minimum & maximum of $f(x)$ be m and M respectively then $\frac{M}{m} = -7$

D) Number of solutions of the equation $f(x^3 + 14x^2 + 13x - 5) = f(x^3 - x^2 + 1)$ is 1

42. Let $AB: y = x$ be a chord to the parabola $y^2 + 4x = 4$ and P be any variable point on the ellipse $x^2 + 9(y-4)^2 - 9 = 0$. If Δ_1 and Δ_2 be the minimum and maximum area of triangle ABP respectively then

A) $\Delta_2 = 2\sqrt{2}(4 + \sqrt{10})$

B) $\Delta_2 - \Delta_1 = 8\sqrt{5}$

C) $\frac{\Delta_2}{\Delta_1} = \frac{13 + 4\sqrt{10}}{3}$

D) $\Delta_2 - \Delta_1 = 5\sqrt{5}$

43. Let $P(-14, 27)$ be a point, L be the line with slope $\frac{5}{12}$ that contains the point $A \equiv (24, -1)$ and let M be the line perpendicular to line L that contains the point $B \equiv (5, 6)$. The original co-ordinate axes are erased and the line L is made the x -axis and line M the y -axis. In the new co-ordinate system A is on the positive x -axis and B is on the positive y -axis, then in the new co-ordinate system

A) abscissa of P is negative

B) abscissa of P is $-\frac{123}{13}$

C) ordinate of P is negative

D) ordinate of P is $\frac{526}{13}$

44. Given three distinct planes $P_1: ax + by + cz = 1$, $P_2: bx + cy + az = 1$ and $P_3: cx + ay + bz = 1$ then which of the following is/are correct

A) if the planes intersect at single point $(3, 3, 3)$ then $a + b + c = \frac{1}{3}$

B) if $a + b + c = 0$, then the planes will not intersect at a single point.

C) if $a + b + c \neq 0$ then the lines $\frac{x-1}{a} = \frac{y}{b} = \frac{z+1}{c}$ and $\frac{x+1}{c} = \frac{y}{b} = \frac{z-1}{a}$ will never intersect

D) the lines $\frac{x-1}{a} = \frac{y-1}{b} = \frac{z-1}{c}$ and $\frac{x+1}{b} = \frac{y+1}{c} = \frac{z+1}{a}$ will never intersect

SECTION – II

(Numerical Value Answer Type)

This section contains 6 questions. The answer to each question is a Numerical values comprising of positive or negative decimal numbers (place value ranging from Thousands Place to Hundredths place).
Eg: 1234.56, 123.45, -123.45, -1234.56, -0.12, 0.12 etc.

Marking scheme : +3 for correct answer, 0 in all other cases.

45. Let $L = \lim_{x \rightarrow \infty} (-1)^x \sin\left(\frac{\pi}{\sqrt{2}} \sqrt{2x^2 + x + 2}\right) \cdot \sin\left(\frac{\pi}{4} + \frac{\pi}{4x}\right)$ where $x \in N$ then $\left(\frac{2-L}{L}\right)$ is

equal to

46. $I = \int_0^{\pi} \frac{x \sin x}{15 + \sin^2 x} dx = \frac{\pi}{16} \ln\left(\frac{a}{b}\right)$; $a, b \in N$, then minimum value of $a + b =$

47. S_1 and S_2 be two circles intersecting at $P(6, 4)$ and both are tangent to x-axis and line $y = mx$ ($m > 0$). If product of radii of circle $S_1 = 0$ and $S_2 = 0$ is $\frac{52}{3}$ find value of m

48. Let $f(x)$ and $g(x)$ be two differentiable functions such that $f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t) dt$ and $g(x) = x - \int_0^1 f(t) dt$ also suppose $\alpha = \min_x |f(x) - g(x)|$ (with minimum taken over all $x \in R$) and β is the minimum distance between the curves $f(x)$ and $g(x)$ then $[\alpha + \beta] = \dots$ (where $[\cdot]$ is GIF)

49. $f(x) = x^3 + ax^2 + bx + c$ is an increasing function and $f(f(2)) = 2$, $f(f(f(3))) = 3$ and $f(1) = -1$ then $f(4) =$

50. Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. consider a vector $\vec{c} = \alpha\vec{a} + \beta\vec{b}$, $\alpha, \beta \in \mathbb{R}$. if the projection of $\vec{c}$ on the vector $(\vec{a} + \vec{b})$ is $3\sqrt{2}$, then the minimum value of $(\vec{c} - (\vec{a} \times \vec{b})) \cdot \vec{c}$ equals ____

SECTION-III (MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-I & List-II). The options for the correct match are provided as (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

51&52 MATCHING:

In the following, There are 64 combinations like $A-P-I$, $A-P-II$, ..., $D-S-IV$ are possible, out of which only some combinations are correct.

Column 1	Column 2	Column 3
A) $\frac{x-1}{1} = \frac{y+1}{-1} = \frac{z}{1}$	P) $x + y - z = -2$	I) Line is in the plane
B) $\frac{x+1}{-1} = \frac{y-1}{-1} = \frac{z}{1}$	Q) $x - y = -2$	II) Line is intersecting the plane but not perpendicular
C) $\frac{x+1}{-1} = \frac{y+1}{1} = \frac{z}{1}$	R) $y - z = 1$	III) Line is outside and parallel to the plane
D) $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z}{1}$	S) $x + z = -1$	IV) Line is perpendicular to the plane

In column 1, equation of line in 3D is given, in column 2, equation of plane is given and in column 3, a possible relation between line in column 1 and plane in column 2 is given.

51. Which of the following is a **CORRECT** combination?

- A) $A-R-II$ B) $B-P-I$ C) $C-Q-III$ D) $D-S-IV$

52. Which of the following is an **INCORRECT** combination?

- A) $A-Q-II$ B) $B-R-III$ C) $C-R-III$ D) $D-S-II$

53&54 MATCHING:

Column -I gives the two curves equation whose number of common tangents and equation of a common tangent if any is given in 2nd & 3rd column respectively. Match the three columns (One or more than one answer)

COLUMN - I	COLUMN - II	COLUMN - III
I) $y^2 = 4x$; $5x^2 + 5y^2 - 16 = 0$	i) 2	P) $2y - x + 4 = 0$
II) $5x^2 + 5y^2 + 30x + 44 = 0$, $5x^2 + 5y^2 - 16 = 0$	ii) 1	Q) $x - 2y + 4 = 0$
III) $y^2 = 4x$; $5x^2 + 5y^2 + 30x + 44 = 0$	iii) 0	R) $x + 2y + 4 = 0$
IV) $5x^2 + 5y^2 - 16 = 0$ $x^2 + y^2 - 6x - 6y - 576 = 0$	iv) 4	S) None

53. Which of the following is correct combination ?

A) (I) $\rightarrow$ (iv) $\rightarrow$ (P) B) (I) $\rightarrow$ (i) $\rightarrow$ P C) (II) $\rightarrow$ iv $\rightarrow$ P D) (IV) $\rightarrow$ (iii) $\rightarrow$ S

54. Which of the following is correct combination ?

A) (II) $\rightarrow$ iv $\rightarrow$ S B) (II) $\rightarrow$ iv $\rightarrow$ Q C) (II) $\rightarrow$ ii $\rightarrow$ R D) (III) $\rightarrow$ i $\rightarrow$ R



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Max.Marks:183

Name of the Student: _____

H.T. NO:

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31-05-20_SR.N-SUPER CHAINA/N-CHAINA_Jee-Adv_GTA-16(P-1)_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEAMTICS : TOTAL SYLLABUS

PHYSICS

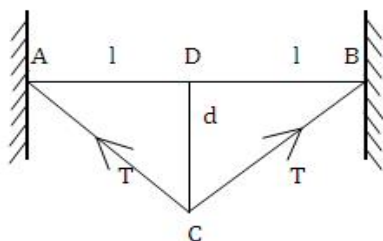
Max. Marks: 61

SECTION – I
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

1. A ring of radius R has charges q and $2q$ distributed uniformly on its two halves. There is a point P on its axis at a distance of $2\sqrt{2}R$ from its center. Which of the following options about the electric field and potential at P is/are correct?
 A) Potential is Kq/R
 B) Component of field along the axis at P is $\frac{2\sqrt{2}Kq}{9R^2}$
 C) Component of field perpendicular to axis at P is $\frac{2Kq}{9R^2}$
 D) Component of field perpendicular to axis at P is $\frac{2Kq}{27\pi R^2}$
2. A wire AB of length $2l$ and cross-section area a is stretched without tension between fixed points A and B . The wire is pulled at the centre into shape ACB as shown in the figure, such that $d \ll l$, The tension in the string is T and the strain produced in it is ϵ . If the Young's modulus of the material of wire is Y , then



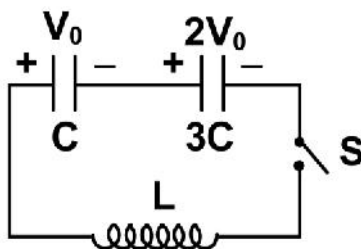
A) $\epsilon = \frac{d^2}{l^2}$

B) $\epsilon = \frac{d^2}{2l^2}$

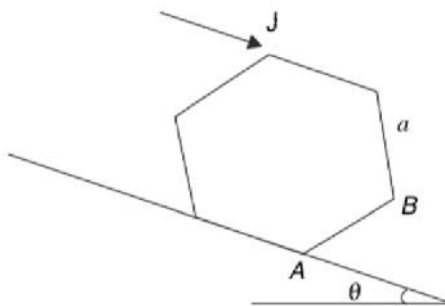
C) $T = aY \left(\frac{d^2}{l^2} \right)$

D) $T = aY \left(\frac{d^2}{2l^2} \right)$

3. Two capacitors of capacitance C and $3C$ are charged to potential difference V_0 and $2V_0$ respectively and connected to an inductor of inductance L as shown in the figure. Initially the current in the inductor is zero. Now the switch S is closed.

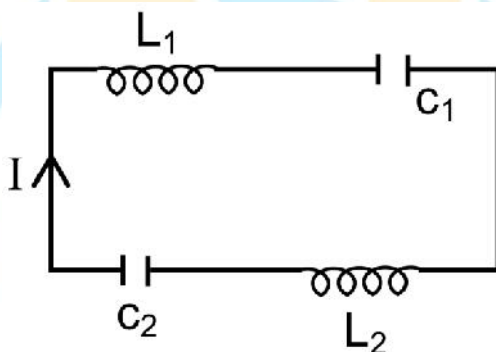


- A) The maximum current in the inductor is $\frac{3V_0}{2} \sqrt{\frac{3C}{L}}$
- B) The maximum current in the inductor is $\frac{V_0}{2} \sqrt{\frac{3C}{L}}$
- C) Potential difference across capacitor of capacitance C when the current in the circuit is maximum is $\frac{V_0}{2} \sqrt{\frac{3C}{L}}$
- D) Potential difference across capacitor of capacitance $3C$ when the current in the circuit is maximum is $\frac{4V_0}{3}$
4. A pencil has mass M and a hexagonal cross-section with each side of hexagon 'a'. It is kept on a rough inclined having inclination θ . Friction is large enough to prevent sliding. Moment of inertia of pencil about one of its edges perpendicular to the cross-section is I . If the pencil moves, it rolls about one of its edges without sliding. Refer cross-section view in diagram. Impulse J , parallel to the incline, is applied at the upper edge, only in option C and D below. (Given $\cos 18^\circ = 0.95$)



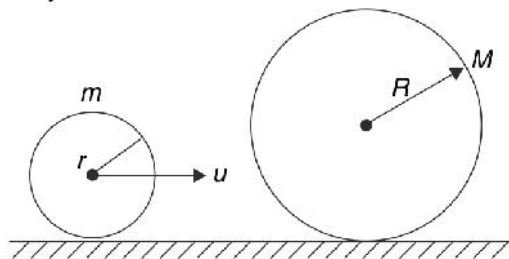
- A) The pencil cannot remain stationary for θ greater than 30° .
- B) The pencil will remain stationary for any θ less than or equal to 45° .
- C) If $\theta = 12^\circ$, minimum value of J required so that edge B hits the surface is $\sqrt{\frac{mgI}{30\alpha}}$
- D) If $\theta = 12^\circ$, minimum value of J required so that edge B hits the surface is $\sqrt{\frac{mgI}{20\alpha}}$

5. Figure shows LC circuit with two inductors and two capacitors. When the circuit was set-up (not at $t = 0$), the capacitors were not charged. The current in circuit is given by $I = 12\sin(2t + \pi/3)$, where ' t ' is time in seconds. Given $L_1 = 3H$, $L_2 = 2H$, $C_1 = 0.2F$. Select the correct statements:



- A) $C_2 = 1/15 F$
- B) $C_2 = 0.15 F$
- C) At $t = 7\pi/12$ s energy in inductor (L_1) is maximum
- D) At $t = \pi/3$ s energy in capacitor (C_2) is increasing
6. A smooth ball of mass M and radius R is lying on a smooth horizontal table. A smaller ball of radius r ($< R$) and mass m travelling horizontally on the table with velocity u hits the larger ball. Collision is elastic. During the interaction of the balls the larger ball does not lose contact with the table at any instant and the line of impact lies in the

vertical plane. Which of the following options is/are correct? (Given $\sin \theta = \frac{(R-r)}{(R+r)}$)



- A) The velocity of the smaller ball after collision is $\frac{(m-M)}{(m+M)}u$
- B) The velocity of the larger ball after collision is $\frac{2mu \cos^2 \theta}{(m+M)}$
- C) Maximum possible interaction force between the balls during collision is $Mg / \sin \theta$
- D) The impulse of interaction force between the spheres is $2Mmu/(M+m)$

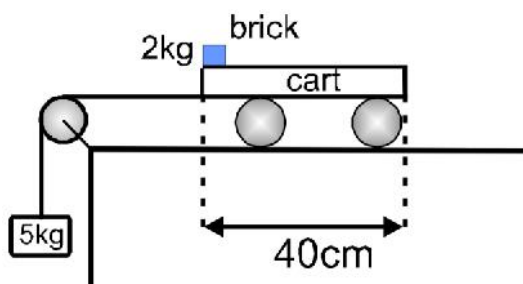
7. Which of the following is/are conservative force(s)?

- A) $\vec{F} = 2r^3 \hat{r}$ B) $\vec{F} = -\frac{5}{r} \hat{r}$ C) $\vec{F} = \frac{3(x\hat{i} + y\hat{j})}{(x^2 + y^2)^{3/2}}$ D) $\vec{F} = \frac{3(y\hat{i} + x\hat{j})}{(x^2 + y^2)^{3/2}}$

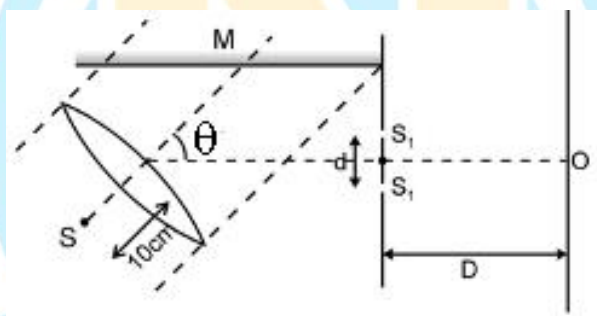
SECTION-II (INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).
Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

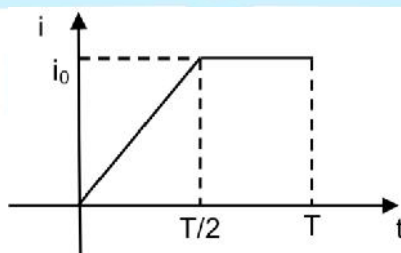
8. A cart of mass 3 kg is pulled by a 5 kg object as shown in figure. Length of cart is 40 cm and the table below it is frictionless. There is a small brick of mass 2 kg on the cart which falls at 0.8 sec after the start of motion. If the coefficient of friction between the cart and the brick is $x/10$, find the value of x . (Take $g = 10 \text{ m/s}^2$)



9. Three particles, A, B and C, of equal masses are initially at the vertices of an equilateral triangle of side $2\sqrt{3}$ m in a horizontal plane. They start moving with same speed 2 m/s such that A always moves towards B, B always moves towards C and C always moves towards A. After collision, A stops, and velocity of C is reversed without change in speed. Distance between B and C one second after the collision is x m. Find x.
10. In Young's Double slit experiment, distance between the slits is d and that between the slits and screen is D. Angle between principle axis of lens and perpendicular bisector of S_1 and S_2 is 37° . The point source S is placed at the focus of lens and aperture of lens is much larger than d. Assuming only the reflected light from plane mirror M is incident on slits, distance of central maxima from point O will be X then find $4X/D$. Point O is the point where perpendicular bisector of S_1S_2 meets the screen. Assume all rays to be paraxial for lens.



11. One cycle of an alternating current is shown in the graph. R.M.S value of current from $t = 0$ to $t = T$ is $i_0\sqrt{2/x}$. Find x.



12. A gas consisting of rigid di-atomic molecules at pressure $P_0 = 10^5$ N/m² and temperature 267 K was slowly compressed adiabatically to $1/5$ of its initial volume. The mean kinetic energy of rotation of molecules in final state is $n \times 10^{-21}$ J approximately. Find the value of integer n. ($k = 1.38 \times 10^{-23}$ J / K, $5^{0.4} = 1.90$)

SECTION – III

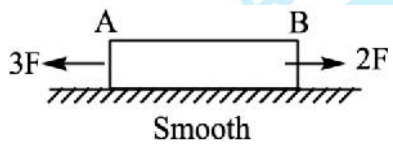
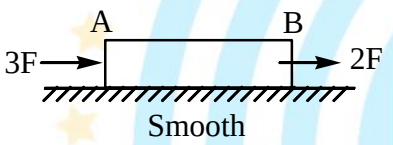
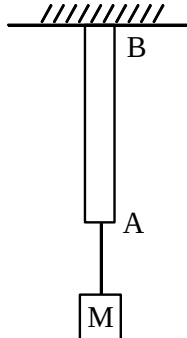
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q,13, Q,15 and Q,15 by appropriately matching the information given in the three columns of the following table.

Four identical rods (AB) each of mass m , length l , cross sectional area 'A' and Young's modulus Y are taken in column-A in different situations. Mark the most appropriate option.

	Column-A		Column-B		Column-C
A)	 Smooth	I)	There may be compressive longitudinal stress at some cross-section in the rod	P)	There is net elongation in the rod
B)	 Smooth	II)	There may be tensile longitudinal stress at some cross-section in the rod	Q)	There is net compression in the rod
C)	 The rod is rotating about end A with constant angular velocity ω on a smooth horizontal surface	III)	Magnitude of longitudinal strain decreases continuously from A to B	R)	Strain energy density continuously increases from A to B
D)		IV)	Magnitude of longitudinal strain increases continuously from A to B	S)	Strain energy density continuously decreases from A to B

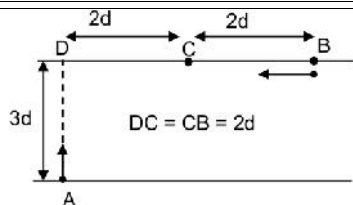
End B is fixed to the ceiling and a block of mass M is connected to end A by a massless spring				
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13. Which of the following does not match with 'II' in Column-B and 'P' in Column-C?
 A) D B) C C) B D) A
14. Which of the following matches with 'IV' in Column-B and 'R' in Column-C?
 A) D B) C C) B D) A
15. Which of the following is correct for all given options in Column-A?
 A) II B) P C) III D) S

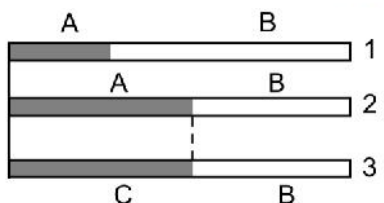
Answer Q,16, Q,17 and Q,18 by appropriately matching the information given in the three columns of the following table.

Column- A gives some situation in physics, Column- B and Column- C list some probable relations between the variables defined in Column- A. Based upon the correct matching between the columns, answer the three questions given below.

Column-A	Column-B	Column-C
A)When light is incident on air-glass interface at angles of incidence α, β and γ , the angles of refraction are β, γ and ϕ respectively. Given that light is going from air to glass. Angle of deviation in three cases are a, b and c.	I) $a > b$	P) $b > c$
B)Two swimmers (P and Q) start to swim at same time from A and B as shown in the figure. The two swimmers reach at C simultaneously. A and b are velocities of swimmers P and Q with respect to water and c is velocity of water in river.	II) $a < b$	Q) $b < c$



C) Three composite rods, initially of equal length. When we decrease the temperature of rods by ΔT , the relation between final length of rods become $L_1 < L_2 < L_3$.

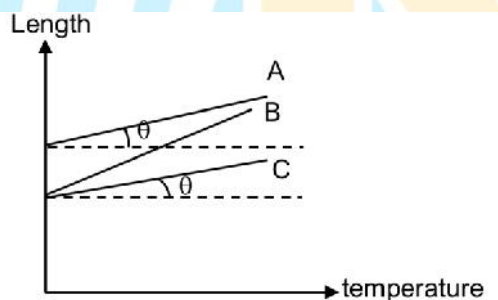


III) $a = b$

R) $b = c$

The coefficient of linear expansion of A, B and C are a , b and c .

D) Length versus temperature graph of three uniform rods A, B and C is given below. The coefficients of linear expansion are a , b and c for A, B and C respectively.



IV) $a < c$

S) $a > c$

16. Which of the following in Column-C matches with all options in Column-A?

A) S B) R C) Q D) P

17. For which of the following cases in Column-A, Option IV in Column-B is correct?

A) D B) C C) B D) A

18. Which of the following options in Column-A matches with exactly same entries in Column-B and Column-C?

A) A & B B) B & C C) C & D D) D & A

CHEMISTRY

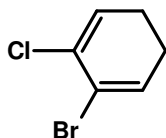
Max. Marks: 61

SECTION – I
(MULTIPLE CORRECT ANSWER TYPE)

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

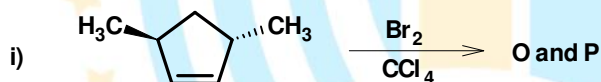
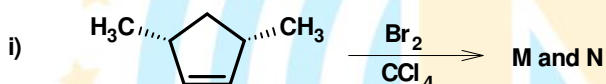
Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

19. The IUPAC name(s) of the following compound is (are):



- A) 1-chloro-2-bromocyclohexa-1,3-diene
- B) 1-bromo-2-chlorocyclohexa-1,3-diene
- C) 2-bromo-3-chlorocyclohexa-1,3-diene
- D) 2-chloro-1-bromocyclohexa-1,3-diene

20. The correct statement (s) for the following addition reaction is (are):



- A) O and P are identical molecules
- B) (M and O) and (N and P) are two pairs of diastereomers
- C) (M and O) and (N and P) are two pairs of enantiomers
- D) Bromination proceeds through trans-addition in both the reactions

21. Extraction of metal from **cassiterite** involves:

- A) carbon reduction of an oxide ore
- B) self reduction of a sulphide ore
- C) Removal of iron impurity
- D) Electro refining

22. With respect to the periodic properties the **incorrect** statement(s) is (are):
- A) Electron affinity: $F > O > Cl$ B) Ionisation energy: $Mg < Al < Si$
- C) Electronegativity: $Si < P < S$ D) Ionic size: $K^+ > Ca^{2+} > Mg^{2+}$
23. The correct statement(s) about V^{2+} and Co^{3+} is (are)
- A) V^{2+} is a reducing agent
- B) Co^{3+} is an oxidising agent
- C) Both V^{2+} and Co^{3+} exhibit d^6 configuration
- D) When Co^{3+} is used as oxidising agent, the cobalt ion attains d^5 electronic configuration
24. To a saturated solution of $AgCl$ containing sufficient amount of solid $AgCl$, NH_3 is gradually added such that its concentration becomes 0.2 M. Which of the option(s) is/are correct about the given solution?
- [Given: K_{sp} of $AgCl = 10^{-10}$, K_f of $Ag(NH_3)^+$ & $Ag(NH_3)_2^+$ are 10^3 & 10^8 respectively]
- A) $[Cl^-] = 2 \times 10^{-2} M$
- B) Heating of $PhCHO$ with final solution gives silver mirror.
- C) $[Ag(NH_3)^+] = 10^{-6} M$
- D) Addition of $PhCOCH_2OH$ to final solution gives silver mirror.
25. Which of the following is/are correct option(s)/statement(s) ?
- A) The reduction potential of H^+ / H_2 is 0.00V.
- B) The discharge of a species at any electrode during electrolysis depends on its reduction or oxidation potential.
- C) Conductivity (κ) increases not as fast as the concentration when the concentration of electrolyte is increased.
- D) $Ag | AgCl (K_{sp} = 10^{-10}) | KCl(10^{-2} M) || KBr(10^{-1} M) | AgBr(K_{sp} = 10^{-13}) | Ag$ is not a concentration cell.

SECTION-II
(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

26. Find the number of alkyl chlorides which on reaction with alc. KOH, heat gives following compound as major product.



27. The sum of number of $d\pi-p\pi$ and S-S bond(s) in tetrathionate ion is:
28. Based on molecular orbital theory, the number of paramagnetic species among the following is:
(Assume that the energy of molecular orbitals doesn't change on addition or removal of electron)
 $\text{O}_2, \text{O}_2^-, \text{O}_2^{2+}, \text{N}_2, \text{N}_2^{2-}, \text{B}_2, \text{P}_2, \text{S}_2, \text{C}_2^{-1}$
29. In an adsorption experiment, a graph between $\log \frac{x}{m}$ & $\log P$ is found to be linear with a slope of 45°C with $\log P$ axis in the low pressure range. The intercept on the $\log \frac{x}{m}$ axis was found to be 0.602. Calculate the amount of the gas adsorbed (in g) per gram of adsorbent under a pressure of 2 atm in this condition ($\log 2 = 0.301$).
30. An accident in a nuclear laboratory resulted in deposition of a certain amount of radioactive material of half-life, 20 days at 12 midnight of 31st Dec 2019. Tests revealed that the radiation was 128 times more than the permissible level required for safe operation of the laboratory, so it was closed. After how many half-lives the laboratory can be opened for work?

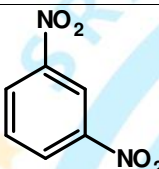
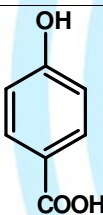
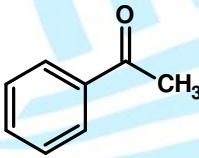
SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q,31, Q,32 and Q,33 by appropriately matching the information given in the three columns of the following table.

Column 1, 2 and 3 contain starting material, reaction conditions and the molecular formula of the products respectively.

	Column-I		Column-II		Column-III
A)		1)	i) $\text{HNO}_3, \text{H}_2\text{SO}_4, 60^\circ$ ii) Na_2S iii) $\text{NaNO}_2, \text{HCl}, 0^\circ \text{C}$ iv) CuBr/Δ	P)	C_6H_6
B)		2)	i) NaOI ii) acidic workup iii) $\text{NaOH}, \text{CaO}, \Delta$ iv) $\text{CH}_3\text{Cl}, \text{AlCl}_3 (\text{Anhyd})$ v) $2\text{Cl}_2, h\nu$ vi) aq. KOH	Q)	$\text{C}_6\text{H}_6\text{NF}$
C)		3)	i) $(\text{NH}_4)_2\text{S}$ ii) $\text{NaNO}_2, \text{HCl}, 0^\circ - 5^\circ \text{C}$ iii) HBF_4, Δ iv) Sn/HCl	R)	$\text{C}_7\text{H}_6\text{O}$
D)		4)	i) $\text{Br}_2, \text{H}_2\text{O}$ ii) $\text{Zn-dust}, \Delta$ iii) Mg (THF) iv) H_2O	S)	$\text{C}_6\text{H}_4\text{NO}_2\text{Br}$

31. Find the correct combination :
 A) B-2-R B) A-3-Q C) D-2-P D) C-4-P
32. Identify the incorrect combination :
 A) A-1-S B) B-3-Q C) D-2-R D) B-2-S
33. Column-I reactant [D], can be distinguished from its correct matched product [R] of column-III (formed by suitable set of reagents in column-II) by :
 A) Silver mirror test B) Neutral FeCl_3 test
 C) Victor mayer test D) Lucas test

Answer Q,34, Q,34 and Q,35 by appropriately matching the information given in the three columns of the following table.

Column-I	Column-II	Column-III
i) $K_3[Fe(CN)_6] + KOH + H_2O_2 \longrightarrow K_4[Fe(CN)_6] + H_2O + O_2$	P) Redox reaction	I) n- factor = 4
ii) $Cr(OH)_3 + NaOH + H_2O_2 \rightarrow Na_2CrO_4 + H_2O$	Q) Equivalent mass $\neq \frac{M}{2}$ of a reactant	II) Paramagnetic gaseous product
iii) $Sb_2O_3 + KIO_3 + 2HCl + 6H_2O \rightarrow 2HSb(OH)_6 + HOI + KCl$	R) Equivalent mass of a reactant $= \frac{M}{3}$	III) Product solution used for quantitative determination of I_2
iv) $S_8 + 12NaOH \rightarrow Na_2S + 6H_2O + Na_2S_2O_3$	S) Disproportionation reaction	IV) Final solution is yellow coloured

34. Which of the following is correct match for reaction (i) given in column-I ?

- A) i-P-III B) i-Q-III C) i-P-II D) i-S-IV

35. Which is not a correct match for reaction (ii) & (iii) given in column-I ?

- A) ii-Q-II B) ii-P-IV C) iii-P-I D) ii- R- IV

36. Which of the following is correct option ?

- A) i-P,Q-II,III B) ii-P,Q,R-II,IV C) iii-P,R-I,III D) iv-P,Q,S-III

MATHEMATICS

Max. Marks: 61

SECTION – I

(MULTIPLE CORRECT ANSWER TYPE)

This section contains 7 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases

37. Given $f(t) = \int_{-1}^t \left[\sin(x^3 - 9x^2 + 18x) + x^2 - 6x + \frac{11}{3} \right] dx$, Then which of the following are true?

- A) For $t \in [-3, 12]$, $f(t) = 0$ has at least two real roots
 B) For $t \in [\sqrt{11}, 11]$, $f(t) = 0$ has at least one real root
 C) The maximum value of 't' for which $f(t) = 0$ is less than 7
 D) The maximum value of 't' for which $f(t) = 0$ is greater than 7

38. For curve $y = x + e^x$,

- A) $\left(\frac{d^2y}{dx^2} \right) \cdot \left(\frac{d^2x}{dy^2} \right) < 0 \forall x \in \mathbb{R}$ B) $\left(\frac{d^2y}{dx^2} \right) \cdot \left(\frac{d^2x}{dy^2} \right) > 0 \forall x \in \mathbb{R}$
 C) $\frac{d^3x}{dy^3} > 0 \forall x \in \mathbb{R}$ D) $\frac{dy}{dx} \left(\frac{d^2x}{dy^2} \right) < 0 \forall x \in \mathbb{R}$

39. A is a 3×3 square matrix such that $A^4 = O$ and $AB = A + B$ then

- A) A is singular matrix B) B is singular matrix
 C) $A + B$ is singular matrix D) $A - B$ is singular matrix

40. $\alpha, \beta, \gamma, \delta$ are roots of the equation $i^4 z^4 + i^3 z^3 + i^2 z^2 + iz + 1 = z^2$ and z_1, z_2, z_3 lie in the

set $S = \left\{ z; \operatorname{Re} \left(\frac{z-1}{z+1} \right) = 0 \right\}$ then which of the following is correct ?

- A) $|\alpha + \beta|^2 + |\beta + \gamma|^2 + |\gamma + \alpha|^2 \geq 3$ B) $|z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2 \geq 3$
 C) $|\alpha + z_2|^2 + |\beta + \alpha|^2 + |\beta + z_2|^2 \geq 3$ D) $|\alpha + z_1|^2 + |\beta + z_2|^2 + |\gamma + z_3|^2 \geq 3$

41. $P(x, y)$ is a point on a Cartesian plane, z_1 and z_2 are complex numbers such that

$$\frac{|xz_1 - yz_2|^2 + |yz_1 + xz_2|^2}{|z_1|^2 + |z_2|^2} = 2x + 5 \text{ and for } a, b \in \mathbb{N}, \text{ which of the following is correct?}$$

A) for $z = a + ib$, $|1 + z + z^2 + \dots + z^n| \geq |z|^n - \frac{1}{|z|}$

B) for $z = a + ib$, $|1 + z + z^2 + \dots + z^n| \leq |z|^n$

C) locus of P is a circle

D) locus of P is a straight line

42. Normal at point $P(\theta)$ to the ellipse $\frac{x^2}{14} + \frac{y^2}{5} = 1$, intersects it again at the point $Q(2\theta)$.

Which of the following are factors of $-\sec 2\theta$?

A) 3

B) 9

C) 2

D) 5

43. Which of the following is/are incorrect ?

A) Number of ways 100 can be written as sum of natural numbers using each digit (0, 1, 2, ..., 9) exactly once, lies in the set $\{5K; k \in \mathbb{N}\}$

B) Number of ways 100 can be written as sum of natural numbers using each digit (0, 1, 2, ..., 9) exactly once, lies in the set $\{6K; k \in \mathbb{N}\}$

C) Let P be a prime number greater than 4 and $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{p-1} = \frac{m}{n}$ (in lowest form). Then, $m + n$ is a multiple of P

D) Let P be a prime number greater than 4 and $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{p-1} = \frac{m}{n}$ (lowest form). Then, m is a multiple of P

SECTION-II

(INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases.

44. $\left[\int_0^{\pi} \frac{x^2 \cos x}{(1 + \sin x)^2} dx \right] = \underline{\hspace{2cm}}$ ([.] denotes G.I.F)
45. Let A be the region, $x \geq 0, y \geq 0, x \leq 1, y \leq 1$. The ratio in which area of A is divided by the curve $\sqrt{x} + \sqrt{y} = 1$ in two parts is $\frac{m}{n}$ ($\frac{m}{n} > 1$) then m is $\underline{\hspace{2cm}}$ (m,n are co-prime)
46. $\int_0^{\frac{1}{\sqrt{2}}} \left(\int_0^x \frac{1+t^4}{(1-t^4)^{\frac{3}{2}}} dt \right) dx = \frac{\pi}{k}$, then $\frac{k}{2} = \underline{\hspace{2cm}}$
47. If $\sqrt{a-x}, \sqrt{x}, \sqrt{a+x}$ are three distinct consecutive terms of an A.P., which are integers and a is an even integer. The least value of $\left[\frac{a}{10} \right]$ is $\underline{\hspace{2cm}}$ ([.] denotes G.I.F)
48. Point P (1,2,3) moves parallel to the line $\frac{x-2}{1} = \frac{y-1}{8} = \frac{z-4}{9}$ and reaches P' such that perpendicular distance of P and P' from plane $x + 2y - z = 0$ is equal. If co-ordinates of P' are (α, β, γ) then $|\alpha + \beta + \gamma| = \underline{\hspace{2cm}}$

SECTION – III

(SINGLE CORRECT ANSWER TYPE)

This section contains 6 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Answer Q,49, Q,50 and Q,51 by appropriately matching the information given in the three columns of the following table.

a,b are non negative real numbers such that $a + b \leq 50$ and $3a + b \leq 90$.

$P(a,b), Q(a,b), R(a,b), S(a,b)$ are values, dependant on a and b equal to

$a^2 + b^2 + 180a + 120b, 4a + b, 15a + b, a^2 + b^2$ respectively. $P_{\max}, Q_{\max}, R_{\max}, S_{\max}$ are the maximum values of $P(a,b), Q(a,b), R(a,b), S(a,b)$ respectively. Column-II and Column-III give the corresponding values of a and b at which these maximum values occur. Match the following.

Column-I		Column-II		Column-III	
i)	$P_{\max} = 8500$	P)	$a = 0$	A)	$b = 0$
ii)	$Q_{\max} = 120$	Q)	$a = 30$	B)	$b = 20$
iii)	$R_{\max} = 450$	R)	$a = 20$	C)	$b = 30$
iv)	$S_{\max} = 2500$	S)	$a = 50$	D)	$b = 50$

49. Which of the following is correct ?

- A) i-Q-C B) ii-P-B C) iii-R-C D) iv-P-D

50. Which of the following is not correct ?

- A) i-P-D B) i-R-C C) iii-P-C D) ii-Q-A

51. Which of the following is correct ?

- A) iv-R-A B) iii-Q-A C) ii-Q-C D) i-Q-B.

Answer Q,52, Q,53 and Q,54 by appropriately matching the information given in the three columns of the following table.

A dice is thrown, if result is a 6, four balls are drawn from Bag A (Bag A has 6 white & 8 black balls) and if result is not 6, three balls are drawn.

Let E be the event that result of dice is a 6, B_l, ω_l be events that "last ball is black",

"last ball is white " respectively. B_{l-1}, ω_{l-1} be events that ball drawn just before last

ball was Black, white respectively. (Balls are drawn one by one without replacement)

COLUMN-I		COLUMN-II		COLUMN-III	
A)	$p(E)$	P)	$p\left(\frac{\omega_l}{E}\right)$	1)	$p\left(\frac{\omega_l}{E}\right)$
B)	$p(\omega_l)$	Q)	$P\left(\frac{B_{l-1}}{\omega_l}\right)$	2)	$p\left(\frac{B_{l-1}}{B_l}\right)$
C)	$p(B_l)$	R)	$p\left(\frac{E}{B_l}\right)$	3)	$p\left(\frac{E}{\omega_l}\right)$
D)	$p\left(\frac{\omega_l}{B_{l-1}}\right)$	S)	$p\left(\frac{\bar{E}}{\omega_l}\right)$	4)	$p\left(\frac{\bar{E}}{\omega_l}\right)$

52. Which of the following is correct. (corresponding values are equal) ?
 A) BP3 B) CR2 C) BP1 D) DP3
53. Which of the following is correct. (corresponding values are equal) ?
 A) AR3 B) ARP C) BR2 D) DQ2
54. Which of the following is a matching corresponding to three different values of probability ?
 A) DR3 B) DS4 C) BS4 D) CQ2

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Max.Marks:183

Name of the Student: _____

H.T. NO:

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31-05-20_SR.N-SUPER CHAINA/N-CHAINA_Jee-Adv_GTA-16(P-2)_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEAMTICS : TOTAL SYLLABUS

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

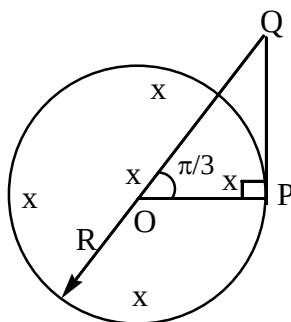
This section contains **7 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

1. Time varying magnetic field $B = \alpha + \beta t$ Tesla, lies in cylindrical region of radius R .

A triangular loop OPQ, lies as shown in figure. Resistance per unit length of loop is λ .

Heat produced in loop in time t is:



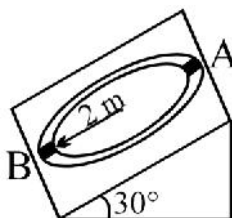
A) $\frac{\pi^2 R^3 \beta^2 t}{36(\sqrt{3}+1)\lambda}$

B) $\frac{\pi^2 R^4 \beta^2 t}{36\sqrt{3}(\sqrt{3}+1)\lambda}$

C) $\frac{\pi^2 R^4 \beta^2 t}{36(\sqrt{3}+1)\lambda}$

D) $\frac{\pi^2 R^3 \beta^2 t}{36\sqrt{3}(\sqrt{3}+1)\lambda}$

2. A small block of mass m is released from rest, from position A, inside the frictionless circular groove of radius 2 m on a fixed inclined plane as shown in figure. The contact force between the block and inclined plane at point B is:



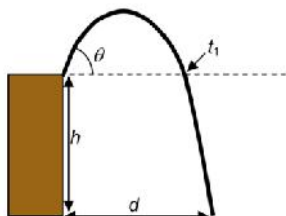
A) $\sqrt{3}mg/2$

B) $5mg/2$

C) $\sqrt{7}mg$

D) mg

3. A projectile is launched at an angle θ above the horizontal from a cliff of height h above the ground. After a time t_1 has elapsed since the launch, the projectile passes the level of the cliff top moving downward. It eventually lands on the ground a horizontal distance d from its launch site. Value of $\tan \theta$ in terms of h , t_1 and the acceleration due to gravity (g) is:



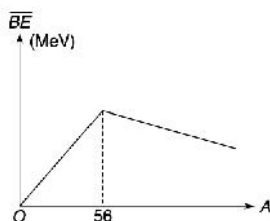
A) $\tan \theta = \frac{t_1^2 g \left[1 + \sqrt{1 + \frac{8h}{gt_1^2}} \right]}{4d}$

B) $\tan \theta = \frac{t_1^2 g \left[1 + \sqrt{1 + \frac{4h}{gt_1^2}} \right]}{4d}$

C) $\tan \theta = \frac{t_1^2 g \left[1 + \sqrt{1 + \frac{4h}{gt_1^2}} \right]}{2d}$

D) $\tan \theta = \frac{t_1^2 g \left[1 + \sqrt{1 + \frac{8h}{gt_1^2}} \right]}{2d}$

4. The graph of binding energy per nucleon (BE) versus mass number (A) for various nuclei has been approximated as made up of two linear components as shown in figure. The slope of line for $0 < A < 56$ is 0.16 and slope of line for $A > 56$ is -0.012. The binding energy of ${}^4_2\text{He}$ nucleus is 11 times the value predicted by the shown graph due to its very stable structure. Estimate the minimum mass number (A) of a nucleus for it to exhibit alpha decay.



A) 92

B) 111

C) 158

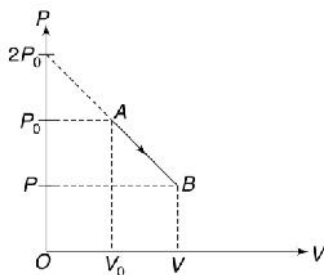
D) 216

5. A uniformly charged non-conducting rod is suspended vertically at its end. The rod can swing freely in the vertical plane without any friction. The linear charge density on the rod is $\lambda C/m$ and it has a uniformly distributed mass M . Length of the rod is L .

A uniform horizontal field E is switched on so that the rod just manages to make itself horizontal. What is the maximum angular speed of the rod during its motion?

A) $\sqrt{\frac{3(\sqrt{3}-1)g}{L}}$ B) $\sqrt{\frac{3(\sqrt{2}-1)g}{2L}}$ C) $\sqrt{\frac{3(\sqrt{2}-1)g}{L}}$ D) $\sqrt{\frac{2(\sqrt{3}-1)g}{L}}$

6. One mole of an ideal gas is expanded from the state A (P_0, V_0) to final state B having volume V . The process follows a path represented by a straight line on the P-V diagram as shown. Up to what volume (V) the gas shall be expanded so that final internal energy is half the maximum internal energy during the process.



A) $\frac{\sqrt{2}+1}{\sqrt{2}}V_0$ B) $\sqrt{2}V_0$ C) $(\sqrt{2}+1)V_0$ D) $\frac{\sqrt{2}+1}{2}V_0$

7. A spaceship is orbiting the earth in a circular orbit at a height equal to radius of earth ($R = 6400$ km) from the surface of the earth. An astronaut is on a spacewalk outside the spaceship. He is at a distance of $l = 200$ m from the ship and is connected to it with a simple cable which can sustain a maximum tension of 10 N. Assume that the spaceship, the astronaut and the center of earth are always in a line. Mass of astronaut along with all his accessories is 100 kg. Estimate the tension in the cable. (Given $g = 9.8 \text{ m/s}^2$)

A) 1 N B) 0.1 N C) 0.01 N D) The cable will break

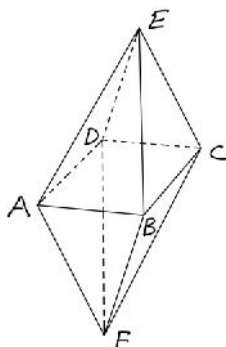
SECTION-II

(ONE OR MORE OPTIONS CORRECT TYPE)

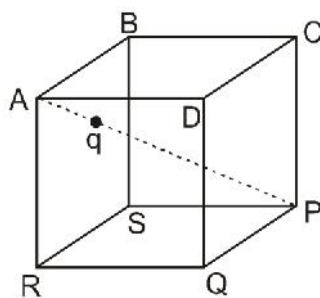
This section contains 7 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +4 for all correct options, 0 if not attempted and -2 in all wrong cases.

8. Consider 12 identical resistors of resistance R each are connected symmetrically in the shape of a square bi-pyramid ABCDEF. Which of the following options about equivalent resistance is/are correct?



- A) The effective resistance between E and F is $R/2$
 B) The effective resistance between A and C is $R/2$
 C) The effective resistance between A and B is $R/2$
 D) The effective resistance between A and B is $5R/12$
9. A charge q is placed on the diagonal AP of a cube at a distance $AP/3$ from the point A. Choose the correct options.

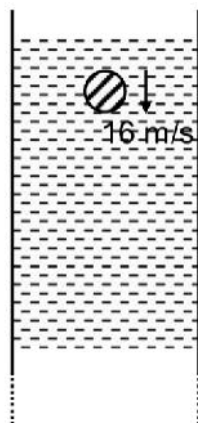


- A) the sum of electric flux passing through the surfaces ABCD and PQRS is $q / 3\epsilon_0$
 B) the sum of electric flux passing through the surfaces ABCD and PQRS is $q / 8\epsilon_0$
 C) the flux through both the surfaces ABCD and PQRS are same
 D) the flux through the surfaces ABCD is larger than the flux through surface PQRS

10. Which of the following statements is/are correct?

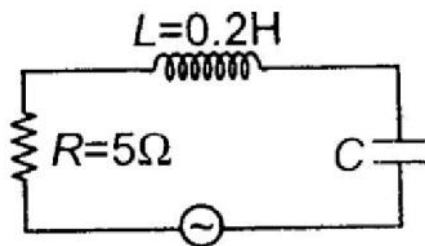
- A) If a moving charged particle enters into a region of uniform magnetic field from outside, it does not complete a circular path.
- B) If a moving charged particle traces a helical path in a uniform magnetic field, the axis of the helix is parallel to the magnetic field.
- C) The power associated with the force exerted by a magnetic field on a moving charged particle is always equal to zero.
- D) If in a region a uniform magnetic field and a uniform electric field both exist, a charged particle moving in this region cannot trace a circular path.

11. In a tall viscous liquid column in gravity free space, a solid ball is projected with 16 m/s as shown at $t = 0$. Velocity of the ball reduces to 1 m/s at $t = 4$ s. F_1 is average viscous force acting on the ball between $t = 0$ to $t = 2$ s and F_2 is average viscous force on the ball between $t = 2$ s and $t = 4$ s. Which of the following options is/are correct?

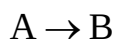


- A) Velocity of ball at $t = 1$ s is 8 m/s
- B) Velocity of ball at $t = 3$ s is 2 m/s
- C) $F_1 / F_2 = 4$
- D) $F_1 / F_2 = 3$

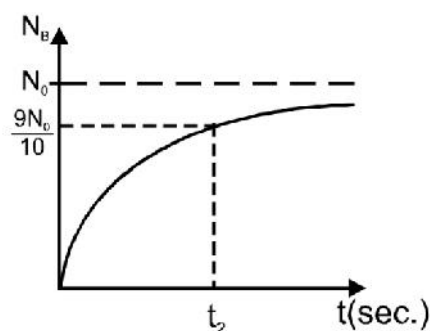
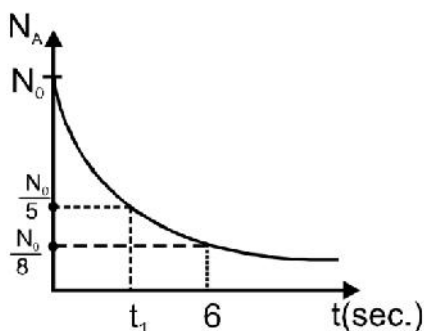
12. An alternating voltage $E = 6\sin(20t) + 8\cos(20t)$ is applied to a series resonant circuit as shown. The correct statement(s) for the given circuit is/are:



- A) The current in the circuit is 2 A
 B) Power dissipated in the circuit is 10 W
 C) The quality factor of the circuit is 0.8
 D) The capacitance C is 12.5 mF
13. In a decay process A decays to B



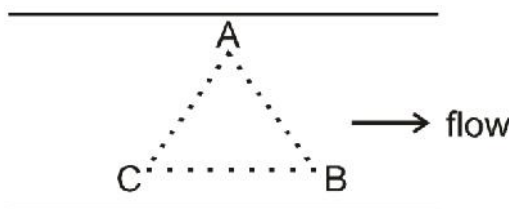
Two graphs of number of nuclei of A and B versus time is given. Which of the following options is/are correct?



- A) $t_2 - t_1 = 4$ B) $t_2 - t_1 = 2$ C) $t_1 = 2\log_2 5$ D) $t_2 = \log_2 100$

14. A, B and C are three rafts floating in a flowing river such that they always form an equilateral

triangle. A swimmer whose swimming speed is constant(w.r.t river) swims from A to B, then B to C and finally C to A along straight lines. If the time taken to go from B to C is t_0 , which of the following is/are correct?



- A) Time taken to go from A to B is $\sqrt{3}t_0$
- B) Time taken to go from A to B is t_0
- C) Time taken to go from C to A is $2\sqrt{3}t_0$
- D) Time taken to go from C to A is t_0

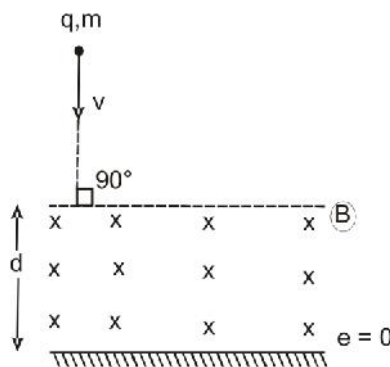
SECTION – III (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 15 and 16:

A small ball of mass m and charge q is projected with velocity v perpendicular to the magnetic field above a plane surface as shown. $d = mv/2qB$. Gravity is absent everywhere. The lower surface is smooth and perfectly inelastic as shown. Answer the following questions for the given situation.



15. Time spent by the ball in the magnetic field is:

- A) $2\pi m / 3qB$ B) $3\pi m / 4qB$ C) $\pi m / 3qB$ D) $3\pi m / 2qB$

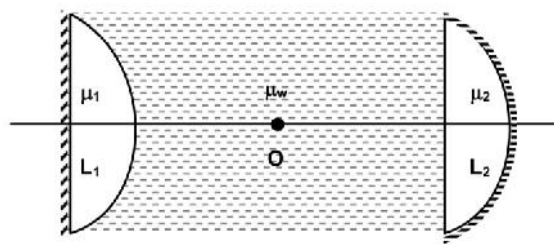
16. Displacement of the ball during the time spend in the magnetic field is:

- A) $(3 - \sqrt{3})d$ B) $(2 - \sqrt{3})d / 2$ C) $2(2 - \sqrt{3})d$ D) $\sqrt{3}d$

Paragraph For Questions 17 and 18:

A cylindrical tube filled with water ($\mu = 4/3$) is closed at its both ends by two thin silvered plane convex lenses as shown in the figure. Refractive index of lenses L_1 and L_2 are 2.0 and 1.5 while their radii of curvature are 5 cm and 9 cm respectively.

A point object is placed somewhere at a point O on the axis of cylindrical tube. It is found that the object and images coincide with each other.



17. The position of object w.r.t. lens L_1 is

- A) 8 cm B) 10 cm C) 12 cm D) 14 cm

18. The position of object w.r.t. lens L_2 is

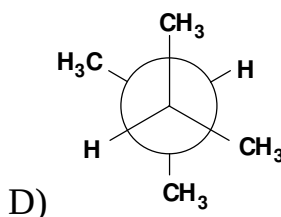
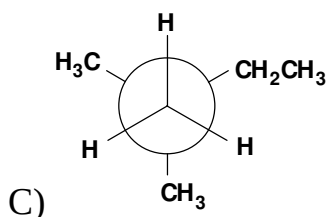
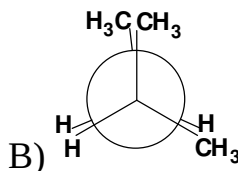
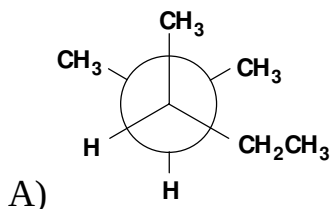
- A) 8 cm B) 10 cm C) 12 cm D) 14 cm

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

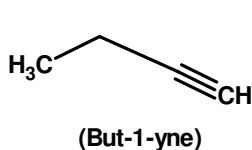
This section contains **7 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme +3 for correct answer , 0 if not attempted and -1 in all other cases.

19. Which of the following Newmann's projections represent conformations of 2,3-dimethyl butane ?



20.



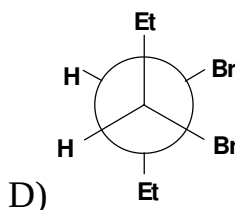
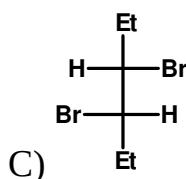
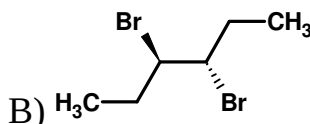
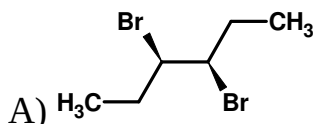
1) NaNH_2

2) $\text{C}_2\text{H}_5\text{I}$

3) Na , Liq NH_3

4) Br_2 , CCl_4

Major product is ?



21. Which of the following combination can produce blue coloured octahedral complex ?

A) $\text{Ni}^{2+}(\text{aq})$ and excess NH_4OH

B) $\text{Cu}^{2+}(\text{aq})$ and excess NH_4OH

C) $\text{Co}^{2+}(\text{aq})$ and excess NH_4SCN

D) $\text{Co}^{2+}(\text{aq})$ and excess $\text{NaNO}_2 / \text{CH}_3\text{COOH}$

- Reaction coordinate diagram for the reaction $A \rightarrow 3B$ and $A \rightarrow 2C$. The diagram shows two competing pathways from reactant A. The upper pathway leads to product 3B with an activation energy $E_a = 10 \text{ cal/mol}$. The lower pathway leads to product 2C with an activation energy $E_a = 20 \text{ cal/mol}$. The energy level of reactant A is higher than the energy level of product 2C, and lower than the energy level of product 3B.

A) 15 cal/mole B) $\frac{130}{11}$ cal/mole C) 12.5 cal/mole D) 12 cal/mole

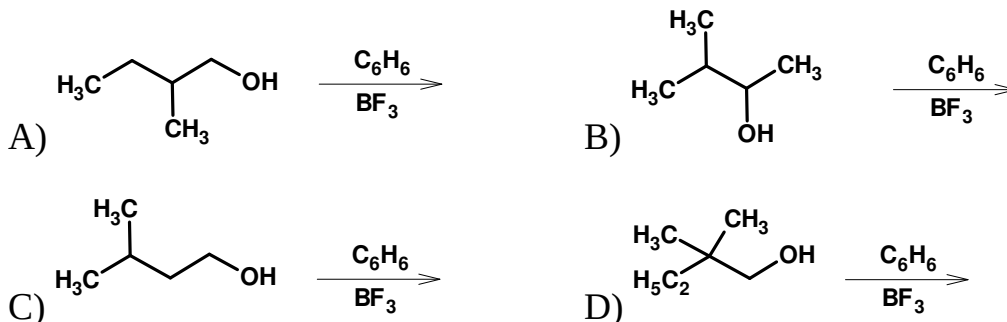
SECTION-II

(ONE OR MORE OPTIONS CORRECT TYPE)

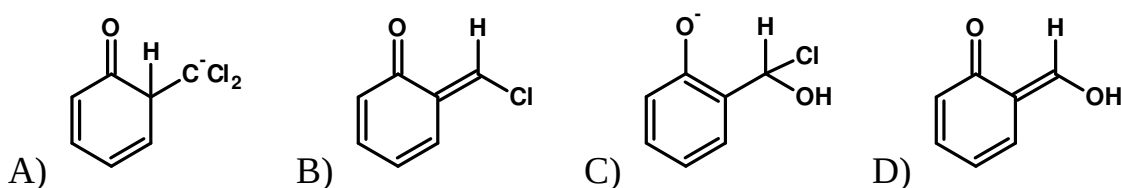
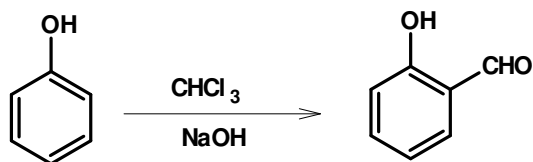
This section contains 7 multiple choice questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +4 for all correct options, 0 if not attempted and -2 in all wrong cases.

26. Which of the following give tert-pentylbenzene as the major product?



27. Which of the following intermediate product(s) is (are) formed during given Reimer-Tiemann reaction?



28. Choose the **correct** steps involved in extraction of Zinc from zinc blende:

- A) Concentration of the ore by froth floatation process by using NaCN as depressant
- B) Roasting of concentrated ore to convert it into ZnO.
- C) Reduction of ZnO by excess carbon above boiling point of zinc.
- D) Refining of the metal by distillation

29. Which of the following complex(es) can have at least two enantiomeric pairs?

- A) $[\text{Co}(\text{en})(\text{H}_2\text{O})_2(\text{NH}_3)_2]\text{Cl}_3$
- B) $[\text{Pt}(\text{gly})_2\text{Cl}_2]$
- C) $[\text{Co}(\text{gly})_3]$
- D) $[\text{Cr}(\text{NH}_3)_3\text{Cl}_2(\text{H}_2\text{O})]\text{Cl}$

30. The standard reduction potential of $Cl^- / AgCl / Ag$ is 0.2 V and that of Ag^+ / Ag electrode is 0.8 V. Which of the following is a true statement ? (Take $\frac{RT \times 2.303}{F} \approx 0.06$)
- A) The solubility of AgCl in pure water is 10^{-5} moles/litre.
- B) The solubility of AgCl in water is more than that of AgBr.
- C) The maximum amount of AgCl that can dissolve in $10^6 L$ of 0.1 M $AgNO_3$ is 1 m mole.
- D) In presence of 0.1 M NH_3 , the solubility of AgCl is less than in 0.1 M KCN.
31. The boiling point of benzene is 353.23 K. When 1.80 g of a non-volatile, non-dissociative & non-associative solute was dissolved in 90 g of C_6H_6 , the boiling point of benzene is raised to 354.11 K. Given : K_b & K_f of C_6H_6 is 2.53 & 5.12 $K \cdot kg \cdot mol^{-1}$. Which of the following is/are correct choice(s) ?
- A) The molar mass of solute is $57.5 g \cdot mol^{-1}$.
- B) The molar mass of solute is $57 g \cdot mol^{-1}$.
- C) The depression in freezing point for the same solution is 1.68 K.
- D) The depression in freezing point for the same solution is 1.78 K.
32. Critical constants for a real gas are given as $T_c = 180 K, V_c = 0.123 L / mole, P_c = 45 atm$. The correct statement for the given real gas is
- A) The actual volume of a single gas molecule is $\left(\frac{0.123 \times 10^{-3}}{6.023 \times 10^{23}} \right) cc$.
- B) $b = 3 \times 0.123 L \cdot mol^{-1}$.
- C) The Boyle's temperature is more than 180 K.
- D) Gas cannot be liquefied at 200 K.

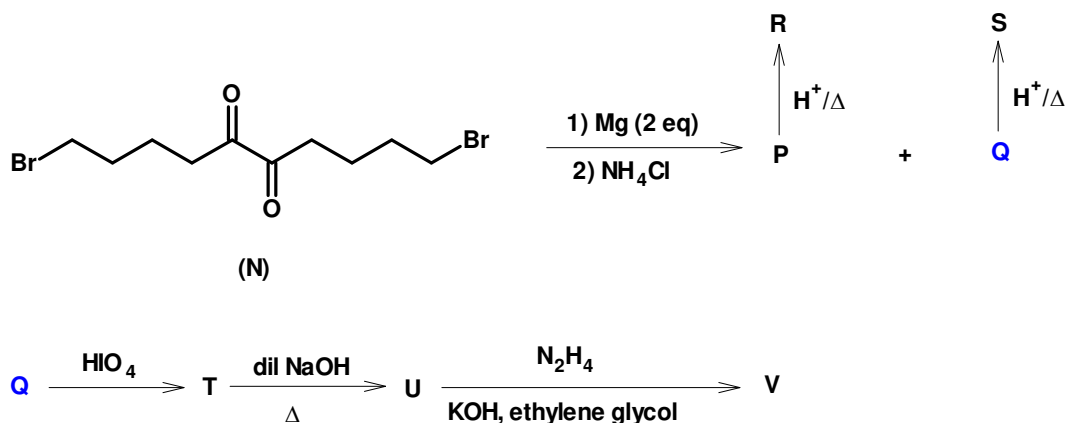
SECTION – III (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 33 and 34:

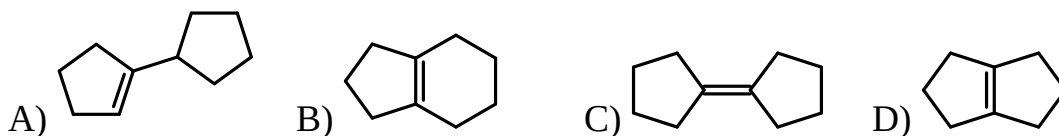
The reaction of compound **N** with Mg (2eq) in $(\text{C}_2\text{H}_5)_2\text{O}$ followed by addition of NH_4Cl gives **P** and **Q**. **P** and **Q** are constitutional isomers. The compound **P** and **Q** possess same molecular formula. The compound **Q** on treatment with HIO_4 gives **T**. The reaction of **T** with $\text{dil NaOH} / \Delta$ produces **U** which on reduction by $\text{NH}_2 - \text{NH}_2 / \text{KOH}$, Ethylene glycol gives **V**.



33. Relation between **[R]** and **[S]** is

- A) Enantiomer B) Diastereomer C) Identical D) Structural isomers

34. The product **[V]** is



Paragraph for Questions 35 and 36:

A black ore (**P**) on oxidative fusion with KOH gives a green coloured compound (**Q**) which turns into a purple coloured compound **R** on reacting with $O_3(g)$. **R** in dilute alkali reacts with ethene gas and forms (**P**) at $0^\circ C$.

35. Choose the **incorrect** statement for (**Q**) and (**R**):

- A) are coloured due to LMCT
- B) have same geometry around central atom
- C) are diamagnetic
- D) contain different number of π -bonds around central atom

36. Choose **incorrect** statement for **R**:

- A) On thermal decomposition produces O_2 gas
- B) Isostructural with $KClO_4$
- C) Disproportionate in presence of acid to give (**P**) and (**Q**)
- D) Not used as a primary standard

SECTION – I
(SINGLE CORRECT CHOICE TYPE)

This section contains **7 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE is correct**.

Marking scheme +3 for correct answer, 0 if not attempted and -1 in all other cases.

37. The polynomial $(x^{2020} + x^{2021} + 2)^{2021} = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ value of $-2a_0 + a_1 + a_2 - 2a_3 + a_4 + a_5 - 2a_6 + \dots$ is
- A) 2 B) -2^{2022} C) 1 D) -2
38. For which of the following ordered pairs (a,b) , two tangents of parabola $y^2 = 4ax$ are normal to circle $x^2 + y^2 - 2ax - 2by + c = 0$
- A) (1,2) B) (2,1) C) (2,5) D) (5,2)
39. Number of real solutions of the equation $[8x^3] + [4x^2] + [2x] = \{x\} - 1$ where $[.]$ & $\{.\}$ denotes G.I.F & fractional part respectively is
- A) 0 B) 1 C) 2 D) 3
40. The area of the triangle formed by points of intersection of plane $ax + by + cz + d = 0$ with x, y, z axes is Δ . If minimum value of $\frac{\Delta(a^2 + b^2 + c^2)}{d^2}$ is $3\sin\theta, \theta \in \left(0, \frac{\pi}{2}\right)$, then θ lies in
- A) $\left(0, \frac{1}{3}\right)$ B) $\left(\frac{1}{3}, \frac{2}{3}\right)$ C) $\left(\frac{2}{3}, 1\right)$ D) $\left(1, \frac{4}{3}\right)$
41. Number of real values of "x" for which the three vectors $\vec{a} = (\sin^{-1} x)\hat{i} + (\cos^{-1} x)\hat{j}$, $\vec{b} = (\sin^{-1} x)\hat{j} + (\cos^{-1} x)\hat{k}$, $\vec{c} = (\sin^{-1} x)\hat{i} + (\cos^{-1} x)\hat{k}$ are coplanar is
- A) 0 B) 1 C) 2 D) Infinite
42. Volume of the tetrahedron formed by the planes with equation $y + z = 0, z + x = 0, x + y = 0, x + y + z = 2$ is
- A) $4/3$ B) $8/3$ C) $16/3$ D) $32/3$

43. From point $P(h,k)$ tangents PA and PB are drawn to circle $x^2 + y^2 - 2x - 4y + 1 = 0$ such that slope of chord AB is 2, then

A) $h + 2k = 5$

B) $h^2 + k^2 - 2h - 4k = 5$

C) Area of $\triangle APB$ is $\frac{4}{\sqrt{3}}$

D) area of $\triangle APB$ is $2\sqrt{3}$

SECTION-II

(ONE OR MORE OPTIONS CORRECT TYPE)

This section contains 7 multiple choice equations. Each question has four choices (A) (B), (C) and (D) out of which **ONE** or **MORE THAN ONE** are correct.

Marking scheme: +4 for all correct options, 0 if not attempted and -2 in all wrong cases.

44. bag A, B, C consists of 'n' balls each ($n \geq 3$). "A" contains "n-2" white and "2" red balls. "B" contains "1" white and rest red. "C" contains "2" white and rest red. A dice is thrown to select the bag. If a prime number comes on the dice we take a ball from "Bag A". if a non prime number comes, bag "B" or "C" is chosen with equal probability and then a ball is taken from it. Then which of the following are true?

A) Probability of selecting a red ball is $\frac{2n+2}{4n}$

B) Probability of selecting a white ball is $\frac{2n-1}{4n}$

C) Probability of selecting a red ball is less than $\frac{1}{2}$

D) Probability of selecting a white ball is less than $\frac{1}{2}$ and greater than $\frac{5}{12}$

45. $f(x) = \frac{9x^2 + 18x \sin x - 9\cos^2 x + 5}{x + \sin x}$, set of values of k for which $f(x) = k$ has exactly two solutions is S. S contains

A) (0,1)

B) (1,2)

C) $(0, \infty)$

D) $(-1,0)$

46. 'CHANDLER' and 'JANICE' are two given words. A letter is selected from both the words. Then which of the following is/are correct?

- A) No. of ways in which there is atleast one vowel among them is 40
- B) No. of ways in which there are two distinct vowels among them is 4
- C) No. of ways in which both letters are same is 16
- D) No. of ways in which both letters are same and are vowels is 4

47. $ax + (a+1)y + az = 0$, $(a+1)x + a(y+z) = 0$ and $a(x+y) + (a+1)z = 0$ are the three plane equations. Then which of the following statements is/are true?

- A) No two planes are parallel for any $a \in \mathbb{R}$
- B) If $a = \frac{-1}{3}$, the three planes form a triangular prism
- C) If $a = \frac{-1}{3}$, the three planes have a common line of intersection
- D) If $a = 0$, the three planes have a common line of intersection

48. Which of the following are factors of $\frac{1}{1 - \cos \frac{\pi}{9}} + \frac{1}{1 - \cos \frac{5\pi}{9}} + \frac{1}{1 - \cos \frac{7\pi}{9}}$

- A) 2
- B) 3
- C) 5
- D) 7

49. Let $F(n) = \begin{vmatrix} 10+n & 9 & 7 \\ 10 & 9+n & 7 \\ 10 & 9 & 7+n \end{vmatrix}$, $D(n)$ = number of positive integral divisors of n , then

which of the following is correct

- A) $FoD(11) > DoF(11)$
- B) $DoDoF(11) = 5$
- C) $DoFoDoDoD \dots D(1) = 4$
- D) $FoD \equiv DoF$

50. Let $A = \{z_1 : z_1^{16} = 1, z_1 \in \mathbb{C}\}$, $B = \{z_2 : z_2^{72} = 1, z_2 \in \mathbb{C}\}$ and $P = \{z_1 z_2 : z_1 \in A, z_2 \in B\}$ are three sets of complex roots of unity (where $\mathbb{C}$ denotes set of complex numbers) and $n(A)$ represents cardinality of A , then

- A) $n(A \cap B) = 8$
- B) $n(A \cap B) = 4$
- C) $n(P) = 144$
- D) $n(P) = 72$

SECTION – III (PARAGRAPH TYPE)

This section contains **2 groups of questions**. Each group has 2 multiple choice questions based on a paragraph. Each question has 4 choices A), B), C) and D) for its answer, out of which **ONLY ONE** is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases.

Paragraph For Questions 51 and 52:

Let $f(x) = \begin{cases} a|x| + b; & x \geq 1 \\ x + c; & 0 \leq x < 1 \\ x^2 + d|x| - 1; & x < 0 \end{cases}$ is a differentiable function from $\mathbb{R}$ to $\mathbb{R}$, then

51. $\lim_{x \rightarrow a^-} f([x^2 - x]) + \lim_{x \rightarrow d^+} f(\{x^2 - x\})$ is: (where $[.]$ and $\{.\}$ denotes G.I.F. fractional part)

- A) 0 B) 1 C) -1 D) -2

52. Which of the following statement is true?

- A) $y = f(x)$ is symmetrical with respect to a point (α, β)
 B) $y = f(x)$ is symmetrical with respect to line $x = k$
 C) $f(x)$ satisfies rolle's theorem in the interval $[0, k]$ for some $k \in \mathbb{R}^+$
 D) $f(x)$ satisfies rolle's theorem in the interval $[-k, 0]$ for some $k \in \mathbb{R}^+$

Paragraph For Questions 53 and 54:

Consider the function $f(x)$ and $g(x)$, both defined from $\mathbb{R} \rightarrow \mathbb{R}$ where

$$f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t) dt \text{ and } g(x) = x - \int_0^1 f(t) dt, \text{ then}$$

53. Minimum value of $f(x)$ is

- A) 0 B) 1 C) $\frac{3}{2}$ D) Doesn't exist

54. The area bounded by $y = -g(x)$ and the curve $y = f(x)$ (in sq. units) is ____

- A) $\frac{9}{4}$ B) $\frac{64}{9}$ C) $\frac{9}{8}$ D) $\frac{32}{27}$



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ICON CENTRAL OFFICE, MADHAPUR-HYD

Sec: Sr.IPLCO_IC_ISB_LIIT

Time: 09:00AM TO 12:00 Noon

GTA-8

Date: 01-05-16

Max.Marks: 270

KEY SHEET

PHYSICS

1	True	2	False	3	True	4	True	5	True	6	BC
7	AB	8	C	9	C	10	BC	11	ABC	12	A
13	AC	14	BC	15	B	16	C	17	B	18	C
19	A-P-WY; B-QRS-WZ; C-PRS-WZ; D-PS-X	20	A-S-Y; B-R-X; C-Q-W; D-P-Z								

CHEMISTRY

21	TRUE	22	FALSE	23	TRUE	24	TRUE	25	TRUE	26	B
27	B	28	D	29	C	30	D	31	D	32	ABC
33	ABD	34	A	35	A	36	C	37	C	38	C
39	A-Q,Y B-S,W C-P,X D-R,Z	40	A-R,W B-P,Z C-S,X D-Q,Y								

MATHS

41	False	42	TRUE	43	TRUE	44	TRUE	45	TRUE	46	BCD
47	ABD	48	AC	49	ABC	50	AC	51	AD	52	ACD
53	AC	54	AC	55	C	56	A	57	D	58	A
59	A-R,Y B-S,X C-P,Z D-Q,W	60	A-RY B-PW C-SX D-QX								



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ICON CENTRAL OFFICE, MADHAPUR-HYD

Sec: Sr.IPLCO_IC_ISB_LIIT
TIME : 02:00 PM TO 05:00 PM

JEE ADVANCED
NEW MODEL-IV_P2

DATE : 01-05-16
MAX MARKS : 246

KEY & SOLUTIONS

PHYSICS

1	100	2	007	3	010	4	008	5	CD
6	BC	7	BC	8	AD	9	A	10	D
11	D	12	B	13	B	14	C		
15	a-A; b-D; c-C; d-B	16	a-B; b-B; c-D; d-A						

CHEMISTRY

17	6	18	4	19	3	20	5	21	BCD
22	BC	23	ACD	24	ABD	25	D	26	C
27	A	28	B	29	B	30	C		
31	a-C; b-D c-B; d-A	32	a-AD; b-AB c-BC; d-A						

MATHEMATICS

33	219	34	47	35	480	36	0	37	ACD
38	ABCD	39	AD	40	C	41	B	42	A
43	C	44	B	45	C	46	B		
47	a-C b-A c-D d-B	48	a-B, b-D, c-A, d-A						



TG ~ @bohring_bot

NARAYANA
IIT ACADEMY
INDIA



Sec: SR-IIT-N-SUPER CHAINA

GTA-18(P-1)

Dt: 14.06.20

Time: 3 Hrs.

2019(P-1)

Max.Marks:186

KEY SHEET PHYSICS

1	A	2	A	3	A	4	C	5	AC
6	AC	7	BD	8	CD	9	AC	10	AC
11	BD	12	ABCD	13	14.00	14	75.00	15	96.00
16	50	17	0.25	18	0.01				

CHEMISTRY

19	A	20	C	21	B	22	D	23	AB
24	ABCD	25	ABCD	26	ABCD	27	ABCD	28	ABCD
29	ABCD	30	ABCD	31	-810.6	32	99.20	33	10
34	4.00	35	4.00	36	3.00				

MATHS

37	B	38	D	39	A	40	C	41	BD
42	ACD	43	AB	44	AC	45	AB	46	ACD
47	ABC	48	AD	49	20	50	4.5	51	1.75
52	0.14 TO 0.15	53	0.5	54	1.57 TO 1.58				



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NARAYANA
IIT ACADEMY
INDIA



Sec: SR-IIT-N-SUPER CHAINA

GTA-18(P-II)

Dt: 14.05.20

Time: 3 Hrs.

2019(P-II)

Max.Marks:186

KEY SHEET

PHYSICS

1	ACD	2	ABC	3	CD	4	CD	5	BC
6	AC	7	C	8	CD	9	5	10	3
11	0	12	0	13	2	14	3	15	D
16	B	17	C	18	C				

CHEMISTRY

19	AB	20	D	21	A	22	C	23	CD
24	A BC	25	ACD	26	ABD	27	6	28	6
29	2	30	3	31	6	32	2	33	C
34	C	35	A	36	B				

MATHEMATICS

37	BC	38	A	39	AC	40	ABCD	41	ABC
42	AC	43	ACD	44	BCD	45	1.25	46	15
47	1.25	48	2	49	2020	50	9	51	B
52	D	53	B	54	A				



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ICON CENTRAL OFFICE, MADHAPUR-HYD

Sec: Sr.IPLCO/IC/ISB/UIT

JEE-ADVANCE_GTA-5

DATE : 21-04-16

Time: 3 Hours

New Model-VII_P1-Model

MAX MARKS : 240

KEY & SOLUTIONS

MATHS

1	2	3	4	5	6	7	8	9	10
C	B	B	D	B	AC	AC	ABC	ABC	ABCD
11	12	13	14	15	16	17	18	19	20
ABD	AB	ABD	AC	AB	ABD	081	400	000	004

PHYSICS

21	22	23	24	25	26	27	28	29	30
D	A	A	C	D	D	BD	AC	BD	CD
31	32	33	34	35	36	37	38	39	40
AC	BC	C	BC	B	B	050	010	100	025

CHEMISTRY

41	42	43	44	45	46	47	48	49	50
C	D	C	D	C	AB	AC	ABCD	ACD	AC
51	52	53	54	55	56	57	58	59	60
B	AD	ACD	D	CD	A	002	421	114	180



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NARAYANA
IIT ACADEMY
INDIA



Sec: SR-IIT-N-SUPER CHAINA

Time: 3 Hrs.

GUTA-1(P-2)

Dt: 30.04.20

Max.Marks:186

KEY SHEET PHYSICS

1	AB	2	BC	3	AC	4	BC	5	BCD
6	ABD	7	ABD	8	AC	9	40.00	10	8.00
11	4.00	12	0	13	3.10	14	0.04	15	A
16	B	17	A	18	D				

CHEMISTRY

19	AC	20	ABC	21	AB	22	ABCD	23	AB
24	ABCD	25	ACD	26	ACD	27	4	28	53.33
29	5	30	6	31	2.85 TO 2.86	32	3	33	A
34	A	35	A	36	C				

MATHS

37	ACD	38	C	39	AC	40	AD	41	ACD
42	ABC	43	ABD	44	ABD	45	3	46	34
47	1.73	48	3	49	8	50	18.00	51	A
52	B	53	D	54	B				



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NARAYANA
IIT ACADEMY
INDIA



Sec: SR-IIT-N-SUPER CHAINA
Time: 3 Hrs.

GTA-16
2017(P-1)

Dt: 31.05.20
Max.Marks:183

KEY SHEET

PHYSICS

1	ABD	2	BD	3	AC	4	AC	5	AC
6	AC	7	ABC	8	4	9	4	10	3
11	3	12	7	13	C	14	A	15	A
16	D	17	A	18	B				

CHEMISTRY

19	C	20	BD	21	ACD	22	AB	23	AB
24	ABC	25	BCD	26	3	27	7	28	6
29	8	30	7	31	D	32	D	33	A
34	C	35	A	36	D				

MATHS

37	AB	38	AD	39	ABCD	40	ABC	41	AC
42	AB	43	ABC	44	3	45	5	46	6
47	2	48	3	49	D	50	C	51	B
52	C	53	A	54	D				



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NARAYANA
IIT ACADEMY
INDIA



Sec: SR-IIT-N-SUPER CHAINA
Time: 3 Hrs.

GTA-16
2017(P-2)

Dt: 31.05.20
Max.Marks:183

KEY SHEET

PHYSICS

1	D	2	C	3	A	4	B	5	C
6	A	7	C	8	ABD	9	AD	10	ABCD
11	ABC	12	BCD	13	BCD	14	BD	15	A
16	A	17	B	18	A				

CHEMISTRY

19	D	20	B	21	A	22	C	23	B
24	B	25	C	26	ABC	27	ABCD	28	BCD
29	BC	30	ABCD	31	AD	32	CD	33	C
34	C	35	C	36	C				

MATHS

37	D	38	C	39	B	40	D	41	C
42	C	43	A	44	B	45	ABCD	46	AB
47	AC	48	AB	49	AC	50	AC	51	C
52	D	53	B	54	D				